

BASE24 Release 6.0 Version 8

Implementation Guide



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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol “|” in the outside margin.

Release 6.0 Version 8 (M. Ahrens, D. Barton, R. Duvall, S. Eschliman, D. Grote, K. Koeppen, B. Kurzenberger, R. McCall, C. Meyers, C. Rink, R. Schieuer, S. Schmitt, J. Steffes, L. Watson, K. Wieler, J. Wishart)
– Nov-2007

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1: Introduction

The ***BASE24 Release 6.0 Version 8 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 Release 6.0 version 7 systems to BASE24 Release 6.0 version 8.

This section provides an overview of the planning requirements for BASE24 users migrating to Release 6.0 version 8. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for Release 6.0 version 8, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to Release 6.0 version 8. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Version 8 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to Release 6.0 version 8.

BASE24 Release 6.0 products include the following:

- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value-Added Products:
 - BASE24-card
 - BASE24-customer service
 - BASE24-InfoBase
 - BASE24-inventory

Who should use this guide?

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to Release 6.0 version 8. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to Release 6.0 version 8, users must be on Release 6.0 version 7. Refer to the ***BASE24 Release 6.0 Version 7 Implementation Guide*** to assist in planning the tasks necessary to get to Release 6.0 version 8.

Using this guide, you can do the following:

- Determine what ACI services are available to assist in the migration to Release 6.0 version 8.
- Develop a project plan to migrate to Release 6.0 version 8.
- Determine staffing and training needs.
- Gain a detailed understanding of the Release 6.0 version 8 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 Release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on an XPNET 3.0 or 3.1 system.

FIX

FIX 2.28 (or newer) is required to apply any new BASE24 fix files produced after 19 October 2006. FIX 2.28, like the previous version of FIX, will allow application of fix files to a specified file name in a concatenated list of fix files. This option can be combined with the FIRSTSCN and LASTSCN options to apply only fixes to the specified file name in a specified SCN range. As of 19 October, FIX 2.28 (or the most current version of FIX) was automatically shipped with each software order. Please replace your existing FIX object with the most current version.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to Release 6.0 version 8. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for Release 6.0 version 8 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's Release 6.0 version 7 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of Release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from Release 5.3 to the latest version of Release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of Release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from Release 6.0 version 1 to Release 6.0 version 2. No changes are made to these programs other than necessary fixes. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from Release 6.0 version 2 to Release 6.0 version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from Release 6.0 version 3 to Release 6.0 version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from Release 6.0 version 4 to Release 6.0 version 5. No changes are made to these programs other than necessary fixes.

The *UC09 subvolumes (e.g., AT60UC09, BA60UC09, PS60UC09) contain the conversion program for users migrating from Release 6.0 version 5 to Release 6.0 version 6. No changes are made to these programs other than necessary fixes.

The *UC0A subvolumes (e.g., AT60UC0A, BA60UC0A, PS60UC0A) contain the conversion program for users migrating from Release 6.0 version 6 to Release 6.0 version 7. No changes are made to these programs other than necessary fixes.

The *UC0B subvolumes (e.g., AT60UC0B, BA60UC0B, PS60UC0B) contain the conversion program for users migrating from Release 6.0 version 7 to Release 6.0 version 8. No changes are made to these programs other than necessary fixes.

If a Release 5.3 user wants to convert to the latest version of Release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 version 1 user wants to convert to Release 6.0 version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 version 2 user wants to convert to Release 6.0 version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 version 3 user wants to convert to Release 6.0 version 4, the conversion programs on *UC07 should be run.

If a Release 6.0 version 4 user wants to convert to Release 6.0 version 5, the conversion programs on *UC08 should be run.

If a Release 6.0 version 5 user wants to convert to Release 6.0 version 6, the conversion programs on *UC09 should be run.

If a Release 6.0 version 6 user wants to convert to Release 6.0 version 7, the conversion programs on *UC0A should be run.

If a Release 6.0 version 7 user wants to convert to Release 6.0 version 8, the conversion programs on *UC0B should be run.

If a Release 6.0 version 1 user wants to convert to Release 6.0 version 8, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user should run the conversion programs on *UC06, *UC07, *UC08, *UC09 and on *UC0A. Finally, the user should run the conversion programs on *UC0B.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the Release 6.0 version 8 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 Release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 Release 6.0 network. This document is included as a Microsoft Word document and is shipped with the Release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to Release 6.0 version 8. This document is included as a HP NSK TGAL document and is shipped with the Release 6.0 version 8 software. This document is named AAREAD0A and is located on the BA60UC0A subvolume.

HELP24 and HELP24 InfoLink

HELP24 and HELP24 InfoLink are excellent sources of assistance and information. The following access options appear on the support page of the ACI Worldwide Inc Internet website (<http://www.aciworldwide.com/support>).

- A current listing of HELP24 phones worldwide.
- An option to send an e-mail message to HELP24.
- An option to sign up for HELP24 InfoLink.
- An option to access HELP24 InfoLink once you have received a user ID and password.

Once you can log on to HELP24 InfoLink, you can do the following:

- Access the HELP24 database to open a new Clarify case.
- Access installation and fix release procedures online.
- Access all BASE24, BASE24-es, and NET24-XPNET documentation online.
- Search the site for information about BASE24.

The **HELP24 eSupport Users Guide** contains procedures for opening a new Clarify case, accessing the Clarify database, and accessing documentation on-line. Once you log on to HELP24, the path to the **HELP24 eSupport Users Guide** is Support Center> InfoLink > HELP24 eSupport Users Guide. Once you log on to HELP24, the path to the installation procedures is Support Center> InfoLink > Product Info > BASE24> BASE24 6.0 Install Info and Guides. The menu displayed gives you three options:

Pre-Install Checklists- for BASE24 6.0 and ITS

Install Guides—Procedures for installing a software release or version for the first time.

Retail Install Guides—Procedures for installing Retail 3.1 or Check Auth 3.2.

Once you log on to HELP24, the path to the Fix Information is Support Center> InfoLink > Fix Information > BASE24

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with Release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A software routine within existing code that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

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2: Functionality

This section contains an overview of the BASE24 Release 6.0 version 8 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the Release 6.0 version 8 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled in Release 6.0 version 8. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value Added Products

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall Release 6.0 version 8 offering.

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BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in Release 6.0 version 8, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Changes for all BASE24 Release 6.0 version 8 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

- Refresh Support of PauseACK Improvements

This enhancement was done to the base Refresh process to ensure that when a refresh has completed successfully and performs a broadcast notify, the Refresh process will more efficiently handle the response it receives from the processes it notified.

- Skipping HISO SAF

The purpose of this enhancement is to resolve a situation where HISO is skipping certain records and sending SAF records to the wrong DPC. The resolution involves re-ordering fields in the key to the SAF, to ensure HISO does not miss a record or send one to the wrong DPC due to the timestamps in the key being too close together.

PCI - Audit Enhancement

This enhancement provides additional auditing of user access to cardholder data as required by PCI DSS V1.1. Before this enhancement, audit logs were written when cardholder data was added, updated, and deleted. This enhancement adds audit logging for access to cardholder data (record reads).

Prerequisites: None

PCI - Initial User Password

This enhancement requires the supervisor to set a password for each new user and on password reset of an existing user to adhere to PCI DSS V1.1 requirements. As currently required, these passwords must meet the password complexity requirements and the user will only be able to use these passwords for initial login and then be required to change the initial or reset password to a new password upon first use.

Prerequisites: None

PCI - Masking on Pathway

This enhancement allows the setting of Pathway screen masking of cardholder information on Pathway screens considered sensitive by PCI DSS V1.1. These include: Primary Account Number (PAN), Cardholder Name, Service Code, and Expiration Date. Since access to this cardholder data can be restricted on need to know basis through proper user policies and procedures, this is not a requirement of PCI DSS V1.1, but this enhancement provides extra protection of cardholder data.

Prerequisites: None

PCI - Masking on Reports

This enhancement allows the option of masking cardholder information on reports considered sensitive by PCI DSS V1.1. These include: Primary Account Number (PAN), Cardholder Name, Service Code, and Expiration Date. Since access to this cardholder data can be restricted on a need to know basis through proper user policies and procedures, this is not a requirement of PCI DSS V1.1, but this enhancement provides extra protection of cardholder data.

Prerequisites: None

Optional Enhancements

There are no BASE24 Base Release 6.0 version 8 enhancements that are optional.

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BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the Release 6.0 version 8 offerings. All BASE24-atm users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

Log ATM Totals to EMS

This project enhances the BASE24-atm authorization process so that it logs ATM total and balancing messages to the EMS log if the TLF file is unavailable.

Multi-language Indicator for Statement Print

This enhancement provides host systems and switch interchanges (where applicable) with an indicator specifying which language was chosen by the consumer at the ATM. This indicator may then be used by the issuer to determine which language the statement requested should be printed in. Both the NCR 5xxx Device Handler and the Diebold 10xx/478x Device Handler were also enhanced to populate a new token field with the language preference of the consumer.

Prerequisites: The NCR 5xxx Device Handler and/or Diebold 10xx/478x Device Handler is required.

NCR NDC+ Device Handler Global Space Issue

This project analyzed the NCR NDC+ Device Handler and made the necessary changes to reduce its use of standard global space.

Prerequisites: The NCR 5xxx Device Handler is required.

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Optional Enhancements

The following BASE24-atm Release 6.0 version 8 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Agilis 2.0 Enhancements

This project enhances the Entibold ATM Configuration module to support the following:

- "ESC ZW - Select Entry Map" command
- "ESC ZX - Defined Entry Map" command
- "SD - Supplemental Dispense" state

In addition, this project will add any new status codes added as part of Agilis 2.0.

Prerequisites: The Diebold 10xx/478x Device Handler is required.

ATC Enhancements

This enhancement is to ATM Authorization to support an ATC Out-of-Range validation as part of a pre-screening level check. The enhancement logic will be performed on all contactless transactions and compare the ATC value received in a request to the ATC value maintained on the CAF. If the variance between the two exceeds a new ATC range parameter (ATC Limit field) defined in the CPF, the transaction can either be declined or ignored.

This enhancement will reduce the chances of fraud by checking the variance amount for which the ATC can vary between what is processed in the transaction message to that stored on the CAF.

For those customers that manage the ATC on the CAF file, this enhancement will also support the ability to add the ATC to a token field which can be provided to and managed by an external host database. This enhancement will provide customers that support a full file CAF refresh the capability of modifying their CAF refresh format to support updating the ATC with the last known ATC used to authenticate the dynamic card verification cryptogram.

ATM Dynamic Currency Conversion

This project enhances BASE24-atm to enable acquirers to convert a withdrawal amount to a cardholder's home currency before sending a transaction to an interchange, rather than send it in the original currency and have the issuer perform the conversion. This functionality is referred to as Dynamic Currency Conversion or DCC. Advantages of DCC include greater visibility of charges for the consumer and the opportunity for the acquirer to derive revenue for the conversion.

Prerequisites: The Diebold 10xx/478x or the NCR 5xxx Device Handlers are required. This enhancement applies only to transactions that are sent to an issuer that allows DCC.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Card before Cash on Fast Cash

This project enhances BASE24-atm to allow cash to be dispensed before the card is ejected for Fast Cash transactions, as was previously supported only for non-Fast Cash transactions. This enhancement helps customers in some environments to not take the cash and forget their card.

Prerequisites: The Diebold 10xx/478x or the NCR 5xxx Device Handlers are required.

Configurable AUTH for Rear Balancing Transactions

This enhancement gives the BASE24-atm Authorization process the ability to deny rear administrative transactions that contain a PIN, if so configured. This enhancement is needed since PCI rules no longer allow customers to enter PINs into the rear keypad which is not PCI compliant.

Prerequisites: The Diebold 10xx/478x or the NCR 5xxx Device Handlers are required.

Diebold 912 AKDS Signature Support

This enhancement supports those situations where Diebold Agilis software is operating on ATM hardware that use EPPs that are signature-based (NCR and Wincor) and utilize the Diebold 912 protocol. This functionality is becoming more important as multi-vendor solutions become more prominent.

Prerequisites: The Diebold 10xx/478x Device Handler is required.

Enhanced Mobile Top-Up Routing Support

The purpose of this project is to uplift the Auth CSM that facilitates Mobile Top-up transactions in the Link network. Since every Mobile Top-up customer in the Link network requires this CSM, it will save these customers considerable time by eliminating the need to continually retrofit it.

NCR NDC+ Additional M-Status Codes

The purpose of this project is to add new Envelope Dispenser related M-Status code handling to the NCR NDC+ Device Handler.

Prerequisites: The NCR 5xxx Device Handler is required.

NCR NDC+ Increase of Secondary EMV AIDs

The purpose of this project is to increase the maximum number of secondary AIDs that can be configured and downloaded to an NCR ATM from 4 to 12. There are now customers needing at least 8 and NCR supports up to 256.

For more information on EMV processing, please refer to “EMV Support” in the BASE24 Base Optional Enhancements of the BASE24 Release 6.0 Implementation Guide (Version 2).

Prerequisites: The NCR 5xxx Device Handler is required.

NCR Standard APTRA Edge NDC+ Support

This enhancement provides support for NCR's new APTRA Edge specification as a new NCR device extension. The new specification allows for configuration parameters to be sent down to the individual ATM thus eliminating considerable customization that now needs to be done as part of an APTRA Edge project. Initially, this new specification will be used by processors due to the numerous iterations that exist due to multiple customers, but eventually this specification could be used by individual banks.

Prerequisites: The NCR 5xxx Device Handler is required.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Preferred Transaction

This project enhances BASE24-atm to provide the functionality to support the concept of a "my favorite transaction". This enhancement is broken into three discreet functions; the handling of the actual "my favorite transaction", the setup of the variables of the "favorite transaction" and the inquiry on the "favorite transaction" variables. This functionality is only available by utilizing the BASE24-eps IFX Device Handler.

Prerequisites: The BASE24-eps IFX Device Handler is required.

Shared NDC+ BNA Support

This new module provides the support to allow customers to purchase Wincor ATM software (or any other vendor that utilizes the Shared NDC+ specification) that operates ATMs that have Bunch Note Acceptors. These ATMs can be manufactured by Wincor, NCR, or Diebold among others.

Prerequisites: The NCR 5xxx Device Handler is required.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Shared NDC+ CPM Support

This new module provides the support to allow customers to purchase Wincor ATM software (or any other vendor that utilizes the Shared NDC specification) that operates ATMs that have Check Processing Modules. For example, these ATMs can be manufactured by Wincor, NCR, or Diebold.

Prerequisites: The NCR 5xxx Device Handler is required.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Tidel Dynamic Key Management

This new Tidel extension provides support for key synchronization as well as PIN and MAC encryption key management. Key synchronization is defined as the keys must be changed when the number of PIN verify or MAC verify/generate transactions exceeds a predetermined limit, the number of PIN verify or MAC verify/generate failures exceeds a predetermined limit, the number of MAC consecutive failures exceeds a predetermined limit, or twenty-four hours has passed since the last MAC or PIN key change.

Prerequisites: The Tidel Device Handler is required.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Tidel MACing Support and Triple-DES Update

This project will enhance the Tidel Device Handler module to support message authentication or MACing. MACing in an ATM environment involves the ATM creating a MAC code by running the ATM transaction through an encryption process. This code is then revalidated by BASE24 to ensure that no data has been altered during transmission thereby reducing possible fraud. In addition the Tidel Device Handler is being enhanced to support the newest Tidel ATMs which utilize the industry standard two key triple-DES method.

Prerequisites: The Tidel Device Handler is required.

Track 2 Context Data Removal

In 2004, Visa, MasterCard, Discover, American Express, and JCB announced the Payment Card Industry Data Security Standard (PCI) to protect consumer data in acquirer and issuer transactions. The enhancement to BASE24 will protect consumer data as it is carried internally in the BASE24-atm Standard Internal Message (STM). The data protected will be discretionary data from track 1 and/or track 2.

This enhancement applies to those transactions that are authorized by BASE24, a host system, or an interchange. BASE24-atm authorization was also enhanced to remove the track 1 and track 2 discretionary data from the STM response if the transaction is authorized by either BASE24 or a Host. Additionally, response messages received from an interchange may have the discretionary data removed. Additional logic was also added to accept a response message where the discretionary data in track 2 is no longer the same as the original transaction (i.e., it has been masked).

Wincor AKDS support of NCR ATMs

This enhancement in Transaction Security Services and the BASE24-atm NCR 5xxx Device Handler supports the multi-vendor environment where Wincor software is running on NCR ATMs. Even though both Wincor and NCR utilize digital signatures, the Wincor implementation is slightly different from NCR's which necessitated the change.

Prerequisites: The NCR 5xxx Device Handler and the Shared NDC AKDS Extension are required. BASE24-eps Transaction Security Services (TSS) Release 1.0 Version 07.4 is also required.

BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the Release 6.0 version 8 offerings. All BASE24-pos users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Immediate Availability PTDS1 Static Data Changes

This enhancement addresses static data changes for terminal specific data. When static data is changed, a device handler will not retrieve this data from disk until a warm boot occurs. To improve performance, when static data changes occur, SPDH has been enhanced to check the timestamp of the records in the PTDD1 and PTDS1 file, and if different then remove the PTDS1 record from memory which will allow the device handler to reread the record from disk. This will allow changes to terminal records to be made available immediately for online transaction processing.

Lesser Amount for Pre-Auth Purchase with Cash Back

This enhancement will enable BASE24 to support Pre-Authorization Purchase with Cash Back transactions where the issuer has authorized the transaction for a lesser amount. To support this condition, both the Purchase amount and Cash Back amount will be reduced by the lesser amount, which is the difference between the total transaction amount in the request versus the amount approved. This enhancement will allow customers to certify with specific interchanges that require this processing.

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Optional Enhancements

The following BASE24-pos Release 6.0 version 8 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

ATC Enhancements

This enhancement is to POS Authorization to support an ATC Out-of-Range validation as part of a pre-screening level check. The enhancement logic will be performed on all contactless transactions and compare the ATC value received in a request to the ATC value maintained on the CAF. If the variance between the two exceeds a new ATC range parameter (ATC Limit field) defined in the CPF, the transaction can either be declined or ignored.

This enhancement will reduce the chances of fraud by checking the variance amount for which the ATC can vary between what is processed in the transaction message to that stored on the CAF.

For those customers that manage the ATC on the CAF file, this enhancement will also support the ability to add the ATC to a token field which can be provided to and managed by an external host database. This enhancement will provide customers that support a full file CAF refresh the capability of modifying their CAF refresh format to support updating the ATC with the last known ATC used to authenticate the dynamic card verification cryptogram.

APACS 70-2 Enhancements

BASE24-pos APACS device handler has been enhanced to support several new services, which includes support for Auxiliary Data record, Contactless RFID, Secure Internet Validation, and APACS 70 October 2004 compliance.

This enhancement to the APACS device handler adds support for two new record types: Electronic Commerce (type 01), and Extended Terminal Capabilities and Attributes (type 06). Electronic Commerce Auxiliary Data records are needed for support of Secure Internet Validation (SIV). Extended Terminal Capabilities and Attributes Auxiliary Data records are required for Contactless chip and magnetic stripe transactions. The APACS device handler will map data passed into BASE24 via Auxiliary Data records to the appropriate BASE24 token.

The APACS device handler has been enhanced to support Contactless transactions for both chip and magnetic stripe cards. Information regarding the Contactless transaction will be sent by the terminal in an Extended Terminal Capabilities and

Attributes Auxiliary Data record. The BASE24 APACS device handler enhancement supports the processing Auxiliary Data based upon the information received, map the appropriate Point of Service Entry Mode value to the PSTM.

The APACS device handler was enhanced to support Secure Internet Validation (SIV), or 3D Secure. 3D Secure is a method of authenticating the cardholder in an e-commerce transaction. Visa's method is known as Verified by Visa (VbV). MasterCard's method is known as SecureCode. Support for e-commerce Auxiliary Data records is required for SIV support. The APACS device handler will map Auxiliary Data record data (from an e-commerce record) to the appropriate BASE24 token. For Visa transactions, Auxiliary Data will be mapped to the TRANS-STAIN-XID-TKN token. For MasterCard transactions, Auxiliary Data will be mapped to the AUTHN-DATA-TKN.

The APACS device handler also was enhanced to support APACS 70 October 2004 compliance. To be compliant with APACS 70 (October 2004), the APACS device handler was enhanced to support the following:

- Support additional values used to identify new types of e-commerce transactions.
- Support for POS-DATA1 token.
- Support additional values to provide more accurate information to the card schemes.
- Support for field 15 in EFT request messages.
- Support a Reason On-Line code '04' to identify an IC Fallback transaction.
- Support the correct value for Balance Enquiry transactions.
- Support for account funding transactions.
- Remove logic that uses the EMV terminal type in e-commerce transactions (70.1)
- Treat the CID as a mandatory data element in EMV transactions.
- Support for two new subfields in the IC request data,
- Support additional validation on hexadecimal fields.
- Support to ensure the CVM Results field is present in an EMV transaction.
- Support for the CVM Results field in Off-Line Repeat Request messages.
- Support clarification of the Card Details.

Prerequisites: The BASE24-pos APACS POS Device Handler is required

CAP Authentication

This enhancement to BASE24 provides support for the Chip Authentication Program (CAP) which facilitates consumer authentication of internet based transactions using chip cards, a one time password, and EMV technology. CAP will use the smart chip card's ability to validate the customer's PIN, and if valid, will generate an application cryptogram that will act as a dynamic one-time password. The CAP enhancement will facilitate MasterCard's CAP authentication program, and Visa's equivalent program using the same technology.

Dynamic CAF

This enhancement enables BASE24-pos to better manage the processing of the Application Transaction Counter (ATC) as well as other dynamic data that is maintained in the Cardholder Authorization File (CAF). This data is used in both CAP as well as Dynamic Card Verification (DCVV). Data that is considered dynamic could be altered if a full file refresh is performed. To ensure that data is not lost, a new file has been created as an extension to the current CAF that is designed to store dynamic data.

Prerequisites: Requirement includes either Contactless Chip Card – Magnetic Stripe Format or TSS CAP.

Hypercom 3DES in Logon and Other Enhancements

This enhancement to the Hypercom device handler adds supports for double length PIN keys in an "0810" Logon Response message. Double length keys will be carried in Private use Field 63. If Field 63 is not present, the single length keys in Field 62 will be used by the terminal. If Field 63 is present, the key fields of Field 62 will be ignored.

Double length keys require TSS support, which requires use of an HSM. However, Hypercom MACing is highly proprietary and currently supported by BASE24 via software only. Therefore, double length MAC key support will not be possible at this time. To support double length PIN keys in the "0810" Logon Response message, TSS support will be added to the Hypercom device handler.

This enhancement will support those conditions where for EMV transactions approved at the terminal while in Offline mode, the terminal will forward the transaction to BASE24 as part of a batch upload and the Transaction Certificate will now be included in the EMV Request token and logged to the PTLF. This enhancement is to enable the HPDH to map the Transaction Certificate (from field DE55, Tag 9F27 – CID, Tag 9F26 – Application Cryptogram) to the EMV Request token. From there, it will be logged to the PTLF.

This project includes an enhancement to the Hypercom device handler which will add support for two new fields. The two fields defined with in a table include a 4 byte Check Service Flags field, and a 1 byte Options 8 field, for a total of 5 bytes. These

fields will now be supported for download sequences where the Hypercom device handler processes data from the Terminal Configuration Table that resides on the HPTC.

A project was also completed that enhanced the Hypercom device handler to support Contactless Chip and Magnetic Stripe transactions. Hypercom's v3.44 outlined the message specifications related to how the Point of Service (POS) Entry Mode is to be interpreted with field 55 for Contactless Magstripe and Contactless EMV. The Hypercom device handler was enhanced to support differentiating processing of Contactless Magstripe and EMV by the data sent in field 55.

Prerequisites: The Hypercom POS device handler and Hypercom Device Handler EMV Extension (PS-DP390 and PS-CC974)

POS Completion Message Configuration

This enhancement provides BASE24-pos customers with the ability to only send completions to the host when BASE24-pos performed the authorization. BASE24-pos will be enhanced to support new COMPL REQ values in the Issuer Processing Code File (IPCF) which will be used to indicate that completions are to be sent to the host only if BASE24-pos performed the authorization.

Customers will now be able to configure the system to only send all BASE24-pos authorized transactions, including declines, or to only send approved BASE24-pos authorized transactions.

Prerequisites: BASE24 ISO Host Interface

SPDH Message Data Element Encryption

The SPDH Message Data Encryption enhancement will enable customers to protect these data fields by employing either *full message encryption* or *configurable message encryption*. Message data will be single-DES or triple-DES encrypted under the Data Encryption Key (FID I, DID j), which currently supports 16-byte (single-length), 32-byte (double-length) and 48-byte (triple-length) keys.

Full message encryption will allow the customer to encrypt all optional data fields in an ACI Standard POS Message, with the exception of those listed in Table 1.2.1. Full message encryption will be enabled by setting the Data Encryption Type field in the POS Terminal Data File (PTD) to "03" (full message encryption).

Configurable message encryption will allow the customer to encrypt specific optional data fields in an ACI Standard POS Message, with the exception of those listed in Table 1.2.1. Configurable message encryption will be enabled by setting the Data Encryption Type field in the PTD to "04" (configurable message encryption). The optional data fields to be encrypted will be specified in the ACNF request and response field map records. If an optional data field is to be encrypted, an "E" will be placed in the one-byte MSG field corresponding to the optional data field's FID for the

specific transaction type. Although specific optional data fields can be encrypted, specific optional data subfields cannot. All subfields under FIDs 6, 7, 8 or 9 will be encrypted, or none.

Prerequisites: Must license BASE24-pos Standard POS Device Handler (PS-DP140) and POS Data Encryption (PS-AE520)

Stored Value BDM Support

Enhancement is to uplift Stored Value BDM Refresh to 6.0.

The Stored Value Bulk Database Maintenance module provides BASE24-stored value customers with the ability to add and delete a range of stored value cards within BASE24. An extension is available that will update the Card Account File (CAF) and the Primary Balance File (PBF). The BDM process is designed to reduce the amount of data entry required to maintain a large quantity of stored value cards. The BDM file can be created either using BASE24-card's Card Operator File, by specifying a range of cards in the BDM start-up message, or by submitting a file with a starting card or account number and then the number of records.

Prerequisites: BASE24-stored value BDM Process (STV-AE010)

Stored Value Support in CRT Authorization

This enhancement added Stored Value transaction support to CRT Authorization. Additional transactions will be made available to CRT operators to support stored value/prepaid card implementations. New transactions will provide the capability to perform card activations, replenish funds to a card/account, and also allow the redemption of funds from a card/account.

Prerequisites: BASE24-stored value (STV-AE000) and BASE24 CRT Authorization (BP-DP130)

Track 2 Context Data Removal

In 2004, Visa, MasterCard, Discover, American Express, and JCB announced the Payment Card Industry Data Security Standard (PCI) to protect consumer data in acquirer and issuer transactions. The enhancement to BASE24 will protect consumer data as it is carried internally in the BASE24-pos Standard Internal Message (PSTM). The data protected will be discretionary data from track 1 and/or track 2.

This enhancement requires changes to BASE24-pos Standard POS device handler to ensure that track 1 and track 2 discretionary data does not reside in the temporary context areas of the PTD. This enhancement will apply to those transactions that are authorized by BASE24, a host system, or an interchange. BASE24-pos authorization was also enhanced to remove the track 1 and track 2 discretionary data from the PSTM response if the transaction is authorized by either BASE24 or a Host. Additionally, response messages received from an interchange may have the discretionary data removed. Additional logic was also added to accept a response message where the discretionary data in track 2 is no longer the same as the original transaction (i.e., it has been masked).

Prerequisites: The Standard POS Device Handler is required

Visa II Device Handler Enhancements

This enhancement to the Visa II POS device handler adds support for Gift cards and Prepaid debit card transactions. Additional enhancements include support for Visa's Card-Level Results along with returning the results properly to the terminal.

The Visa II device handler also included enhancements to support Extended Address Verification Data as required by American Express. Extended Address Verification Data contains detailed information about the cardholder, including billing and shipping names, addresses and phone numbers. By including Extended Address Verification Data in request messages, transaction acquirers can receive better interchange discount rates.

Prerequisites: The Visa II POS Device Handler (PS-DP420).

BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 9.

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General Enhancements

There are no general enhancements to Interchange Interfaces.

The following BASE24 enhancements will also be available within several of the interchange interfaces. All BASE24 interface users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

PCI - Audit Enhancement

This enhancement provides additional auditing of user access to cardholder data as required by PCI DSS V1.1. Before this enhancement, audit logs were written when cardholder data was added, updated, and deleted. This enhancement adds audit logging for access to cardholder data (record reads).

Prerequisites: None

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Optional Enhancements

The following BASE24 enhancements are optional. All BASE24 Interface users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

PCI - Masking on Pathway

This enhancement allows the setting of Pathway screen masking of cardholder information on Pathway screens considered sensitive by PCI DSS V1.1. These include: Primary Account Number (PAN), Cardholder Name, Service Code, and Expiration Date. Since access to this cardholder data can be restricted on need to know basis through proper user policies and procedures, this is not a requirement of PCI DSS V1.1, but this enhancement provides extra protection of cardholder data.

Prerequisites: None

PCI - Masking on Reports

This enhancement allows the option of masking cardholder information on reports considered sensitive by PCI DSS V1.1. These include: Primary Account Number (PAN), Cardholder Name, Service Code, and Expiration Date. Since access to this cardholder data can be restricted on a need to know basis through proper user policies and procedures, this is not a requirement of PCI DSS V1.1, but this enhancement provides extra protection of cardholder data.

Prerequisites: None

Track 2 Context Data Removal

In 2004, Visa, MasterCard, Discover, American Express, and JCB announced the Payment Card Industry Data Security Standard (PCI) to protect consumer data in acquirer and issuer transactions. The enhancement to BASE24 will protect consumer data as it is carried internally in the BASE24-atm Standard Internal Message (STM/PSTM). The data protected will be discretionary data from track 1 and/or track 2.

This enhancement applies to those transactions that are authorized by BASE24, a host system, or an interchange. BASE24-atm authorization was also enhanced to remove the track 1 and track 2 discretionary data from the STM response if the transaction is authorized by either BASE24 or a Host. Additionally, response messages received from an interchange may have the discretionary data removed. Additional logic was also added to accept a response message where the discretionary data in track 2 is no longer the same as the original transaction (i.e., it has been masked).

BASE24 Value Added Products

All BASE24 value added product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Value Added Products, refer to Section 10.

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General Enhancements

There are no general enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory Voucher Manager

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Optional Enhancements

There are no optional enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory Voucher Manager

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3: Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 Release 6.0 version 8. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to Release 6.0 version 8. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 Release 6.0 requires a minimum of HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files. Customers choosing to implement BASE24-es Transaction Security Services (known as TSS) with BASE24 Release 6.0 must review the BASE24-es Hardware and Software Requirements for the BASE24-es release they will deploy.

5. UPGRADE TO XPNET 3.0 or 3.1

Customers should note that XPNET 3.0 is the minimum requirement for BASE24 Release 6.0.

6. PROJECT SCOPE MEETING

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 version 8.

7. MIGRATION PLANNING

Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 version 8.

9. DEFINE REQUIREMENTS DEFINITION

The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in Release 6.0 version 8 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with Release 6.0 version 8 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 Release 6.0 version 8 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions and outlines can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 Release 6.0 version 8 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in Release 6.0 version 8, a host program will need to be developed to interpret the new token data that can be extracted with Release 6.0 version 8.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in Release 6.0 version 8 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 version 8 software will need to be installed at the customer site. Licensed Release 6.0 version 8 software modules will be provided to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested or as directed by the interchange.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system should be performed as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to Release 6.0 version 8. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer's project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

Publications

The BASE24 documentation set has been updated to include the Release 6.0 version 8 enhancements. The updated BASE24 documentation is available in three formats.

- **HELP24 eSupport InfoLink.** All files for the manuals in the BASE24 6.0 documentation set are located on the HELP24 eSupport InfoLink database. You have the option of copying the files for individual manuals from the HELP24 eSupport InfoLink database to your PC.
- In addition, the HELP24 eSupport InfoLink database contains documentation that is added or revised between ZIP files or CD-ROMs.
- **Software Management Electronic Distribution.** ZIP or ZIP Executable files that contain the same contents as the CD-ROM are available through the ACI electronic distribution system.
- **CD-ROM.** The CD-ROM contains all PDF files for the manuals in the documentation set.

The BASE24 6.0 documentation set contains the most current version of each document. If a BASE24 manual has not been updated for Release 6.0, the manual for the prior release appears on the Release 6.0 menu with a note identifying the prior release.

Documentation for XPNET 3.0 and XPNET 3.1 is available in a separate set titled NET24-XPNET 3.1 that is available in the same formats as the BASE24 documentation.

BASE24 customers can download files from the HELP24 eSupport InfoLink database or order the documentation ZIP files or CD-ROM from their ACI Customer Manager.

Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 version 8.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

The course descriptions, outlines and current education schedule can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

HELP24

Please use ACI Worldwide's eSupport site to report problems or ask questions about BASE24 Release 6.0 version 8. eSupport is an Internet Web based application that interfaces with the HELP24 Clarify database. Using eSupport, you can search our KnowledgeBase Solutions for information and resolutions to problems you are experiencing. You can also directly open Clarify cases. An eSupport option, InfoLink, is your link to a wealth of information that can help you in your day to day business. It allows you on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. Contact your local Help24 office for an eSupport logon.

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows us to troubleshoot issues in a secure, online environment. Compared to the dial-up environment we have used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows you to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option that provides the ability to transfer files from your PC to ours, or vice versa. This feature can be used in the event the file(s) are too large to email.

For more information, please refer to the WebEx User's Guide located under Product Info on Infolink. Contact your local HELP24 office to schedule your WebEx demonstration.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks is estimated and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to Release 6.0 version 8 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate Release 6.0 version 8 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to Release 6.0 version 8. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in Release 6.0 version 8. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

4: BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from Release 6.0 version 7.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the Base modules changes that affect all BASE24 users, along with the associated conversion and implementation requirements that all BASE24 users must address.
- Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 10 to identify the changes that BASE24 users need to consider when upgrading to Release 6.0 version 8.

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Standard Information Format

A standard format is used in Sections 4 through 10 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to Release 6.0 version 8. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, only those implementing the enhancement).

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to Release 6.0 version 8.

- General Changes
- PCI – Audit Enhancement
- PCI – Initial User Password

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described in section 4.

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General Changes

- Changes for all BASE24 Release 6.0 version 8 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The Cardholder Authorization File (CAF) has been modified to include an additional field on screen 13 for the CAP Token Validation Support. A new screen, screen 21, has been added for the Preferred Transaction enhancement. The Cardholder Authorization File screen, requester and server have been modified to process all of the new fields and screen. For further information, see ATC Enhancements, CAP Token Validation Support in Section 6 and Preferred Transactions in Section 5 where the impacts of these enhancements are described.
- The Card Prefix File (CPF) has been modified to include additional fields on screen 3 for ATC Enhancements, screen 4 for the Enhanced Mobile Top-Up Routing enhancement, screens 11 and 13 for the CAP Token Validation Support enhancement. The Card Prefix File screen, requester and server have been modified to process all of the new fields. For further information, see ATC Enhancements, CAP Token Validation Support and Enhanced Mobile Top-Up Routing in Section 6 where the impacts of these enhancements are described.
- The Interchange Configuration File (ICF) has been modified to include an additional field on screen 2 for the PCI – Masking on Reports. The Interchange Configuration File screen, requester and server have been modified to process this new field. For further information, see PCI – Masking on Reports later in Section 4.
- The Enhanced Interchange Configuration File (ICFE) has been modified to include an additional field on screen 2 for the PCI – Masking on Reports. The Enhanced Interchange Configuration File screen, requester and server have been modified to process this new field. For further information, see PCI – Masking on Reports later in Section 4.

- The Institution Definition File (IDF) has been modified to include an additional field on screen 1 for the Dynamic CAF and screen 4 for PCI – Masking on Reports. A new IDF screen, screen 43, has been added for the Preferred Transaction enhancement. The Institution Definition File screen, requester and server have been modified to process the new fields and screen. For further information, see PCI – Masking on Reports later in Section 4, Preferred Transaction in Section 5 and Dynamic CAF in Section 6 where the impacts of these enhancements are described.
- The Store and Forward File (SAF) has been modified. No Pathway changes are required for this, but it might be necessary to convert a SAF file. For further information, see Skipping HISO SAF in Section 4.
- The Security File (SEC) has been modified to include an additional field on screen 1 for the PCI – Masking on Pathway. The Security File screen, requester and server have been modified to process this new field. For further information, see PCI – Masking on Pathway later in Section 4.
- Refresh Support of PauseACK Improvements
This enhancement was done to the base Refresh process to ensure that when a refresh has completed successfully and performs a broadcast notify, the Refresh process will more efficiently handle the response it receives from the processes it notified. This enhancement only affects those customers who have implemented the Release 6.0 version 4 enhancement, Dual Site Support, and who have enabled broadcast notification by setting the LCONF param, BROADCAST-NOTIFY, to a value of 2.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLBCNST, DDLFCAF, DDLFCPF, DDLFICF, DDLFICFE, DDLFIDF, DDLFSAF, DDLGSEC,
- Inline changes to BA60AFT. RQCAFS, RQCPFS, RQICFS, RQICFES, RQIDFS, RQSECS, SCRNCFAF, SCRNCPF, SCRNICF, SCRNICFE, SCRNIIDF, SCRNNSEC, SVCAFS, SVCNPF, SVICFS, SVICFES, SVIIDF and SVISECS.
- Inline changes to BA60REFR.REFRG and REFRS
- Updated event (log) message manual BA60LOGM.BAREFR

Note: The list of files impacted by the merged changes is also included in the **Software Impacts** section of the applicable enhancement(s).

A complete list of the merged files can be found in the SCN file BA60UD0B.SCNMERGE.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Cardholder Authorization File (CAF)

For customers converting from Release 5.3 to Release 6.0 version 8, the CAF conversion program must be run to convert the CAF. Running the conversion program is required for any BASE24 customer. Users must create the new CAF file using the BADDLFUP file.

No separate conversion program was created to move the CAF from version 7 to version 8. Customers must do a full file refresh using the new CAF layout. This will initialize the CAP fields.

The CAF conversion program on the BA60UC04 subvolume can be used to convert the CAF records from the Release 5.3 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVCAFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cafconv.t,name $cafm,nowait/ cnvcafm -  
force -list -listreplace
```

Release 5.3 users should update the CNVCAFR obey file on the BA60UC04 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any BASE24 customer. Users must create the new CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the Release 5.3 format to the Release 6.0 version 8 format. The CPF conversion program on the BA60UC0B subvolume can be used to convert the CPF records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm –  
force –list –listreplace
```

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVCPFR obey file on the BA60UC0B subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Interchange Configuration File (ICF)

The ICF conversion program must be run to convert the ICF. Running the conversion program is required for any BASE24 customer utilizing the ICF in transaction processing.

The ICFE conversion program on the BA60UC04 subvolume can be used to convert the ICF records from the Release 5.3 format to the ICF/ICFE Release 6.0 version 8 format. This program will also split the Release 5.3 ICF into a Release 6.0 version 8 ICF and ICFE file, depending on the switch type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVICFES file. Information on compiling and running this program are found in the following section for the Enhanced Interchange Configuration File (ICFE).

The ICF conversion program on the BA60UC0B subvolume can be used to convert the ICF records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVICFS file. Running this conversion program is required for any BASE24 customer utilizing the ICF in transaction processing. Users must create the new ICF file using the BADDLFUP file

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#icfcnv.t,name $icfm,nowait/ cnvcpfm -force  
-list -listreplace
```

Release 6.0 version 7 users should update the CNVICFR obey file on the BA60UC0B subvolume with the appropriate information. The CNVICFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Enhanced Interchange Configuration File (ICFE)

The ICFE conversion program must be run to convert the ICF records from the Release 5.3 format to the ICF/ICFE Release 6.0 version 8 format or to convert the ICFE records from the Release 6.0 version 7 format to the Release 6.0 version 8 format.

The ICFE conversion program on the BA60UC04 subvolume will split the Release 5.3 ICF into a Release 6.0 version 8 ICF and ICFE file, depending on the switch type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVICFES file. This ICFE conversion program required for any BASE24 customer utilizing the ICF/ICFE in transaction processing. Users must create the new ICF/ICFE file using the BADDLFUP file.

The ICFE conversion program on the BA60UC0B subvolume can be used to convert the ICFE records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVICFES file. Running this conversion program is required for any BASE24 customer utilizing the ICFE in transaction processing. Users must create the new ICFE file using the BADDLFUP file

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#icfecnv.t,name $icfem,nowait/ cnvicfem -  
force -list -listreplace
```

Release 5.3 users should update the CNVICFER obey file on the BA60UC04

subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVICFER obey file on the BA60UC0B subvolume with the appropriate information. The CNVICFER should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Institution Definition File (IDF)

The IDF conversion program must be run to convert the IDF. Running the conversion program is required for any BASE24 customer. Users must create the new IDF file using the BADDLFUP file.

The IDF conversion program on the BA60UC04 subvolume can be used to convert the IDF records from the Release 5.3 format to the Release 6.0 version 8 format. The IDF conversion program on the BA60UC0B subvolume can be used to convert the IDF records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVIDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#idfcnv.t,name $idfm,nowait/ cnvidfm -force  
-list -listreplace
```

Release 5.3 users should update the CNVIDFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVIDFR obey file on the BA60UC0B subvolume with the appropriate information. The CNVIDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Store and Forward File (SAF)

The SAF conversion program must be run to convert the SAF. Running the conversion program is required for any BASE24 customer with transactions in the SAF file when converting from Release 6.0 version 7 to the Release 6.0 version 8. Users must create the new SAF file using the BADDLFUP file.

No conversion of the SAF is provided to convert the SAF records from the Release 5.3 format to the Release 6.0 version 8 format. The Release 5.3 format records must be processed prior to converting to Release 6.0 version 8.

The SAF conversion program on the BA60UC0B subvolume can be used to convert the SAF records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVSAFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#safcnv.t,name $safm,nowait/ cnvsafm -  
force -list -listreplace
```

Release 5.3 users should update the CNVSAFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVSAFR obey file on the BA60UC0B subvolume with the appropriate information. The CNVSAFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Security File (SEC)

The SEC conversion program must be run to convert the SEC. Running the conversion program is required for any BASE24 customer. Users must create the new SEC file using the BADDLFUP file.

The SEC conversion program on the BA60UC04 subvolume can be used to convert the SEC records from the Release 5.3 format to the Release 6.0 version 8 format. The SEC conversion program on the BA60UC0B subvolume can be used to convert the SEC records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVSECS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#seccnv.t,name $secm,nowait/ cnvsecm –  
force –list –listreplace
```

Release 5.3 users should update the CNVSECR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVSECR obey file on the BA60UC0B subvolume with the appropriate information. The CNVSECR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

With the Version 4 enhancement, Dual Site Support, if broadcast notification is enabled, it is the customer's responsibility to determine if any Authorization process(es) have not responded and which ones. The modifications made for the Refresh Support of PauseACK Improvements enhancement does not change this.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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PCI – Audit Enhancement

This section describes the implementation requirements associated with the PCI Audit enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI Audit enhancement.

BASE24 must support the auditing of user access to sensitive information, in accordance with the PCI Data Security Standards.

The PCI Data Security Standards state that the system must implement automated audit trails for all system components to reconstruct events, including: “10.2.1 All individual user access to cardholder data”.

Requirement 10.2.1 applies to any access - read, update, create, or delete. BASE24 auditing needs to be enhanced to address user reads of cardholder data. This is logging of activity by human users – there is no requirement to log accesses that occur as part of transaction processing.

For reads, there is no requirement to generate before and after images, or even to log all the records viewed. The requirement is to log sufficient information to enable an investigator to determine if a user viewed a particular cardholder’s information.

The requirement includes failed queries, so auditing is also performed under the following circumstances:

- A user attempts to view a record belonging to an institution, but does not have the required security authority to access the institution’s information.
- A user attempts to access a file or screen, or to perform an operation on a record, but does not have the required security authority for the function attempted.

The Online Maintenance File (OMF) has always contained information on users who have logged on and logged off the Pathway system as well as information on users who have added, changed, or deleted records in BASE24 files accessed through the BASE24 Pathway system.

Pathway Servers which allow cardholder data to be displayed have been enhanced to write OMF records whenever a user reads a record from the file. The new records are logged with a file maintenance type of “E” (read) or “F” (read next). This includes Pathway Servers which have not previously supported the OMF, because the data displayed on the screen cannot be modified, e.g. the (P)TLF Server. For these Servers, a file maintenance type of “E” (read) is used when a single record is

displayed, e.g. on the (P)TLF Detail Screen, and a file maintenance type of “F” (read next) is used when multiple records are displayed, e.g. on the (P)TLF Summary Screen.

To save disk space, the full record image is not logged to the OMF when a record is read. Instead, for files which include the PAN in its key fields, the OMF record is logged with “common” card data, comprising the Logical Network, the FIID, the PAN and Member Number, and (in some cases) the Account Type. The exception is the National Negative File (NNF), for which the full record image is logged to the OMF, because the record length is only 17 bytes. When multiple records are displayed, multiple occurrences of the “common” card data are included in the OMF record. The POS Referral File (PRF) does not include the PAN in its key fields, so the data logged to the OMF comprises the Date, the Retailer ID and the Referral Code. When multiple records are displayed (on the MRF Screen), multiple occurrences of this data are included in the OMF record.

OMF records are also written when a security violation occurs, because the operator does not have access for the institution (FIID) to which the record belongs. These new records are logged with a file maintenance type of “I” (institution security error). There are two circumstances under which an OMF record is logged with a file maintenance type of “I”:

- When a user attempts to change the FIID being used for file operations, the Requester sends a message to the Security Server with a transaction code of ‘17’ (Validate FIID/Region/Group Access). If the Security Server determines that the user is not permitted access to the FIID, then an error message is returned to be displayed on the screen. The Security Server has been enhanced to log an OMF record as well.
- Some files, e.g. CAF, NEG, may contain records belonging to different FIIDs, but the FIID itself is not part of the record key. So, it is not possible to determine whether the user has security access to the FIID until the record has been read. For these files, the appropriate Server ensures that the FIID of the record in the file matches the FIID in the message sent by the Requester, when a user attempts to read a record, read the next record, or delete a record. If the Server finds that the FIIDs do not match, then an error message is returned to be displayed on the screen. These Servers have been enhanced to log an OMF record as well.

OMF records are also written when a security violation occurs, because the operator does not have access to the file or the screen or authority for the function attempted. These new records are logged with a file maintenance type of “J” (file operation security error). There are two circumstances under which an OMF record is logged with a file maintenance type of “J”:

- When a user attempts to access a new file, the Requester sends a message to the Security Server with a transaction code of ‘10’ (Validate Screen Access). If the

Security Server determines that the user is not permitted access to the file, then an error message is returned to be displayed on the screen. The Security Server has been enhanced to log an OMF record as well.

- When a user attempts to access a new file, the Requester sends a message to the Security Server with a transaction code of '10' (Validate Screen Access). If the Security Server determines that the user is permitted access to the file, then a response is sent to the Requester, containing the operator's security access for each function (RADU) on each screen of the file. If a security violation subsequently occurs when the operator attempts a file operation, then the Requester denies the request and outputs a message to the screen. Requesters have been enhanced to send an additional message to the Security Server with a transaction code of '23' (File Access Error). The Security Server has been enhanced to process a message with a transaction code of '23', by logging an OMF record.

The CRT Authorization Security (CSEC) File is used with BASE24-pos CRT Authorization. It allows institutions to prohibit an operator from performing certain transactions and functions from the Online Transaction (OLT) screen and the POS Referral File (PRF) screen. OMF records are written when a security violation occurs, because the operator does not have authority for the function attempted, based on configuration settings in the CSEC. The new records are logged with a file maintenance type of "J" (file operation security error).

There are several switch-specific Maintenance Log File (MLF) screens impacted by this enhancement. Typically, these screens allow MLF records to be displayed (either individually or as a group), and deleted. For the interchange interfaces that allow MLF records to be read individually (Banknet, JCB ISO, VisaNet), an OMF record is logged with a file maintenance type of "E" (read). For all other interchange interfaces (that allow MLF records to be read as a group), an OMF record is logged with a file maintenance type of "F" (read next). When an MLF record is deleted, an OMF record is logged with a file maintenance type of "D" (delete).

The structure of an MLF record differs between interchange interfaces, so the OMF record is logged with "common" MLF data, comprising the FIID, the Date/Time, the PAN and Member Number, instead of the full record image itself. All interchange interfaces use an OMF application code of either "0E" or "IH" for their maintenance log files. However, it is possible to distinguish between OMF records logged for different maintenance log files, because the interface FIID is present in each OMF record.

Considerations

Items to consider when implementing this enhancement:

- This enhancement covers BASE24-atm, BASE24-pos (including BASE24-pos Parametric Auth) and BASE24-bp (including CRT Auth). No changes to support BASE24-teller, BASE24-telebanking or BASE24-card are included.
- The auditing of user access to cardholder data is not configurable, i.e. there is no mechanism for not auditing user access.
- Protection of authentication data, e.g. Track-2, Card Verification Digits, PINs, is out of scope, and is addressed in other enhancements.
- Protection of displayed cardholder data, e.g. PAN, is out of scope, and is addressed in other enhancements.
- If a user attempts a “read” or “read next” function on the NNF screen, but the PAN does not begin with ‘3’, ‘4’ or ‘5’, then no OMF record is written. The Server identifies the file to be accessed (ANEG, BNEG or INEG) from the first digit of the PAN, and therefore cannot determine the OMF application code if the first digit of the PAN does not contain one of these values.
- If a user attempts a “read” or “read next” function, but the record is not retrieved successfully, then no OMF record is written. This is consistent with current OMF processing, where only successful record adds, updates and deletes are recorded on the OMF.
- No auditing logic has been added to the OLT AFT modules, because the OLT screen is essentially a mechanism for entering transactions, for which knowledge of the full PAN is required.
- No auditing logic has been added to the CRT Administration AFT modules, because these screens essentially provide a mechanism for entering post-event transactions (sales drafts, chargebacks, representments), for which knowledge of the full PAN is required.
- No auditing logic has been added to the data entry screens used to support interchange interface file update messages, because knowledge of the full PAN is required to generate the messages. However, when the log file is perused (via the maintenance log file server), auditing logic is invoked.
- Auditing logic has been added to the SDF (used by the SPAN2 and VisaNet Interchange Interfaces). However, OMF records logged as a result of an added, updated or deleted SDF record are not converted to EBCDIC by the OMF Extract module. This capability has not been added as part of this enhancement, and the same restriction applies to OMF records logged as a result of a read SDF record.

- Auditing logic has been added for security violations where a user does not have the requisite access for the institution (FIID). This logic applies to all files (not just those which contain cardholder data) where the institution access is validated, and the OMF record will contain no indication of the file/record/screen that the operator was attempting to access.
- Auditing logic has been added for security violations where a user accesses a file, but the FIID in the record does not match the FIID under which the operator is logged on. This logic applies only to files which contain cardholder data, and where the FIID is not part of the key to the file.
- Auditing logic has been added for security violations where a user does not have any access to the file. This logic applies to all files (not just those which contain cardholder data) where the file access is validated, and the OMF record will contain no indication of the file that the operator was attempting to access.
- Auditing logic has been added for security violations where a user does not have authority for the operation attempted. This logic applies only to files which contain cardholder data where the operator access is validated, but the OMF record will contain no indication of the operation that was being attempted.
- Most interchange-specific maintenance log file requesters do not validate that the user is permitted to perform the operation attempted, because the operator's file/screen access is "inherited" from the access allowed to the interchange-specific ICF(E) screen (the MLF is typically accessed via a menu from here). There are some exceptions, namely BNET, EURO, JCBI, MACI, VISA. However, since security access cannot be configured explicitly at the file level, no auditing logic has been added for security violations where a user does not have authority for the operation attempted on the MLF.
- The maintenance log files for the MAGI Interchange Interface, the NEN Interchange Interface and the PISO Interchange Interface are out of scope, because these interchange interfaces are due to be "decommissioned" in the near future.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFOMF, DDLGOMFX, DDLGPTH
- Inline changes to BA60DEFS.DDLGDEFS
- Inline changes to BA60AFT.SVISECS, RQCAFS, SVCAFS, RQNEGS, SVNEGS, RQUAFS, SVUAFS
- Inline changes to BA60SRC.BACOUTLS
- Inline changes to BA60EXT.OMFXS

All BASE24-atm customers:

- Inline changes to AT60AFT.RQTLFS, SVTLFS
- Inline changes to AT60EXT.ATOMFXS

All BASE24-pos customers:

- Inline changes to PS60AFT.RQPTLS, SVPTLS
- Inline changes to PS60EXT.PSOMFXS
- Inline changes to BP60AFT.RQNNFS, SVNNFS

All BASE24-pos CRT Authorization customers:

- Inline changes to BP60CRTA.SVCSECS, RQPRFS, RQMRFS, SVPRFS, RQCHFS, SVCHFS
- Inline changes to BP60EXT.BPOMFXS

For more complete details regarding the impact of this enhancement on the switch interfaces, refer to the PCI – Audit enhancement within Section 9 of this document.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNAUDT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract

OMF Extract files will contain additional records, with file maintenance types of “E”, “F”, “I”, “J”, and new application codes:

<u>Application Code</u>	<u>File Name</u>
"30"	TLF
"33"	PTLF
"75"	PRF
"0E" or "1H"	Switch-specific MLFs

User/Operational Impacts

Disk Impacts

Additional disk space will be required to hold the extra records logged to the OMF. A single Logical Network supports up to 10 OMFs per day, and each OMF (which is a Format-1 file) can hold up to 2Gb of data. Operators for some large BASE24 customers are likely to access AFT screens containing cardholder data (particularly the TLF, PTLF and PRF) many times per day. Even with considerable usage, it is extremely unlikely that the OMF capacity (20Gb per day) will be reached.

Performance Impacts

The writing of the additional OMF records will have an impact on system performance. However, the logging is performed by the AFT subsystem, so there will be no direct impact on transaction processing.

Vendor Impacts.

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

PCI - Initial User Password

This section describes the implementation requirements associated with the PCI – Initial User Password enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI – Initial User Password enhancement.

PCI DSS 8.5.3 states that initial passwords must be set to unique values for every user and must be changed immediately after the first use. Although the BASE24 user access system previously required new users to change their passwords during their first logon, their initial passwords were a default value of blanks. This was not compliant with the PCI standard. The PCI – Initial User Password enhancement enables institutions to comply with the standard.

When a new user is added to the BASE24 system using the Security File screen, the system administrator is now required to enter an initial password for the user. The default value of blanks is no longer allowed. Furthermore, the initial password must meet the complexity requirements specified by the institution, if complex passwords are enabled. The administrator is required to enter the password twice, reducing the chance of typographical errors. The new user is required to change the password when logging on to the BASE24 system for the first time. The new password, too, must meet the same complexity requirements.

The system administrator is no longer permitted to change an existing user's password to blanks (if, for example, the user has forgotten the password). The Shift-F8 function key (set user's password to blanks) on the Security File screen has been disabled. Instead, the administrator must update the password on the Security File screen, again within the complexity requirements. The user is subsequently required to change the password when logging on to the BASE24 system.

When a new BASE24 system is started, SUPER/SUPER is the only user recognized by the system. Previously, the initial password for SUPER/SUPER was all zeroes and the administrator was encouraged (but not required) to change it to a confidential password. With this enhancement, the initial password remains zeroes, but the administrator is required to change it when logging on to the BASE24 system for the first time. The password must meet the complexity requirements specified by the institution, if complex passwords are enabled.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.RQSECS, SCRNSSEC and SVISECS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNPSWD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

ACI Worldwide, Inc.

Skipping HISO SAF

This section describes the implementation requirements associated with the Skipping HISO SAF enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Skipping HISO SAF enhancement.

The processing implemented in this enhancement is to prevent HISO from positioning beyond SAF records that get added with duplicate timestamps causing SAF records to be left in the file. There is also an edit to prevent HISO from sending SAF records to subsequent DPCs.

Software Impacts

All BASE24-atm customers:

- Inline changes to DDLFSAF and BA60HISO.BAHISOS.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNHSAF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

If the HISO SAF cannot be emptied before installation of this enhancement, there is a conversion file that can be used to create a SAF in the correct format.

Host/Co-Network Impacts

No changes were made to the host version of the SAF record (SAFX). This will eliminate the need for any host changes to support this enhancement.

User/Operational Impacts

Not applicable.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide, Inc.

Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with Release 6.0 version 8, along with the changes, conversion requirements, and implementation requirements. These enhancements are the following:

- PCI – Masking on Pathway
- PCI – Masking on Reports
 - BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements for future reference.

ACI Worldwide, Inc.

PCI – Masking on Pathway

This section describes the implementation requirements associated with the PCI Masking on Pathway enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI Masking on Pathway enhancement.

Sensitive information that is currently displayed on BASE24 screens must be “masked”, in accordance with the PCI Data Security Standards.

There are essentially four related requirements that affect BASE24:

- Definition of sensitive information. In the PCI Data Security Standard, sensitive information is defined as the Primary Account Number (PAN), the Cardholder Name, the Service Code, the Expiration Date.
- Definition of masking. Masking is defined in section 3 of the PCI Data Security Standard. The PAN must be masked when displayed (the first six and last four digits are the maximum number of digits to be displayed). This requirement does not apply to employees and other parties with a specific need to see the full PAN.
- Masking templates. The customer should be able to define a masking template, for the PAN. The masking template defines left-most and right-most characters to display, and also the minimum number of characters padded. For example, “display leftmost four characters and rightmost three characters, with a minimum of six characters padded”. The default mask should be to display the rightmost four digits and fill the remaining digits with asterisks.
- Screen masking. All screens that display sensitive information as identified above must optionally mask the sensitive data. The customer will be able to designate individuals (by user id) that are or are not authorized to view sensitive information in the clear. By default, users will see masked values.

Except for a few screens, the PAN is the only one of the data elements defined as “sensitive information” to be displayed on the BASE24 screens. A new LCONF parameter has been implemented to define the PAN masking requirements for the BASE24 screens. A new Security File (SEC) parameter has been implemented to define (by User ID) whether users can view sensitive data in the clear. The default value ensures that PANs are masked.

A few screens display the card’s expiration date and/or the cardholder’s name in addition to the PAN. If the PAN is configured for masking for the user, then the other sensitive data will be masked completely.

The new LCONF parameter is called AFT-PAN-DIGITS. The LCONF parameter contains a 3-digit numeric value, used to determine how to mask PANs displayed on Pathway screens, as described in the following table:

Digit Number	Description	Default Value
1	The maximum number of leftmost digits to be displayed unmasked.	0
2	The minimum number of digits to be masked.	9
3	The number of rightmost digits to be displayed unmasked.	4

The new field added to the Security File (SEC) record is called MASKING-FLG, and specifies whether sensitive data on screens should be displayed in the clear for the AFT operator. All BASE24 AFT modules that include sensitive data, e.g. a cardholder PAN, use this field in conjunction with the new LCONF parameter to determine how sensitive data should be presented to the user.

The MASKING-FLG is retrieved when the Security File is read at logon. It is then passed around the AFT subsystem, so that individual Requesters and Servers can determine whether the user is permitted to view sensitive data.

The actual masking of the PAN is performed by a utility procedure, which is invoked by Requesters and Servers associated with screens and files that contain cardholder information.

In general, sensitive information on records which can only be read or deleted (i.e. not updated or added) via the AFT subsystem, e.g. TLF, PTLF, is masked by the Server. However, sensitive information on records which can also be updated or added via the AFT subsystem, e.g. CAF, NEG is typically masked by the Requester.

In most cases, the Requester stores the “unmasked” PAN when the PAN is displayed “masked”, as it is often necessary to return the “unmasked” PAN to the Server in subsequent file operations.

There are several switch-specific Maintenance Log File (MLF) screens impacted by this enhancement. Records on these screens can typically only be read or deleted via the AFT subsystem, so the PAN is masked by the appropriate Server. However, there is another consideration with some of these files. When a user configured for masking attempts to delete a record from the perusal screen, the PAN (and the expiration date) on the screen will be masked. To determine the record to delete, some Servers currently compare data (including a timestamp and the PAN) in the message received from the Requester against the record read from the file. This comparison would no longer work, because the PAN would be masked when a delete request is received from the Requester. To resolve this issue, additional changes have been applied, and the extent of these changes depends on the previous comparison criteria:

- The PAN has been replaced with the System Trace Audit Number (STAN) and other message data in the matching criteria. This issue affects the AMEX GNS Interchange Interface, the EPS-Net Interchange Interface, and the MDS Interchange Interface, and means that the MLF DDL and the interchange interface code itself is impacted by this enhancement.
- The PAN has been excluded from the matching criteria (the STAN is already included in the matching criteria). However, the timestamp field is now being logged with greater accuracy, to reduce the likelihood of the wrong record being deleted. This issue affects the DDSI Interchange Interface and the FDRI Interchange Interface, and means that the interchange interface code itself is impacted by this enhancement.

In addition, there are some Servers that have used the PAN in the matching criteria, and have relied mainly on the timestamp. However, the number of digits in the timestamp field used in the comparison has been increased, to reduce the likelihood of the wrong record being deleted. This issue affects the MACI Interchange Interface, the PBLX Interchange Interface, and the SHZM Interchange Interface, but does not mean that the interchange interface code itself is impacted by this enhancement.

Considerations

Items to consider when implementing this enhancement:

- This enhancement covers BASE24-atm, BASE24-pos (including BASE24-pos Parametric Auth) and BASE24-bp (including CRT Auth). No changes to support BASE24-teller, BASE24-telebanking or BASE24-card are included.
- Protection of authentication data, e.g. Track-2, Card Verification Digits, PINs, is out of scope, and is addressed in other enhancements.
- This enhancement does not support PANs longer than 19 digits.
- The “masking” configuration parameter for a user applies to all screens, i.e. a user can either view all sensitive data in the clear or have all sensitive data masked.
- The “masking” logic does not apply to administrative PANs, e.g. in the ADMN file, which are used by merchants.

- No “masking” logic has been added to the RPT AFT modules, because sensitive data will have been masked (if required) when the data was written to the RPT file.
- No “masking” logic has been added to the OLT AFT modules, because the OLT screen is essentially a mechanism for entering transactions, for which knowledge of the full PAN is required. Also, the OLT screen has to manage up to seven PANs (and associated expiration dates) at any one time. The only practical way of achieving this is to store both masked and unmasked data in the TQF, which would require the TQF record length to be increased. The cost of this modification is thought to outweigh the benefits.
- No “masking” logic has been added to the CRT Administration AFT modules, because these screens essentially provide a mechanism for entering post-event transactions (sales drafts, chargebacks, representments), for which knowledge of the full PAN is required.
- No “masking” logic has been added to the data entry screens used to support interchange interface file update messages, because knowledge of the full PAN is required to generate the messages. Also, the messages are sent directly to the interchange interface (and then to the interchange), and are never processed by the log file server. However, when the log file is perused (via the log file server), “masking” logic is invoked.
- The maintenance log files for the MAGI Interchange Interface, the NEN Interchange Interface and the PISO Interchange Interface are out of scope, because these interchange interfaces are due to be “decommissioned” in the near future.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLGPTH, DDLGSEC
- Inline changes to BA60AFT.RQMEGAS, SCRNSSEC, RQSECS, SVISECS, RQCAFS, SVCAFS, RQNEGS, SVNEGS, RQUAFS, SVUAFS
- Inline changes to BA60SRC.BACOUTLS

All BASE24-atm customers:

- Inline changes to AT60AFT.RQTLFS, SVTLFS

All BASE24-pos customers:

- Inline changes to PS60AFT.RQPRTL, SVPTL
- Inline changes to BP60AFT.RQNNFS, SVNNFS

All BASE24-pos CRT Authorization customers:

- Inline changes to BP60CRTA.SCRNPRF, RQPRFS, RQMRFS, SVPRFS, SCRNCHEF, RQCHFS, SVCHFS, SCRNCRED, RQCRED

For more complete details regarding the impact of this enhancement on the switch interfaces, refer to the PCI – Masking on Pathway enhancement within Section 9 of this document.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNMASK.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support these changes. Details of the new parameters are provided in the LCONFBA file located on the BA60MISC subvolume.

AFT-PAN-DIGITS

This param contains a 3-digit numeric value, and is used to determine how to mask PANs displayed on Pathway screens. This is an optional param.

- The first digit specifies the maximum number of leftmost PAN digits to be displayed unmasked on AFT screens.
- The second digit specifies the minimum number of PAN digits to be displayed masked on AFT screens.
- The third digit specifies the number of rightmost PAN digits to be displayed unmasked on AFT screens.

Valid values are numeric digits. If the param is not found or is invalid, then the default value of “094” is used.

Database Configuration

Security File (SEC)

The following new field has been added to SEC Screen 1:

This field specifies whether cardholder data should appear in the clear on reports for the institution.

Screen Name:	MASKING FLAG
Screen:	SEC Screen 1
DDL Name:	SEC-BASE-INFO.MASKING-FLG

Valid values are 'Y' and 'N'. The default value is 'Y'.

A full description of this field can be found in the ***BASE24 Files Maintenance Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The SEC conversion program must be run to convert the Security File (SEC). Details of this conversion program can be found in the **General Changes** earlier in Section 4.

For BASE24 AEGN Interface customers, the AEGN ANEG conversion program must be run to convert the ANEG. Details of this conversion program can be found in Section 9.

For BASE24 EURO Interface customers, the EURO ENRLF conversion program must be run to convert the ENRLF. Details of this conversion program can be found in Section 9.

For BASE24 MDS Interface customers, the MDS MLF conversion program must be run to convert the MLF. Details of this conversion program can be found in Section 9.

Host/Co-Network Impacts

Super Extract

Records extracted from the OMF that contain information about Security (SEC) File changes will now include the MASKING FLAG configured on SEC Screen 1.

User/Operational Impacts

Disk Impacts

Not applicable

Performance Impacts

Not applicable

AFT Screens

AFT operators will no longer be able to view sensitive cardholder information in the clear, unless explicitly permitted to do so by the MASKING-FLG parameter on the operator's record in the SEC File.

Vendor Impacts.

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

PCI – Masking on Reports

This section describes the implementation requirements associated with the PCI Masking on Reports enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI Masking on Reports enhancement.

Sensitive information that is currently displayed on BASE24 reports must be “masked”, in accordance with the PCI Data Security Standards.

There are essentially four related requirements that affect BASE24:

- Definition of sensitive information. In the PCI Data Security Standard, sensitive information is defined as the Primary Account Number (PAN), the Cardholder Name, the Service Code, and the Expiration Date.
- Definition of masking. Masking is defined in section 3 of the PCI Data Security Standard. The PAN must be masked when displayed (the first six and last four digits are the maximum number of digits to be displayed). This requirement does not apply to employees and other parties with a specific need to see the full PAN.
- Masking templates. The customer should be able to define a masking template, for the PAN. The masking template defines left-most and right-most characters to display, and also the minimum number of characters padded. For example, “display leftmost four characters and rightmost three characters, with a minimum of six characters padded”. The default mask should be to display the rightmost four digits and fill the remaining digits with asterisks.
- Report masking. All reports that print sensitive information as identified above must optionally mask the sensitive data. Reports will use the masking template concept described above, where the user can define the mask. By default, the mask should be to display only the rightmost four digits.

Except for one BASE24-fhm report, the PAN is the only one of the data elements defined as “sensitive information” to be printed in the BASE24 reports. Three new parameters on the Institution Definition File (IDF), the Interchange Configuration File (ICF) and the Enhanced Interchange Configuration File (ICFE) have been implemented to define the PAN masking requirements for the BASE24 reports. Another new parameter on the IDF and ICF(E) has been implemented to define whether reports should display sensitive data in the clear. The default values will ensure that PANs are masked.

The BASE24-fhm Cardholder Authorization File Activity Report displays the card's expiration date in addition to the PAN. If the PAN is configured for masking on this report, then the expiration date will be masked completely.

The three new IDF and ICF(E) parameters defining the PAN masking requirements each contain a 1-digit numeric value, used to determine how to mask PANs displayed on reports, as described in the following table:

Number	Description	Default Value
1	The maximum number of leftmost digits to be displayed unmasked.	0
2	The minimum number of digits to be masked.	9
3	The number of rightmost digits to be displayed unmasked.	4

The other new IDF and ICF(E) parameter is called MASKING-FLG, and specifies whether sensitive data on reports should be displayed in the clear for the institution or the interchange interface.

All BASE24 report modules that include sensitive data, e.g. a cardholder PAN, use the new fields to determine how sensitive data should be presented to the user.

The new parameters are retrieved from the IDF by the ATM, POS and FHM report programs, which already use the IDF for processing. The Base POS report has been enhanced to retrieve the IDF, so that it has access to the new parameters.

There are several switch-specific transaction detail reports impacted by this enhancement. The new parameters are retrieved from the ICF or ICFE by the switch reports, which already use the ICF(E) for processing.

The actual masking of the PAN is performed by a utility procedure, which is invoked by programs generating reports that contain sensitive cardholder information.

Considerations

Items to consider when implementing this enhancement:

- This enhancement covers BASE24-atm, BASE24-pos (including BASE24-pos Parametric Auth) and BASE24-bp (including CRT Auth). No changes to support BASE24-teller, BASE24-telebanking or BASE24-card are included.
- Protection of authentication data, e.g. Track-2, Card Verification Digits, PINs, is out of scope, and is addressed in other enhancements.
- This enhancement will not support PANs longer than 19 digits.
- This enhancement reduces the length of the REPORTING NAME field on Screen 1 of the ICF(E) from 20 bytes to 16 bytes. This field is used only to display the name of the interchange on switch reports.

- Except for the switch reports (STAT09), it is possible to configure the masking requirements by institution. For an individual institution, PAN masking applies either to all of the reports or to none of the reports. Therefore, it is not possible to configure different masking requirements for different reports for the same institution.
- For the switch reports (STAT09), it is possible to configure the masking requirements by interchange interface process. Therefore, it is possible to configure different masking requirements for different interchanges.
- It is a requirement that “unmasked reports should only be available to user IDs of authorized personnel “. This requirement is out of scope of the BASE24 enhancement. If a bank wants/needs someone to have the reports unmasked, then their NSK logon would have to provide enough security access to allow the reports to be run manually with unmasked data being output.
- The BASE24 file access report (OMF04) may display sensitive data, depending on the type of file being audited in the Online Maintenance File (OMF) record. This report is out of scope, because the position of the sensitive data in the record is dependent on the type of file being audited.
- The BASE24-fhm reports include information on files used by the BASE24 teller (NBF, WHFF), remote banking (CSTT, CACT, CPIT, PIT), and customer service (CCIF, CCMF) products. These are out of scope, so the PANs in these reports will not be masked.
- The switch report for the PISO Interchange Interface is out of scope, because this interchange interface is due to be “decommissioned” in the near future.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFIDF, DDLFICF, DDLFICFE
- Inline changes to BA60AFT.SCRNIDF, RQIDFS, SVIDFS, SCRNICF, RQICFS, SCRNICFE, RQICFES
- Inline changes to BA60SRC.BACOUTLS
- Inline changes to BA60EXT.OMFXS

All BASE24-atm customers:

- Inline changes to AT60RPT.RP06B24S, RP07B24S, RP08B24S

All BASE24-pos customers:

- Inline changes to PS60RPT.B24PS1S, B24PS2S, B24PS5S

All BASE24-pos CRT Authorization customers:

- Inline changes to BP60CRTA.B24RP30S

All BASE24-fhm customers:

- Inline changes to FH60RPT.FHMRPTS

All BASE24 switch customers:

- Inline changes to SW60RPT.RP09SWIS

For more complete details regarding the impact of this enhancement on the switch interfaces, refer to the PCI – Masking on Reports enhancement within Section 9 of this document.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNRMSK.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Institution Definition File (IDF)

The following new fields have been added to IDF Screen 4:

This field specifies whether cardholder data should appear in the clear on reports for the institution.

Screen Name: DATA MASK FLAG
Screen: IDF Screen 4
DDL Name: IDF.BASE.RPT-PAN-DIGITS.MASKING-FLG
Valid values are 'Y' and 'N'. The default value is 'Y'.

This field specifies the number of rightmost PAN digits to be displayed unmasked on reports.

Screen Name: RIGHT UNMASKED DIGITS
Screen: IDF Screen 4
DDL Name: IDF.BASE.RPT-PAN-DIGITS.RIGHT-UNMASKED

Valid values are 0 - 9. The default value is 4.

This field specifies the minimum number of PAN digits to be displayed masked on reports.

Screen Name: MIN MASKED DIGITS
Screen: IDF Screen 4
DDL Name: IDF.BASE.RPT-PAN-DIGITS.MIN-MASKED

Valid values are 0 - 9. The default value is 9.

This field specifies the maximum number of leftmost PAN digits to be displayed unmasked on reports.

Screen Name: MAX LEFT UNMASKED DIGITS
Screen: IDF Screen 4
DDL Name: IDF.BASE.RPT-PAN-DIGITS.MAX-LEFT-UNMASKED

Valid values are 0 - 9. The default value is 0.

A full description of these fields can be found in the ***BASE24 Files Maintenance Manual***.

Interchange Configuration File (ICF)

The following new fields have been added to ICF Screen 2:

This field specifies whether cardholder data should appear in the clear on reports for the interchange.

Screen Name: DATA MASK FLAG
Screen: ICF Screen 2
DDL Name: ICF.BASE.RPT-PAN-DIGITS.MASKING-FLG

Valid values are 'Y' and 'N'. The default value is 'Y'.

This field specifies the number of rightmost PAN digits to be displayed unmasked on reports.

Screen Name: RIGHT UNMASKED DIGITS
Screen: ICF Screen 2
DDL Name: ICF.BASE.RPT-PAN-DIGITS.RIGHT-UNMASKED

Valid values are 0 - 9. The default value is 4.

This field specifies the minimum number of PAN digits to be displayed masked on reports.

Screen Name: MIN MASKED DIGITS
Screen: ICF Screen 2
DDL Name: ICF.BASE.RPT-PAN-DIGITS.MIN-MASKED

Valid values are 0 - 9. The default value is 9.

This field specifies the maximum number of leftmost PAN digits to be displayed unmasked on reports.

Screen Name: MAX LEFT UNMASKED DIGITS
Screen: ICF Screen 2
DDL Name: ICF.BASE.RPT-PAN-DIGITS.MAX-LEFT-UNMASKED

Valid values are 0 - 9. The default value is 0.

A full description of these fields can be found in the ***BASE24 Files Maintenance Manual***.

Enhanced Interchange Configuration File (ICFE)

The following new fields have been added to ICFE Screen 2:

This field specifies whether cardholder data should appear in the clear on reports for the interchange.

Screen Name: DATA MASK FLAG
Screen: ICFE Screen 2
DDL Name: ICFE.BASE.RPT-PAN-DIGITS.MASKING-FLG

Valid values are 'Y' and 'N'. The default value is 'Y'.

This field specifies the number of rightmost PAN digits to be displayed unmasked on reports.

Screen Name: RIGHT UNMASKED DIGITS
Screen: ICFE Screen 2
DDL Name: ICFE.BASE.RPT-PAN-DIGITS.RIGHT-UNMASKED

Valid values are 0 - 9. The default value is 4.

This field specifies the minimum number of PAN digits to be displayed masked on reports.

Screen Name: MIN MASKED DIGITS
Screen: ICFE Screen 2
DDL Name: ICFE.BASE.RPT-PAN-DIGITS.MIN-MASKED

Valid values are 0 - 9. The default value is 9.

This field specifies the maximum number of leftmost PAN digits to be displayed unmasked on reports.

Screen Name: MAX LEFT UNMASKED DIGITS
Screen: ICFE Screen 2
DDL Name: ICFE.BASE.RPT-PAN-DIGITS.MAX-LEFT-UNMASKED

Valid values are 0 - 9. The default value is 0.

A full description of these fields can be found in the ***BASE24 Files Maintenance Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The IDF conversion program must be run to convert the Institution Definition File (IDF). Details of this conversion program can be found in the **General Changes** earlier in Section 4.

The ICF conversion program must be run to convert the Interchange Configuration File (ICF). Details of this conversion program can be found in the **General Changes** earlier in Section 4.

The ICFE conversion program must be run to convert the Enhanced Interchange Configuration File (ICFE). Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Super Extract

Records extracted from the OMF that contain information about Institution Definition File (IDF) changes will now include the new parameters configured on IDF Screen 4.

Records extracted from the OMF that contain information about Interchange Configuration File (ICF) changes will now include the new parameters configured on ICF Screen 2.

Records extracted from the OMF that contain information about Enhanced Interchange Configuration File (ICFE) changes will now include the new parameters configured on ICFE Screen 2.

User/Operational Impacts

Disk Impacts

Not applicable

Performance Impacts

Not applicable

Reports

It will no longer be possible to view sensitive cardholder information in the clear on BASE24 reports, unless this is explicitly permitted by the MASKING-FLG parameter on the IDF (for ATM, POS, Base POS and FHM reports) or the ICF and ICFE (for Switch reports).

Vendor Impacts.

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

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5: BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from Release 6.0 version 7.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to Release 6.0 version 8.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to Release 6.0 version 8.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Log ATM Totals to EMS
- NCR NDC+ Device Handler Global Space Issue
- Multi-language Indicator for Statement Print

This area is summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes for all BASE24 Release 6.0 version 8 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The ATM Terminal Data File (ATD) has been modified to include an additional field on screen 2 for Shared NDC+ BNA Support. The ATD screen, requester and server have been modified to include the new field. An additional field has been added to the Diebold 10XX/478X screen 7 and NCR 5XXX screen 7 for the ATM Dynamic Currency Conversion enhancement. The ATD requester and server submodules for the respective device handlers have been modified to include the new fields. For further information, see Shared NDC+ BNA Support and ATM Dynamic Currency Conversion Reports later in Section 5.
- The Terminal Data File (TDF) has been modified to include an additional field on screen 9 for Tidel MACing Support and Triple-DES Update. The screen, requester and server for the Tidel specific screens have been modified to include the new field. For further information, see Tidel MACing Support and Triple-DES Update later in Section 5.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60AFT.RQATDS, SCR NATD1 and SVATDTS

CUSTOMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not Applicable

Database Configuration

Not Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ATM Terminal Data (ATD)

The ATD conversion program must be run to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 version 8 format or to convert the ATD records from the Release 6.0 version 7 format to the Release 6.0 version 8 format.

The ATD conversion program on the BA60UC04 subvolume can be used to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 version 8 format depending on the device type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. This ATD conversion program is required for any BASE24 customer utilizing the ATD/TDF in transaction processing. Users must create the new ATD/TDF files using the BADDLFUP file

The ATD conversion program on the BA60UC0B subvolume can be used to convert the ATD records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATD file. Running this conversion program is required for any BASE24 customer utilizing the ATD in transaction processing. Users must create the new ATD file using the BADDLFUP file

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#atdcnv.t,name $atdm,nowait/ cnvatdm -  
force -list -listreplace
```

Release 5.3 users should update the CNVATDR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVATDR obey file on the BA60UC0B subvolume with the appropriate information. The CNVATDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Terminal Data File (TDF)

The TDF conversion program must be run to convert the TDF.

The ATD conversion program on the BA60UC04 subvolume can be used to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 version 8 format. This program will also split the Release 5.3 TDF into a Release 6.0 version 8 ATD and TDF file, depending on the device type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. Information on compiling and running this program are found in the following section for the ATM Terminal Data File (ATD).

The TDF conversion program on the BA60UC0B subvolume can be used to convert the TDF records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVTDFS file. Running this conversion program is required for any BASE24 customer utilizing the TDF in transaction processing. Users must create the new TDF file using the BADDLFUP file

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#tdfcnv.t,name $tdfm,nowait/ cnvtdfm  
-force -list -listreplace
```

Release 6.0 version 7 users should update the CNVTDFR obey file on the BA60UC0B subvolume with the appropriate information. The CNVTDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Log ATM Totals to EMS

This section describes the implementation requirements associated with the Log ATM Totals to EMS enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Log ATM Totals to EMS enhancement

Currently, when an ATM is being balanced, settlement and balancing messages are passed, via the device handler process, to BASE24-atm Auth where they are then logged to the TLF. If the TLF is unable to be written to, these admin messages are not logged. This enhancement will add support for logging to the EMS these various admin messages so that, if the TLF becomes unavailable, settlement and balancing message information will not be lost.

New EMS messages have been added to log the appropriate admin data, based upon the message type. The following messages have been added to the BA60LOGM.ATAUTH event message documentation file: 0281, 0282, 0283, 0284, 0285, 0286.

Admin messages affected by this enhancement include:

- Terminal Balancing, admin code '05', '06', '09'
- Terminal Cash Adjustment, admin code '01', '02', '03', '04', '07', '08'
- Terminal Settlement, admin code '20', '21', '22'

Software Impacts

All BASE24-atm Auth customers:

- Changes to AT60AUTH.AUTHLIBS
- Updated event (log) message manual BA60LOGM.ATAUTH

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional Event Messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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NCR NDC+ Device Handler Global Space Issue

This section describes the implementation requirements associated with the NCR NDC+ Device Handler Global Space Issue enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR NDC+ Device Handler Global Space Issue enhancement.

The data stack global storage area for the BASE24-atm NCR 5XXX device handler is subdivided into a global primary area and a global secondary area. Global primary data space can store up to 256 words of direct variables and pointers. Global secondary data space has no size limit, but the total of all primary and secondary areas cannot exceed 32k words. Currently, the N50 device handler is approaching the 256 word limit of the global primary space. This makes the addition of new product features, or customer modifications, increasingly difficult. This enhancement addresses this problem by freeing up additional primary data space. This is accomplished by restructuring a number of global variables, moving them to subordinate positions with new indirect structures. Moving these global variables to indirect structures places them within the secondary memory area, which has no explicit limit.

N50G:

Two indirect structures have been created in the N50G file as follows. Many, but not all, of the direct variables have been moved to one or the other of these indirect structures. If the global variable previously ended in an '_g', the '_g' was dropped since the overall indirect structure includes that suffix.

```
struct      .n50_g;
    begin
        :
    end; ! struct n50_g
and . . .
struct      .n50_ssb_g;
    begin
        :
    end; ! struct n50_g
```

In addition, the N50G indirect struct 'misc_flg_g' has been removed, and its subordinate variables moved to one or the other newly created indirect structures.

In all cases, variable names were retained, except for the '_g' suffix, though they are now referenced as part of the overall indirect structure.

For example, the variable match_flg_g has been moved to a subordinate position within the new indirect structure n50_g. The '_g' suffix is dropped, and is now referenced as n50_g.match_flg.

Impacts to remaining files involve changing references to those global variables moved to one or the other subordinate structures.

The main device handler module, as well as all relevant extension code must be recompiled.

Software Impacts

All BASE24-atm NCR 5XXX device handler customers:

- Changes to AT60N50.N50G, N50NCDDS, N50S, N50SSBDS

All BASE24-atm NCR 5XXX, AKDS device handler customers:

- Changes to AT60AN50.N50AKDS

All BASE24-atm NCR 5XXX, Wincor AKDS device handler customers:

- Changes to AT60AWN5.N50WAKDS

All BASE24-atm NCR 5XXX, SSB device handler customers:

- Changes to AT60BN50.N50SSBS, AT60CN50.N50SSCS, AT60DN50.N50SSDS

All BASE24-atm NCR 5XXX, Forced Cutover Audit device handler customers:

- Changes to AT60FN50.N50FCAS

All BASE24-atm NCR 5XXX, EMV device handler customers:

- Changes to AT60IN50.N50EMVG, N50EMVS

All BASE24-atm NCR 5XXX, Wincor EMV device handler customers:

- Changes to AT60IWN5.N50WEMVG, N50WEMVS

All BASE24-atm NCR 5XXX, Dynamic Key Management device handler customers:

- Changes to AT60KN50.N50DKMS

All BASE24-atm NCR 5XXX, Multiple Account Select by Qualifier device handler customers:

- Changes to AT60MN50.N50MADS, N50MANS, N50MAS

All BASE24-atm NCR 5XXX, non-Currency Dispense device handler customers:

- Changes to AT60NN50.N50NCDS

All BASE24-atm NCR 5XXX, Dial-Up customers:

- Changes to AT60PN50.N50DLUS

All BASE24-atm NCR 5XXX, Mobile Top-Up Device handler customers:

- Changes to AT60TN50.N50MTUS

All BASE24-atm NCR 5XXX, Multiple Currency device handler customers:

- Changes to AT60UN50.N50MCUS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNNGBLS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Multi-language Indicator for Statement Print

This section describes the implementation requirements associated with the Multi-language Indicator for Statement Print enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Multi-language Indicator for Statement Print enhancement

This enhancement provides host systems and switch interchanges (*) with an indicator specifying which language was chosen by the consumer at the ATM. This indicator may then be used by the issuer to determine which language to use when formatting the statement data for the response. Both the NCR 5xxx Device Handler and the Diebold 10xx/478x Device Handler were also enhanced to populate a new token field with the language preference of the consumer. Note that the language preference selection is already part of standard product for these two device handlers.

* As of Release 6.0 version 8, the only interchange interface supporting this new field is the SAMA Span2 Interface.

The valid values for this new token field are obtained from lang_cde_tbl_g in C100NAMS/N50NAMS depending on the language selected by the consumer. These tables need to be modified based on the customer's requirements but the values must follow the ISO 639-2 standard for 3-byte language designation (for example "ENG" is English).

The LANG-CDE field within the ATM Data1 Token (AG) was added previously as part of a SAMA Span2 enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLATTKN

All BASE24-atm NCR 5xxx Device Handler customers:

- Inline changes to AT601000.C1000G, C1000S, C100NAMS
- Updated event (log) message manual BA60LOGM. AT1000

All BASE24-atm Diebold 10xx/478x Device Handler customers:

- Inline changes to AT60N50.N50G, N50NAMS, N50S
- Updated event (log) message manual BA60LOGM.AT5070

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNSTMT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

The ATM Data1 Token (AG) now contains a 3-byte language indicator that the host can use when determining what language it should use to format the statement print receipt data.

```
*      A code used to provide additional information
*      regarding the language selected by the cardholder
*      at the ATM.  Valid values are the 3 character code
*      described in ISO 639-2.
*
```

```
02  LANG=CDE                                PIC X(3).
```

User/Operational Impacts

Users should be aware that additional Event Messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with Release 6.0 version 8, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Agilis 2.0 Enhancements
- ATM Dynamic Currency Conversion
- Card before Cash on Fast Cash
- Configurable AUTH for Rear Balancing Transactions
- Diebold 912 AKDS Signature Support
- Diebold EMV Upgrade
- Enhanced Mobile Top-Up Routing Support
- NCR NDC+ Additional M-Status Codes
- NCR NDC+ Increase of Secondary EMV AIDs
- NCR Standard APTRA Edge NDC+ Support
- Preferred Transaction
- Shared NDC+ BNA Support
- Shared NDC+ CPM Support
- Tidel Dynamic Key Management
- Tidel MACing Support and Triple-DES Update
- Track 2 Context Data Removal
- Wincor AKDS support of NCR ATMs

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements for future reference.

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Agilis 2.0 Enhancements

This section describes the implementation requirements associated with the Agilis 2.0 enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Agilis 2.0 enhancement.

The Diebold entry program (ENTIBOLD) has been modified to support several new features of Agilis 2.0. These include support for the Enhanced Envelope Dispenser State 'SD ', Miscellaneous Features 4, and the Key Mapping feature (i.e. control sequences 'ESC-ZW', 'ESC-ZX'). These new features can be used to properly configure the terminal via terminal downloads.

The "Supplemental Dispense" state 'SD ' directs the terminal to perform specific envelope dispenser operations such as dispensing envelopes and displaying particular screens.

The following screen shows the Enhanced Envelope Dispenser State 'SD ':

INTERBOLD STATE TABLE				
Type ____	Envelope Dispenser State			
State Number ____	Group ____			
Unit Number	_____			
Operation	_____			
Dispense Screen	_____			
Post Dispense Screen	_____			
Consumer Response Time	_____			
Good Operation Next State	_____			
Consumer Fault Next State	_____			
Hardware Fault Next State	_____			
Network Fault Next State.....	_____			

=====				
F1= CHANGE STATE TYPE	F2=READ	F3=ADD	F4=DELETE	F5=UPDATE
	F9=NEXT	F10=PRINT		F16=EXIT

Valid values:

Type:	SD
State Number:	000 – 254 and 256 – ZZZ
Group:	0000 – 9999

Unit Number	001
Operation	001 - 002
Dispense Screen	Any Valid Screen Number
Post Dispense Screen	Any Valid Screen Number
Consumer Response Time	000 – 255, 900 – 931, 999
Good Operation Next State	000 – 254, 256 – ZZZ
Consumer Fault Next State	000 – 254, 256 – ZZZ
Hardware Fault Next State	000 – 254, 256 – ZZZ
Network Fault Next State	000 – 254, 256 – ZZZ

In conjunction with support for state 'SD', this enhancement also adds support for Miscellaneous Features 4. The newly formatted Miscellaneous Data screen, showing Miscellaneous Features 4 follows:

```
MISCELLANEOUS DATA      Group[0000]

PARAMETERS: Camera Control [  ]      LUNO [  ]
  Misc Features 1 [  ]  Misc Features 2 [  ]  Misc Features 3 [  ]
  Misc Features 4 [  ]

TIMERS (in seconds):
00: Customer Keyboard Response [  ]      01: Timeout Screen Display ... [  ]
02: Close State Screen Display [  ]      03: Communications Response .. [  ]
04: Envelope Deposit ..... [  ]      05: Currency Removal ..... [  ]
06: Poll Interval ..... [  ]      07: Withdrawal Door Open ..... [  ]
08: Securomatic Deposit ..... [  ]      09: Card Removal ..... [  ]
10: Take Cash Screen Display . [  ]      11: Vandal Shield Close ..... [  ]
15: Card Reader Enable ..... [  ]      18: Swipe Reader Error Display [  ]
20: IDM Deposit ..... [  ]      21: Statement Printer Capture [  ]
23: Dip Card Reader ..... [  ]
```

The following is a brief description of Miscellaneous Features 4. For more information regarding the use of this field, consult the **BASE24 Diebold 10XX/478X Device Support Manual**, and Diebold's **Agilis 91X Terminal Programming Manual**, December, 2005.

Miscellaneous Features 4

Field Size: 3 characters

Valid Range: 000 through 255

Description: The value in the Miscellaneous Features 4 field is the sum of the weights of the features selected. Agilis 91x allows the network to select the features listed in the following table.

Weight	Features
128	Reserved
64	Reserved
32	Reserved
16	Disallow dispense with safe door open
8	Reserved
4	Envelope dispenser configuration control
2	Send Envelope Dispenser status to network
1	Disable Automatic Envelope Dispensing

Envelope Dispenser Configuration Control

The *Envelope Dispenser Configuration Control* feature applies only to terminals equipped with an envelope dispenser (enabled). Selecting this feature allows the network to modify certain configurable features of envelope dispenser operation. For further details, refer to the description of the Send Envelope Dispenser Status to Network feature, and to the description of the Disable Automatic Envelope Dispensing feature.

Send Envelope Dispenser Status to Network

The *Send Envelope Dispenser Status to Network* feature applies only to terminals equipped with an envelope dispenser (enabled). This feature allows the network to enable or disable sending statuses to the network for envelope dispenser faults. This feature applies only to envelope dispenser faults that occur during automatic envelope dispensing or during the processing of a Deposit state (C state). Envelope dispenser faults that occur during the processing of an Envelope Dispenser state (SD state) or an Envelope Dispenser function command (SD function ID) are always reported to the network.

The setting of the Send Envelope Dispenser Status to Network feature has an effect only if the Envelope Dispenser Configuration Control feature is selected. If the Envelope Dispenser Configuration Control feature is *not* selected, Agilis 91x ignores the setting of the Send Envelope Dispenser Status to Network feature.

Disable Automatic Envelope Dispensing

The *Disable Automatic Envelope Dispensing* feature applies only to terminals equipped with an envelope dispenser (enabled). Selecting this feature directs the terminal to disable the automatic dispensing of envelopes at the beginning of deposit operations. Automatic envelope dispensing is described in Section 3.9.2.

The setting of the Disable Automatic Envelope Dispensing feature has an effect only if the Envelope Dispenser Configuration Control feature is selected. If the Envelope Dispenser Configuration Control feature is *not* selected, Agilis 91x ignores the setting of the Disable Automatic Envelope Dispensing feature.

Disallow Dispense with Safe Door Open

The *Disallow Dispense with Safe Door Open* option requires the chest door (safe door) to indicate closed when doing a network authorized dispense operation. Once this feature is enabled, it can only be disabled by clearing terminal memory. Any subsequent Write Command 3 messages must then have this option turned off (set to 0). The MDS status generated when the chest door is open is DI01:3E:02:00.

NOTE

If a terminal does not have a chest door or does not have a chest door switch, Agilis 91x assumes a Door Open condition. In such cases, it is recommended that you do not enable this feature.

In addition, the Diebold entry program has been enhanced to support two new control sequences, used to define and select a particular Key Map. Control sequence 'ESC-ZX' defines a Key Map. This allows any keypad key (0 thru 9, A thru M) to be used as any function key (A thru M). The key pressed is translated via the defined key map, and that translated value is then used by a state or function just as if that value had been the actual key pressed.

The 'ESC-ZX' control sequence has the following format:

```
ESC ZX ttt nnn kk ff ... kk ff
```

The parameter *ttt* specifies the map number of the key map that this control sequence defines. The valid range for ttt is from 001 to AZZ.

The parameter *nnn* indicates the number of key pairs (kk ff) in this control sequence. The valid range for nnn is from 001 to 100.

For each key pair (*kk ff*), the parameter *kk* specifies the physical key that is to be translated to a specified function key value when the key *kk* is pressed. The valid range for *kk* is 30 through 39 (keys 0 through 9), and 41 through 4D (keys A through M).

For each key pair (*kk ff*), the parameter *ff* specifies which function key value is reported when the key *kk* is pressed. The valid range for *ff* is 41 through 4D (A through M).

EXAMPLE

The following example of an ESC ZX control sequence defines a key map that translates five keypad keys (1, 2, 3, 4, and J) to function key values (A, B, C, D, and M).

```
ESC ZX 100 005 31 41 32 42 33 43 34 44 4A 4D
```

Control sequence 'ESC-ZW' directs the terminal to use a previously defined key map (previously defined via the 'ESC-ZX' control sequence).

The 'ESC-ZW' control sequence has the following format:

```
ESC ZW ttt
```

The parameter *ttt* specifies the map number of the key map to be used with the screen in which the ESC ZW is embedded. The valid range for *ttt* is from 001 to AZZ, and 000.

If the value of *ttt* is 000, no key map is selected, including the default key map. Only the keys explicitly enabled by the state or function are enabled. All enabled keys report their original (untranslated) values when pressed. This usage allows the network to suspend key mapping for specific instances.

Key Mapping configuration data (for both 'ESC-ZX' and 'ESC-ZW' control sequences) is downloaded to the terminal using Write Command 2 (screen) messages. For more information on the Key Mapping control sequences refer to the **Agilis 91x Terminal Programming Manual, v2.0, December, 2005**.

Software Impacts

All BASE24-atm Diebold 10XX/478X device handler customers:

- A new Diebold Entry program object (ENTIBOLD).

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNAGIL.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX/478X Configuration File (CNFGIX)

The Diebold 10XX/478X Configuration File (CNFGIX) will be amended to include, as needed, the new Enhanced Envelope Dispenser state 'SD '. In addition, screen records requiring use of the Key Mapping feature must also be amended to include the appropriate control sequences ('ESC-ZW' / 'ESC-ZX'). Current Miscellaneous Data records should be first read, then deleted, prior to adding new Miscellaneous Data records supporting the Miscellaneous Features 4 field.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- Agilis 91x Terminal Programming Manual, v2.0, December, 2005
- BASE24 Diebold 10XX/478X Device Support Manual

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ATM Dynamic Currency Conversion

This section describes the implementation requirements associated with the ATM Dynamic Currency Conversion enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ATM Dynamic Currency Conversion enhancement.

BASE24-atm has been enhanced to support Dynamic Currency Conversion (DCC) through a set of separately licensable add-ons to its Authorization process and its Diebold 10xx/478x and NCR 5xxx device handlers. The new functionality enables a BASE24 acquirer to convert a withdrawal to the cardholder's home currency before sending the transaction to an interchange, rather than send it in the acquirer's currency and have the issuer perform the conversion. DCC benefits the acquirer by enabling that party to mark up the conversion rather than pass the opportunity to the issuer. It also benefits the cardholder by making charges immediately visible for purposes such as business expense accounts. The enhancement is compatible with all devices driven by BASE24 through the supported device handlers, including Diebold and NCR ATMs and other devices such as Wincor terminals that run in Diebold or NCR emulation. The enhancement is compatible with other BASE24 Multiple Currency products such as the Authorization Multiple Currency and Device Handler Multiple Currency Dispense modules, but it does not require any of those modules to be licensed or installed.

Within the BASE24 DCC modules, the "acquirer's currency" is assumed to be the terminal currency that is defined for the ATM at which the withdrawal is made. The "cardholder's currency" is assumed to be the currency defined for the cardholder's Bank Identification Number (BIN), which is the first six digits of the PAN. Acquirers wanting to perform DCC must have access to the relevant BIN information provided by the interchanges, such as the information in Visa's Account Billing Currency File and Table IP0040T1 [Account Range] of MasterCard's Integrated Product Message (IPM) Member Parameter Extract (MPE) file.

For DCC to be applicable, a transaction has to satisfy the following requirements:

- The transaction is a withdrawal.
- The transaction is SPROUTE-routed to an interchange, rather than authorized by BASE24 or a host. Support for multiple currencies within BASE24 is provided by the existing, separately licensable Multiple Currency and Multiple Currency Dispense modules.

- The interchange allows DCC. Whether this is so may depend on the interchange, the region in which the withdrawal takes place, and the region in which the card was issued. This enhancement assumes that BASE24-atm customers are aware of all interchange requirements.
- The acquirer's currency as defined above is different from the cardholder's currency.

Interchanges that support or are likely to support DCC in the near future have strict requirements concerning the information that must be presented to the cardholder before the cardholder chooses to allow the conversion, as well as the information that must be printed on receipts. The BASE24 DCC modules support these requirements through several files, tables, and LCONF assigns and params, which are described below.

The modules can also be configured to allow/not allow DCC and to charge different mark-ups based on the terminal, the BIN, the currency being converted to, and the issuer to which the transaction is expected to be routed.

Except for the requirement that a transaction be SPROUTE-routed, DCC can be performed independently of the actual routing of a transaction, being configured through the issuer identifier and other data in the Dynamic Currency Conversion Data file (see below). This makes it possible, for instance, for an acquirer who routes transactions through a single LINK interchange to display different information at the ATM based on the requirements of the issuers to which LINK may send them.

For terminals dispensing multiple currencies, the BASE24 DCC modules support same-to-same conversion, or conversion in which the dispensed currency and the cardholder's currency are the same. For instance, a U.S. customer at an ATM in Britain may withdraw 100 U.S. dollars, which are "converted" to U.S. dollars at a mark-up of 5%, resulting in a charge of U.S. \$105.00. Without DCC, the same withdrawal would be typically converted to pounds sterling by the acquirer at no mark-up, then converted back to dollars by the issuer at whatever mark-up they deemed appropriate. Note that support for multiple-currency ATMs, with or without DCC, requires the separately licensable Multiple Currency Dispense module for the device handler.

The BASE24 DCC modules do not support currency conversion for transactions in which the terminal currency is the same as the BIN currency, even if the dispensed currency is different. This functionality is provided by the device handler Multiple Currency Dispense modules.

DCC is compatible with surcharging, whether performed on the request side (non-interactive surcharging or interactive request notification surcharging) or on the response side (interactive response notification surcharging). When DCC is

combined with surcharging, the messages presented to the cardholder are as follows:

- For non-interactive surcharging, the transaction amount presented to the cardholder in the DCC offer will include the surcharge amount.
- For interactive request notification surcharging, BASE24 presents the surcharge notification according to the existing logic. If the cardholder approves the surcharge, BASE24 follows it with a DCC offer in which the transaction amount includes the surcharge.
- For interactive response notification surcharging, BASE24 presents a DCC offer in which the transaction amount includes an “estimated” acquirer surcharge calculated from the SURF. If the cardholder chooses to allow DCC, the normal response-side surcharge notification will be replaced with one that includes the required DCC information.

The exact information that is presented in each of the steps above can be configured as described below under **Device Configuration**.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.BADDLMM, BADDLM, DDLBATKN, and DDLFOMF
- New file BA60DDL.DDLFDCCD
- Inline changes to BA60SRC.BASEFM, BASEMM, BASEM, BATKNID, and BATKNCVS
- Inline changes to BA60AFT.BAAFTM, BAAFTMM, COBTKN, MEGATBL, SECTBL, and SVMHLPS
- New files RQDCCDM, RQDCCDS, SCRNDCCD, SVDCCDM, and SVDCCDS
- Inline changes to OK60LIB.OKLIBTS
- Inline changes to BA60LIB.BALIBTG, BLBPROPS, and BALIBTS
- Inline changes to BA60EMTB.BEMTBMM, BEMTBM, BEMTDS, and EMTBRUN
- New files BA60EMTB.DCDEMTTM, DCDEMTTG, and DCDEMTTS
- Inline changes to BA60EXT.OMFXM, OMFXG, and OMFXS

All BASE24 Multiple Currency or Multiple Currency Dispense customers:

- Inline changes to BA60UMCU.BAMCUTLG and BAMCUTLS

All BASE24-atm customers:

- Inline changes to AT60SRC.ATMFM, ATMMM, and ATMM
- Inline changes to AT60AFT.COBRCP
- Inline changes to AT60AUTH.AUTHM, AUTHG, AUTHD, AUTHTBL, AUTHS, and AUTHLIBS

All BASE24-atm Diebold 10xx/478x Device Handler customers:

- Inline changes to AT601000.C1000M, C100DDL5, C1000G, C1000D, C1000S, C1000MXS, SCRN10XX, and SV10XXTS

All BASE24-atm NCR 5xxx Device Handler customers:

- Inline changes to AT60N50.N50M, N50DDL5, N50G, N50D, N50S, N50OMFXS, SCRN5XXX, and SV5XXXTS

All BASE24-atm customers taking the ATM Dynamic Currency Conversion enhancement:

- New subvolume, AT60RATH, with files for the Authorization DCC extension
- DCC customers who do not already license any BASE24 Multiple Currency or Multiple Currency Dispense products must install the Multiple Currency utilities on subvolume BA60UMCU and the Device Handler Multiple Currency Dispense utilities on subvolume AT60UMCD. (The actual extension modules for these products, such as those on subvolumes AT60UATH, AT60U100, and AT60UN50, are NOT required, although they are compatible with DCC.)

All BASE24-atm Diebold 10xx/478x Device Handler customers taking the ATM Dynamic Currency Conversion enhancement:

- New subvolume, AT60R100, with files for the Diebold DCC extension

All BASE24-atm NCR 5xxx Device Handler customers taking the ATM Dynamic Currency Conversion enhancement:

- New subvolume, AT60RN50, with files for the NCR DCC extension

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNDCC.

CUSTMACS Changes

Set the following flag to true to bind the DCC module to the Authorization process:

- atm_dcc_on

Set the following flag to true to bind the DCC module to the Diebold 10xx/478x Device Handler:

- atm_dh_c1000_dcc_on

Set the following flag to true to bind the DCC module to the NCR 5xxx Device Handler:

- atm_dh_n50_dcc_on

LCONF Assign Changes

This section describes the assign records that are required to be added to the LCONF to support this enhancement. Details of the assigns are provided in the BASE24 Logical Network Configuration File manual. A sample of the assigns are included in the LCONFBA file located on BA60MISC subvolume

DCCD

This assign gives the file name of the Dynamic Currency Conversion Data file. The EMT Build process uses the data in this file to build the DCCDEMT.

DCCDEMT

This assign gives the file name of the Dynamic Currency Conversion Data Extended Memory Table. The Authorization DCC module uses the data in this table to determine whether DCC is applicable to a transaction and to build the necessary data for DCC notification and conversion.

ERF

Customers who do not already license any BASE24 Multiple Currency or Multiple Currency Dispense products must ensure that this assign is present and contains the name of the Exchange Rate File. The EMT Build process uses the data in this file to build the ERFEMT.

ERFEMT

Customers who do not already license any BASE24 Multiple Currency or Multiple Currency Dispense products must ensure that this assign is present and contains the file name of the Exchange Rate File Extended Memory Table. The Authorization DCC module uses this table in conjunction with the mark-up rates in the DCCDEMT to calculate the DCC conversion rates.

LCONF Param Changes

This section describes the param records that are required to be added to the LCONF to support this enhancement. Details of the params are provided in the BASE24 Logical Network Configuration File manual. A sample of the params are included in the LCONFBA file located on BA60MISC subvolume

DCC-CRNCY-RATE-SOURCE

This param names the authority from whom the DCC conversion rates were derived, for instance, "BANK OF XXX WHOLESALE RATE". This information may be displayed at the ATM to satisfy cardholder notification requirements.

DCCDEMT-WARMIdentifier

Each param of this form (for example, DCCDEMT-WARM01) gives the symbolic name of a process to which the Device Control Terminal (DCT) is to send a warmboot command when warmbooting the DCCDEMT.

DCCDEMT-MAX-TBL-SIZE

This optional param tells the EMT Build program the maximum size in megabytes of the DCCDEMT. Valid values are 1-127. If the param is not present or is invalid, the EMT Build program uses a value of 127.

ERFEMT-WARMIdentifier

Customers who do not already license any Multiple Currency or Multiple Currency Dispense products should ensure that appropriate params of this form (for example, ERFEMT-WARM01) are present. Each such param gives the symbolic name of a process to which the Device Control Terminal (DCT) is to send a warmboot command when warmbooting the ERFEMT.

ERFEMT-MAX-TBL-SIZE

Customers who do not already license any Multiple Currency or Multiple Currency Dispense products may wish to install this optional param. The param tells the EMT Build program the maximum size in megabytes of the ERFEMT. Valid values are 1-127. If the param is not present or is invalid, the EMT Build program uses a value of 127.

Database Configuration

BASE24-atm Terminal Data File (ATD)

The Diebold 10xx/478x and NCR 5xxx device info (ATD screen 7 for both device types) has been enhanced to include a 16-byte DCC Profile, which is used to map the terminal to the relevant data in the DCCD. A value other than spaces in this field will be part of the key used to access Currency Data records in the DCCDEMT, as described below. A value of spaces indicates that DCC is not performed for transactions originating at the terminal.

Exchange Rate File (ERF)

BASE24 customers who do not already license any Multiple Currency product will need to ensure that this file is installed and filled with the proper data. This file contains data that is used to create the ERFEMT, which is accessed by the

Authorization DCC module during transaction processing. The file contains records mapping each currency that is supported by the network to an exchange rate and an effective date. The Authorization DCC module uses the rate for the dispensed currency and the BIN currency, along with the mark-up rate from the DCCD (see below), to calculate the conversion rate that will be used in DCC. The module also uses the effective date from the relevant ERF record (or the later date, if dates for the dispensed currency and the BIN currency differ). This is placed in the DCC Processing Token as the conversion rate source date, which can be used by the device handler to satisfy cardholder notification requirements.

More information on the ERF can be found in the ***BASE24 Files Maintenance Manual***.

New Data Files

Dynamic Currency Conversion Data file (DCCD)

This file contains data that is used to create the DCCDEMT, which is accessed by the Authorization DCC module during transaction processing. The file contains two types of records:

BIN Currency record –

This record maps a BIN or a range of BINs to an issuer designator and a currency code.

The issuer designator is a two-character code that forms part of the key to the Currency Data record. The Authorization DCC module also places this value in the DCC Processing Token, where it can be used by the device handler DCC module to format issuer-specific screens and receipts. Values are user-defined, but BASE24 assumes values beginning with “V” represent Visa brands and values beginning with “M” represent MasterCard brands.

The currency code should be the ISO-defined currency code associated with the BIN.

If no BIN Currency record exists for the cardholder’s BIN, then DCC will not be performed for the transaction.

Currency Data record –

This record maps a terminal DCC profile, issuer designator, and currency code to a currency description and a mark-up rate.

The currency description is an optional description of the currency, e.g., "UK POUNDS". If spaces are used, the ISO alphabetic currency code will be used in DCC processing, e.g., “GBP”. If the currency code in the record is wild cards

(see below), then spaces must be used, and indicate to use the alphabetic code for the BIN currency.

The mark-up rate indicates the percent mark-up to be used in conjunction with the ERF rates to calculate the DCC conversion rates

The key fields may be wildcarded with asterisks (*) and will be searched by the following hierarchy:

DCC profile	Issuer designator:	Currency code:
Exact match	Exact match	Exact match
Exact match	Exact match	***
Exact match	**	Exact match
Exact match	**	***
*****	Exact match	Exact match
*****	Exact match	***
*****	**	Exact match
*****	**	***

If no Currency Data record is found by the search above, then DCC will not be performed for the transaction.

These files are accessed via the new DCCD requester screen. See the **BASE24 Files Maintenance Manual** for more information on this screen.

Token file (TKN)

The following BASE24 tokens have been added for this enhancement:

- The DCC Status Token (S2) contains core information used to identify and process DCC-applicable transactions. Fields include the terminal DCC profile and the current status and substatus of DCC processing for the transaction.
- The DCC Processing Token (S3) contains information needed to perform DCC notification and conversion. Fields include the issuer designator, BIN currency code and description, neutral (without mark-up) and effective (with mark-up) conversion rates, mark-up percentage, conversion rate source and date, converted amounts, saved (before DCC conversion) currency code and amounts, and surcharge amounts in the BIN currency.

Template Changes

Not applicable

Device Configuration

Diebold 10XX/478X Configuration File AT601000.CNFGIX

The Diebold 10XX/478X Configuration File has been modified to include:

- Load Group 320
- Load Group 321

Load group 320 is the basic DCC load group. It contains the states and screens that are used for DCC notification. It must be loaded on top of group 0 and is compatible with the various groups for interactive request notification surcharging or non-interactive surcharging.

Load group 321 is the DCC load group for interactive response side notification. It contains the states and screens that are used for DCC notification. It must be loaded on top of group 0, group 320, and group 310 (interactive response notification surcharging support).

Diebold 10XX/478X DCC Verbiage tables (AT60R100.C1DCNAMS)

This new file will be included in the Diebold 10XX/478X device handler DCC module subvolume. The file is a TAL source code file that is included when the Diebold DCC module is compiled, similar to the C100NAMS file on the main Diebold device handler subvolume. Contents of the file include the following:

- Verbiage to be displayed on DCC notification screens and receipts.
- Tables that control what screens are displayed during DCC notification and which data fields are displayed in what locations on the screen.

By modifying this file and re-Making the device handler, the BASE24 user can configure DCC notification based on whether and what type of surcharging applies as well as what issuer is expected to authorize the transaction (i.e., the issuer identifier from the DCCD BIN Currency record). See the comments within the file for more information on its contents.

NCR 5xxx Configuration File AT60N50.CNFG5070

The NCR 5xxx Configuration File has been modified to include:

- Load Group 320
- Load Group 321

Load group 320 is the basic DCC load group. It contains the states and screens that are used for DCC notification. It must be loaded on top of group 0 and is compatible with the various groups for interactive request notification surcharging or non-interactive surcharging.

Load group 321 is the DCC load group for interactive response side notification. It contains the states and screens that are used for DCC notification. It must be loaded on top of group 0, group 320, and group 310 (interactive response notification surcharging support).

NCR 5xxx DCC Verbiage tables (AT60RN50.N5DCNAMS)

This new file will be included in the NCR 5xxx device handler DCC module subvolume. The file is a TAL source code file that is included when the NCR DCC module is compiled, similar to the N50NAMS file on the main NCR device handler subvolume. Contents of the file include the following:

- Verbiage to be displayed on DCC notification screens and receipts.
- Tables that control what screens are displayed during DCC notification and which data fields are displayed in what locations on the screen.

By modifying this file and re-Making the device handler, the BASE24 user can configure DCC notification based on whether and what type of surcharging applies as well as what issuer is expected to authorize the transaction (i.e., the issuer identifier from the DCCD BIN Currency record). See the comments within the file for more information on its contents.

Mega Table (MEGATBL) Changes

The DCCD requester name was added to MEGATBL.

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry was added to the SECTBL to support the new DCCD screen. This file will appear on the Base Product Menu.

PATHCONF Changes

The new DCCD server must be defined in the PATHCONF file as SERVER-DCCD. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server.

CRT Access Security Changes

Access to the new DCCD screen must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-atm Terminal Data File (ATD)

The ATD conversion program must be run to convert the ATD. Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

The host or co-network may require modification to process the new DCC Status Token or DCC Processing Token. If the tokens for this enhancement are to be sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

Super Extract Process

If the new tokens for this enhancement are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24-atm users that want to use OMF Extract to accept and process information from the ATD record.

User/Operational Impacts

Operational Impacts:

BASE24 users must understand the concept of EMT Builds and the procedure to create an Extended Memory Table file (EMT). BASE24 users must understand the EMT Build Utility report that is generated by the EMT Build Utility. BASE24 users must also understand the concept of EMT warmboots and the procedure to warmboot an EMT. The Authorization DCC module requires a new ERFEMT to be built and warmbooted whenever a change is made to the ERF, and a new DCCDEMT to be built and warmbooted whenever a change is made to the DCCD. New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Card before Cash on Fast Cash

This section describes the implementation requirements associated with the Card before Cash on Fast Cash enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Optional Enhancement #1 enhancement.

BASE24-atm has been enhanced to provide customers the option of ejecting the card before dispensing cash on Fast Cash transactions through its Diebold 10xx/478x and NCR 5xxx device handlers. The same functionality was previously supported only on non-Fast Cash transactions. This enhancement helps cardholders in some environments not to take the cash and forget their card.

Software Impacts

All BASE24-atm Diebold 10xx/478x Device Handler customers:

- No software changes to this device handler were required as part of this enhancement. The enhancement consisted solely of the configuration changes described below and of testing to ensure that card before cash and Fast Cash were mutually compatible.

All BASE24-atm NCR 5xxx Device Handler customers:

- Inline changes to AT60N50.N50S

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNCBC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX/478X Configuration File AT601000.CNFGIX

The Diebold 10XX/478X Configuration File has been modified to include a new load group 19, which supports card before cash on Fast Cash transactions. This group must be loaded on top of group 5, the basic Fast Cash load group.

NCR 5xxx Configuration File AT60N50.CNFG5070

The NCR 5xxx Configuration File has been modified to include a new load group 19, which supports card before cash on Fast Cash transactions. This group must be loaded on top of one of the existing load groups, such as group 1 for basic Fast Cash support. It is not compatible with the EMV Fast Cash groups (601 and 612).

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts:

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Configurable AUTH for Rear Balancing Transactions

This section describes the implementation requirements associated with the Configurable AUTH for Rear Balancing Transactions enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Configurable AUTH for Rear Balancing Transactions enhancement.

BASE24-atm has been enhanced to enable customers to configure the Authorization process to deny any administrative request that is initiated from the rear of the ATM if the request contains a PIN. Payment Card Industry (PCI) standards forbid personnel to enter PINs through a rear keypad, which is not PCI compliant. This enhancement provides a means to deny such transactions for any customer utilizing the Diebold 10xx/478x or NCR 5xxx device handler.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG, AUTHS and AUTHLIBS.

All BASE24-atm Diebold 10xx/478x Device Handler customers:

- Inline changes to AT601000.C1000S.

All BASE24-atm NCR 5xxx Device Handler customers:

- Inline changes to AT60N50.N50G and N50S

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNCARB.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that are required to be added to the LCONF to support this enhancement. Details of the params are provided in the BASE24 Logical Network Configuration File manual. A sample of the params are included in the LCONFAT file located on BA60MISC subvolume

ATM-REAR-ADMN-TRAN-PIN-ALWD

This param informs the Authorization process whether PINs are allowed on rear administrative transactions. If the param is present and has the value N, the Authorization process will deny any rear administrative request that contains a PIN. If the param is not present or does not have the value N, the Authorization process will not deny a rear administrative request because it contains a PIN.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts:

Engineers analyzing audits of administrative transactions should be aware that the device handlers have been modified to use a transaction code of 98 for all rear administrative requests to enable the Authorization process to distinguish them from other administrative requests. Previously, the processing code for an administrative request was specific to the device handler and was not examined by the Authorization process. The same is still true in the enhanced version, except for the special value given to transaction code 98. Other than this slight difference in the audit, the change should have little if any impact on users. The Authorization process does not log these requests or forward them to any other process.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Diebold 912 AKDS Signature Support

This section describes the implementation requirements associated with the Diebold 912 AKDS Signature Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold 912 AKDS Signature Support enhancement.

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support the Diebold 912 Automated Key Distribution System (AKDS) Signature Support enhancement. Changes to an existing bindable extension, AT60A100.C100AKDS, have been developed to support the Diebold 912 AKDS Signature Support enhancement.

This enhancement enables customers to perform AKDS downloads from the Diebold 10XX/478X device handler to NCR ATMs loaded with Diebold's Agilis 91x XV2.1 firmware.

Prerequisite: The Transaction Security Services (TSS) version 07.4 or greater must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLACNST
- Inline changes to the BA60ASRC.SECAKDS (AKDS Security Utilities)

All BASE24-atm Diebold 10XX/478X device handler customers:

- Minor inline changes to AT601000.C1000G.
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm NCR 5XXX device handler customers using the existing Shared NDC+ Certificate or Signature AKDS functionality:

- Minor inline changes to AT60AW10.C10WAKDG.

All BASE24-atm Diebold 10XX/478X device handler customers using the existing Diebold AKDS functionality, and all BASE24-atm NCR 5XXX device handler customers requiring the new Diebold 912 AKDS Signature Support functionality:

- Inline changes to AT60A100.C100AKDS and AT60A100.C100AKDG

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNDKAD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to support this enhancement. Note that these params already exist for the Diebold 10XX/478X device handler.

ATM-DH-AKDS-TSS-TIMER

This param specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

ATM-DH-1000-AKDS-SERIAL-NUM-VRFY

This param specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

This section describes the fields in the files that may be affected by this enhancement. Note that these fields already exist, and were not added for this enhancement. They are simply being used by the Diebold 912 AKDS Signature Support enhancement, too. The settings of each field will depend on the configuration desired by the institution.

ATM Terminal Definition Files (ATD)

The Hardware field, on screen 1, is now used by the Diebold 912 AKDS Signature Support enhancement.

This field will be updated by the Diebold 10XX/478X device handler during Diebold 912 AKDS Signature processing, and set depending on the value that is returned by the ATM in a Hardware configuration response message.

Screen Name: HARDWARE
Screen: ATD Screen 1
DDL Name: ATDD1-CORE.HDWR
Valid Values are: 0 – Unknown
 1 – Wincor
 2 - NCR
 3 - Diebold
 4 - Fujitsu
 5 – De La Rue
 6 – Triton
 7 – Olivetti
Default Value: 3 if the device type is 30 (Diebold)
 2 if the device type is 22 (NCR)
 0 for all other device types.

The Key Signer field on screen 22 is used to indicate the name to be used in the HOST field of the TSS Host Public Key Security Datasource (HSPKD). If multiple Host Public Keys are required for an individual vendor's ATMs, the Key Signer field can be used to assign a separate HOST name for each of those Public Keys.

Screen Name: KEY SIGNER
Screen: ATD Screen 22
DDL Name: ATDS1-CORE.KEY-SIGNER

This field allows up to 16 alphanumeric characters of free-format text to be entered. Embedded spaces are allowed. The default value is "Host".

The EPP Format field, also on screen 22, is used to indicate whether the EPP, and therefore the ATM, supports Signature- or Certificate-based AKDS processing. This field will be updated by the Diebold 10XX/478X device handler during Diebold 912 AKDS Signature processing, and set depending on the value that is returned by the ATM in a Hardware configuration response message.

Screen Name: EPP FORMAT
Screen: ATD Screen 22
DDL Name: ATDD1-CORE.EPP-FORMAT
Valid Values are: 0 – No RKL Supported
1 – Signature
2 - Certificate
3 – Both
Default Value: 2 if the device type is 30 (Diebold)
1 if the device type is 22 (NCR).
0 for all other device types.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM implementing AKDS.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) version 07.4 or greater must be installed in order to use this enhancement.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, the 'Enable PKI' field must be enabled on the Security Device Configuration window of the ACI Desktop Graphical User Interface. If this field is not enabled for a security device, it is not used for PKI functions. This enables the consolidation of all PKI functions in just one or a few devices.

Setup for Transaction Security Services

Initial setup procedures must be carried out for TSS before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain the EPP's private and Public key pair, along with the Signature of the Public Key. It also must contain NCR's Public Key to be used for Host signature validation. It contains the Serial Number of the EPP along with its signature. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Datasource (HSPKD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair.

The public key must be sent to the Certificate Authority (NCR) to be verified and signed. The Certificate Authority must return BASE24's public key signature, as well as the Certificate Authority's own public key. These values must be entered into the Transaction Security Services database.

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the Diebold 10XX/478X device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will not be automatically performed if the ATM has an AKDS-enabled EPP and a master key has not yet been downloaded to the EPP. Instead, the device handler will generate an event message informing the operator that the master key must be loaded. The master key must be present before a PIN encryption key or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the Diebold 912 AKDS Signature Support enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the Diebold 10XX/478X device handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Diebold 912 AKDS Signature Support enhancement is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Diebold 912 AKDS Signature Support enhancement.

- NCR ATMs running Diebold's Agilis 91x XV2.1 (or later) firmware.
 - An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme.
 - Thales e-Security version 6.02 or greater firmware and the optional DSP plug-in (for RSA functions) provide public key cryptography commands using the RSA scheme

Reference Documents

- Atalla: ***Atalla Key Block Banking Command Reference Manual*** (524130-002, July 2002)
- Thales: ***Host Security Module RG7000 Programmer's Manual*** (1270A514 Issue 4, November 2001)
- Diebold 91x Message Protocol Extensions for NCR Based RSA Initial Key Loading, October 2006.

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Enhanced Mobile Top-Up Routing Support

This section describes the implementation requirements associated with the Enhanced Mobile Top-Up Routing Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Enhanced Mobile Top-Up Routing Support enhancement.

BASE24-atm has been enhanced to provide an alternative to normal mobile top-up processing.

The BASE24-atm Mobile Top-Up Device Handler extension is still required to enable cardholders to replenish, or top up, their mobile telephone accounts at an ATM.

Under standard mobile top-up routing logic, the Device Handler process forwards the transaction to the Transaction Context Manager (TCM) process specified in the SPLIT-TXN- RTG-DEST param. The TCM is responsible for routing the transaction to multiple authorizing entities in order to gain full authorization of the transaction.

Under the enhanced mobile top-up routing logic, the SPLIT-TXN-RTG-DEST param will be set to point directly to the BASE24-atm Authorization process. The Authorization process will send the mobile top-up transaction to the host process configured in the Institution Definition File (IDF) based on the MTU PREFIX ROUTING field on CPF screen 4. The Authorization process has been modified to route all mobile top-up traffic to the configured interchange, if the mobile top-up prefix routing has been configured. The host process configured in the IDF is set to an appropriate interchange interface process that is capable of handling mobile top-up transactions.

The interchange interface process will send the transaction to the interchange. The interchange is responsible for splitting the transaction, into the funds side and the top-up side, and sending it to the appropriate entities for authorizing. The funds side will be sent to an appropriate funds authorizer. Should the BASE24 system be the destination of the funds request, the transaction will come back to BASE24 via an appropriate interchange interface process. When the funds side of the transaction enters BASE24-atm Authorization this time, the transaction will no longer use the MTU PREFIX ROUTING field on CPF screen 4, but will use either the EMV PREFIX ROUTING field on CPF screen 11 or the default PREFIX ROUTING field on CPF screen 1 for authorization (standard transaction processing logic). The transaction will be authorized and returned to the interchange. The interchange will coordinate the merging of the funds response and the top-up response. It will then return a single response to BASE24-atm for routing back to the acquiring device.

Note that in the enhanced mobile top-up routing environment, because the interchange splits and routes the mobile top-up transactions, the Transaction Context Manager (TCM) process cannot be utilized; this enhancement replaces the routing normally done by the TCM process.

The routing of Mobile Top-Up transactions to an interchange in the BASE24-atm Authorization module is part of standard BASE24-atm Authorization processing and is not a separately licensable module. The routing will be configurable using a new Mobile Top-Up Prefix Route field on the ATM segment of the Card Prefix File (CPF).

When configuring enhanced mobile top-up routing, BASE24-atm acquired Mobile Top-Up transactions where the new MTU prefix route field is non-blank, must be configured, via the IDF, to be sent online to an interchange. BASE24 should not be configured to stand-in for the initial mobile top-up transaction because it is not equipped to split the transaction into the two required parts.

The LCONF parameter, SPLIT-TXN-RTG-DEST, will need to be set to the BASE24-atm Authorization process for the enhanced mobile top-up routing environment.

This enhancement assumes BASE24-atm customers utilizing the enhanced mobile top-up routing are already aware of any specific interchange requirements for routing and configuration that are outside the scope of this enhancement.

Note: The BASE changes in the BASE24-atm Mobile Top-Up Integration optional enhancement must be installed prior to implementation of the Enhanced Mobile Top-Up Routing support enhancement. For more information refer to the BASE24 Release 6.0 Implementation Guide (Version 5). A complete list of the files affected by the Release 6.0 version 5 mobile top-up support can be found in the SCN file BA60UD08.SCNMTOP.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCPF
- Inline changes to BA60AFT.SCRNCPF, RQCPFS, and SVCDFS
- Inline changes to BA60EXT.OMFXS

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG, AUTHLIBS, and AUTHS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNEFTP.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be modified in the LCONF to support these changes.

SPLIT-TXN-RTG-DEST

For the enhanced mobile top-up processing environment (e.g., where an interchange network will split the transaction), the SPLIT-TXN-RTG-DEST param will specify the BASE24-atm Authorization process.

Database Configuration

Card Prefix File (CPF)

The following new field has been added to the CPF Screen 4 to configure Enhanced Mobile Top-Up Routing support.

Prefix routing groups are used to group prefixes for different types of authorization processing. The MTU prefix routing group number field can be used to define a routing group to be used for routing MTU transactions to an interchange network for splitting the transaction into a Funds side and a Top-up side. It is only used for BASE24-atm acquired transactions. If non-blank, it must be different from the EMV prefix routing group in the EMV segment of the CPF and different from the default routing group for this prefix as specified in the base segment of the CPF if the acquiring authorization process is also the issuing authorization process.

Screen Name: MTU PREFIX ROUTING
Screen: CPF Screen 4
DDL Name: CPF.ATM.MTU-PREFIX.RTE
Valid Values are: 1 alphanumeric character.

0-9 - A number assigned to a routing group. Any number specified should have corresponding prefix group entry in the IDF.

A - Any prefixes not to be included in a special routing group.

Space - Use either the EMV prefix routing group in the EMV segment of the CPF or the default value for this prefix as set in the base segment of the CPF. A blank in this field indicates the BASE24-atm authorization process is not responsible for routing mobile top-up transactions to an interchange for splitting the transaction. The default value is spaces.

Institution Definition File (IDF)

In an enhanced mobile top-up routing environment, the authorization process sends the mobile top-up transaction to the host process configured in the ATM Routing Table on screen 9 of the Institution Definition File (IDF) based on the MTU PREFIX ROUTING field on CPF screen 4. This host process configured in IDF will be set to a network interface. This network interface will send the transaction to the network interchange. The network interchange is responsible for splitting the transaction, into the funds side and the top-up side, and sending it to the appropriate entities for authorizing.

When configuring enhanced mobile top-up routing for BASE24-atm acquired Mobile Top-Up transactions, where the new MTU prefix route field is non-blank, the IDF must be configured to send the transaction online to an interchange network. BASE24 should not be configured to stand-in for the mobile top-up transaction because it is not equipped to split the transaction into the two required parts.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

A mobile top up transaction message has two processing requirements: the funds portion and the top up portion (includes the mobile telephone number and all processing information). If the SPLIT-TXN-RTG-DEST param is present, the Device Handler process will forward the transaction to the destination specified in the param. For standard mobile top-up processing environments, the Transaction Context Manager (TCM) process is specified in the param. For the enhanced mobile top-up processing environment (e.g., where an interchange network will split the transaction), this param will specify the BASE24-atm Authorization process. If the param is not present, the transaction is declined by the Device Handler.

Note that when the SPLIT-TXN RTG-DEST param is present, the Device Handler process will always send transactions to this process (i.e. usually the Transaction Context Manager process, but for enhanced mobile top-up routing environment, the BASE24-atm Authorization process). After the Device Handler process retrieves the destination name from the SPLIT-TXN- RTG-DEST param it will add the Split Transaction Routing token with the name of the Authorization process in the FUNDS-AUTH-DEST field.

When the BASE24-atm system uses the TCM process, all messages between the Device Handler process and the Authorization process are routed through the Transaction Context Manager process. However, in the enhanced mobile top-up routing environment there will not be a TCM, all messages will be routed between the Device Handler process, the Authorization process, and the interchange.

In the enhanced mobile top-up scenario, the transaction flows from the ATM to the device handler. The device handler sends the transaction to the BASE24-atm authorization process. The authorization process sends the transaction to the host process configured in the IDF based on the MTU prefix route group in the ATM segment of the CPF. This process will actually be set to an interchange interface process name. The interchange interface process will send the transaction to the interchange. The interchange will split the transaction into the funds side and the top-up side. The funds side will be sent to a funds authorizer, in this case the BASE24-atm auth process for authorization. At this time, the transaction will no longer use the MTU prefix route group, but will use either the EMV prefix route group or the default prefix route group from the CPF for authorization. The transaction will be authorized and returned to the interchange. The interchange will coordinate the merging of the funds response and the top-up response. It will return a single response to BASE24-atm.

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts:

Customer service/care functions will need to be modified to handle the unique issues associated with failed or suspect mobile top up transactions.

Operators must be trained to realize they are not using the Transaction Context Manager (TCM) for routing and splitting their Mobile Top-Up transactions.

BASE24-atm acquired Mobile Top-Up transactions where the new MTU prefix route field is non-blank must be configured to be sent online to an interchange network. BASE24 should not be configured to stand-in for the transaction because it is not equipped to split the transaction into the two required parts.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

NCR NDC+ Additional M-Status Codes

This section describes the implementation requirements associated with the NCR NDC+ Additional M-Status Codes enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR NDC+ Additional M-Status Codes enhancement.

The NCR 5XXX Device Handler has been modified to support several additional M-status codes when processing both Solicited and Unsolicited device status messages for the Envelope Dispenser. The following new M-status codes for the Envelope Dispenser are supported in this enhancement:

<u>M-Status</u>	<u>Description</u>
07	Motor failed
08	Communications failed
09	Control electronics failed
10	Exit shutter failed
11	Depository bin overfill
12	Envelope cassette, print head missing
50	Interlock failure

Support for these additional M-status codes includes the following new event messages: 3307-3312, 3350, 3355, 3446, 8307-8312, 8350, 8355, 8446, 10307-10312, 10350, 10355 and 10446.

In addition, the NCR 5XXX Device Handler has been modified to support a large number of previously unsupported M-status codes when processing Unsolicited device status messages for the ATM's Encryptor. The following new M-status codes for the Encryptor are supported in this enhancement:

<u>M-Status</u>	<u>Description</u>
05	Invalid command
07	Communication failure
08	Security module reset
11	Invalid command data
12	Data mismatch
13	No key loaded in DEU
15	Setup failure

16	Invalid keyspace ID format
17	Keyspace already exists
18	Keyspace does not exist
19	Keyspace ID1 wrong attribute
20	Keyspace ID2 wrong attribute
22	Invalid keyspace access right
23	Only two RAR access allowed
24	Secret keyspace create disable
25	Keyspace not parent
26	Key ID2 not specified keyspace
27	Invalid parent format
28	Invalid parent id
29	Parent keyspace programmer
30	Key spaces do not match
30	Key spaces do not match
32	Key ID outside max range
33	Key ID1 has wrong attribute
34	Key ID2 has wrong attribute
37	Invalid ID. No pin encryption
38	Key does not exist
39	Key ID does not specify an IV
40	Invalid key ID, no key
41	Invalid key ID, XOR not valid
48	Invalid attribute field
49	Attribute already in family
50	Invalid parameter field
51	Undefined parameter
52	Variation index outside range
53	Variation index in family
54	Invalid mode for function
55	Invalid left shift
56	Invalid length
57	Invalid access right
58	Invalid POST PIN PAD digit
58	Invalid POST PIN PAD digit
59	Invalid PRE PIN PAD length
60	Mod or exp bit length error

64	Password has no decimal digits
65	Invalid password, no match
66	Keyspace inaccessible
70	PIN verification failure
74	Invalid PIN encrypt function
75	Key integrity error
76	Invalid PIN, no encrypt allow
97	Timeout error
98	Communication error

Refer to the **56XX/Personas Self-Service Financial Terminal 'Diagnostic Status Code Notebook', January 2003**, for more information on NCR's M-status codes.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNMSTAT.

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

- Changes to AT60N50.N50S
- Updated event (log) message manual BA60LOGM.AT5070

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional Event Messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

- 56XX/Personas Self-Service Financial Terminal 'Diagnostic Status Code Notebook', January 2003

ACI Worldwide, Inc.

NCR NDC+ Increase of Secondary EMV AIDS

This section describes the implementation requirements associated with the NCR NDC+ Increase of Secondary EMV AIDS.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR NDC+ Increase of Secondary EMV AIDS enhancement.

The NCR 5XXX EMV Device Handler currently supports downloading a maximum of 4 Secondary Application Identifiers for each defined Primary AID. The device handler also currently supports downloading a maximum of 54 bytes of Data Objects (EMV Tag data). AID data, as well as Data Object data, is downloaded to the terminal in AID records during terminal download.

AID records are created and maintained via page 45 of the DCT. From page 45, select the Access Configuration Menu, then the EMV Configuration Menu. Finally, select the Application Identifier Table. This table contains one record for each AID supported within the system. AID records are stored on the N50 configuration file (CNFG5070, and EMV group 600).

This enhancement increases the number of Secondary AIDs supported, per Primary AID, from 4 to 12. It also adds an additional 14 bytes to the Data Object field of the AID screen. This provides for a maximum 68 bytes of Data Object data.

BASE24-atm
Optional Enhancements
NCR NDC+ Increase of Secondary EMV AIDS

The following screen shows the Application ID table after the application of this enhancement:

```
BASE24-DCT  APPLICATION ID TABLE  TES2  07/06/19  09:00  01 OF 01

EMV GROUP  600          AID NUMBER  0
PRIMARY AID
LOWEST  VERSION NUMBER  0000      DEFAULT APPL LABEL
HIGHEST VERSION NUMBER  0000
NUMBER OF DATA OBJECTS  0        TERMINAL ACTION CODE
DATA OBJECTS

NUMBER OF COMPLETION DATA OBJECTS  0
COMPLETION DATA OBJECTS

SECONDARY AIDS

-----

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:  NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Refer to the BASE24 publications **NCR 5XXX Device Support Manual**, and the **EMV Support Manual** for more information on EMV and the Application ID table.

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

- Changes to AT60N50.N50DDLs, RECIMGCS, SVN50S

All BASE24-atm NCR 5XXX EMV Device Handler customers:

- Changes to AT60IN50.RQEMAIDS, SCNEMAID

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Any existing Application Identifier (AID) Table records in the CNFG5070 configuration file should be read and deleted. AID records with the new format can then be added, and downloaded to the terminal.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Terminal devices must be reloaded with new AID records.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Changes have been made to the AID requester (RQEMAIDS), and N50 server (SVN50S). The new requester object must be SCUPed into the network's POBJ, and the network must point to the new server object. The Application Identifier Table, accessed via page 45 of the DCT, will show the new AID record format.

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- BASE24 NCR 5XXX Device Support Manual
- BASE24 EMV Support Manual

NCR Standard APTRA Edge NDC+ Support

This section describes the implementation requirements associated with the NCR Standard APTRA Edge NDC+ Support.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR Standard APTRA Edge NDC+ Support enhancement.

The NCR 5XXX Device Handler has been modified to support NCR's Aprta Edge Standard NDC Message Format terminal software. Support for Aprta Edge Standard NDC includes a number of changes to the main N50 device handler module. These are generally hooks into a new N50 Edge extension module. This new module is compiled, and bound into the device handler object, at compile time. Inclusion of the new Edge module is dependent upon the setting of a new compiler flag for the Edge module, `atm_dh_n50_edge_on`.

NCR Standard Aprta Edge NDC terminal software vs. NCR 'Reference' Aprta Edge terminal software:

NCR's new version of NCR Edge terminal software is not built specifically to BASE24 specifications, as was the previous 'Reference' version. Aprta Edge Standard NDC Message Format terminal software includes various proprietary properties such as proprietary op codes, and next states and screens. In addition, the 'Standard' version of Aprta Edge is remotely configurable via terminal downloads of XML language instructions. These XML instructions over-ride the existing vanilla XML configuration present on all 'Standard' Edge machines.

With the 'Reference' version of Edge, customers were forced to commit to costly customization work since much of the Aprta Edge application configuration was determined at the time the terminal software was built. The 'Standard' version of Edge eliminates this costly customization work by giving customers the ability to modify their ATM's configuration through terminal downloads of XML language instructions. The downloaded XML instructions overlay the corresponding XML instructions built into the application software.

NCR Standard Aprta Edge NDC terminal software vs. NCR NDC+ terminal software:

The new Edge Standard extension module will be responsible for processing transaction request messages that originate from an Aprta Edge ATM running NCR's Standard Message Format terminal software. Since these messages generally conform to the standard NDC+ message format, basic message processing remains largely unchanged. However, Aprta Edge Standard NDC Message Format software

has its own proprietary Op Codes, and next states and screens, which differ from those presently defined in BASE24's Release 6.0 NCR 5XXX device handler. Therefore, when processing Standard Aprta Edge requests and responses, Op Codes, state and screen numbers, and response codes are appropriately mapped. This mapping, as well as all other Edge specific processing, takes place within the new Aprta Edge extension module so as to minimize the impact on the main device handler module.

A further significant difference between NCR NDC+ terminal software and Edge Standard NDC Message Format software results from the fact that Aprta Edge comes with its own built-in states and screens. This means that while our standard configuration file states and screens can be downloaded to the Standard Aprta Edge machine, they are not required or even used. Instead, Standard Edge NDC Message Format software is configurable through XML data language contained in files resident in the terminal software. By downloading to the terminal specific XML commands, which overlay and over-ride existing configuration instructions, customers can remotely customize their Aprta Edge machines. This is accomplished as part of the regular ATM load, using the existing CNFG5070 configuration file, and specially formatted Screen / Keyboard Data Load Messages. The Standard Aprta Edge terminal will accept XML commands contained within the standard Screen / Keyboard Data Load Message. It is the presence of XML data that causes the Edge terminal to not discard these 'Screen' load messages and, instead, process the XML instructions.

It is anticipated that customers would want the flexibility of being able to process transactions originating from both NDC+ and Aprta Edge terminals, using a single device handler object. Since the basic message format is the same for either terminal type (NDC+ message format), this is possible as long as the device handler knows from which type of terminal the message originated. The device handler can make this determination based upon the device's Software ID and Release Number. The Aprta Edge Standard NDC Message Format application has the following Software ID and Release Number:

Software ID: V351

Release Number: <starts with> 01....

Remote Terminal Configuration:

Aprta Edge XML configuration load messages are constructed using the DCT's existing Screen / Keyboard Data access capability. Therefore, modifications to the DCT system are not required. Customer's wishing to customize their terminals' standard configuration may use the existing DCT customer screen entry capability to create XML load messages, filling the customer screen palette with XML data instead of screen verbiage. Edge machines would normally discard customer screens during a terminal download. However, the presence of XML data within the Screen message causes the Edge application to process the instructions within the message, preventing them from being discarded.

Downloading modifications to the Data Dictionary and/or XML file(s) is accomplished using the standard NDC+ Screen / Keyboard Data Load Message (Message Class '3'), but with a slightly modified format. The general format for the Screen / Keyboard Data Load Message carrying modification instructions is as follows.

Field	No. of Chars	M / O	Description
a	Var	M	Protocol-dependent Header
b	1	M	Message Class = '3' (Data Command)
c	1	M	Response Flag (reserved for future use)
fs	1	M	Field Separator
d	3	M	Logical Unit Number (LUNO)
fs	1	M	Field Separator
e	3	O	Message Sequence Number (reserved for future use)
fs	1	M	Field Separator
f	1	M	Message Sub-Class = '1' (Customization Data)
g	1	M	Message Identifier = '1' (Screen and/or Keyboard Data)
fs	1	M	Field Separator
h	3	*	Screen Number (Record Number)
I	Var	**	Screen Data Field (XML Data)

* For XML data records, the Screen Number is actually the number of the particular load record, and is user defined. It exists only to permit the user to distinguish between various records within the configuration file.

** For XML data records, the Screen Data Field will contain the actual XML instruction(s). Note that this field is defined as variable in length. NCR has defined the maximum size of the entire Screen / Keyboard Data Load Message to be 2000 bytes. However, the actual maximum record size will be significantly less than 2000 bytes as it is limited by the size of the DCT Screen input palette, and because the configuration instructions are written as nested XML code.

The following is an example of an instructional update of the Data Dictionary as it might be constructed on the Screen entry screen of the DCT.

```
<CONFIGURATION_DATA reboot="y">
  <DD_UPDATE>
    <GROUP>CONSUMERFLOW</GROUP>
    <NAME>ENVELOPEDEPOSIT</NAME>
    <VALUE>YES,1,8</VALUE>
  </DD_UPDATE>
</CONFIGURATION_DATA>
```

This command affects the Envelope Deposit transaction. This instruction enables the Envelop Deposit transaction at the ATM (the 'YES' part of the instruction), and places the transaction selection at the 8th position of the first menu (the "1,8" part of the instruction).

Note that the downloaded instruction is transmitted to the ATM as a nested block of XML, starting with <CONFIGURATION_DATA> and ending with </CONFIGURATION_DATA>. The example above also includes the [reboot="y"] clause. This clause must be included whenever a reboot of the ATM is required. Some, but not all, Data Dictionary updates require rebooting. In fact, updates to "CONSUMERFLOW" such as the one in the example above do not require rebooting. The reboot="y" command is included in the example above only to demonstrate the use of the command.

The actual update instructions for the Data Dictionary are included within the <DD_UPDATE> </DD_UPDATE> parameters. This example contains a single update instruction, but additional Data Dictionary update commands can be included within a single load message, by inserting them between the <CONFIGURATION_DATA> </CONFIGURATION_DATA> parameters. However, the entire update command must be contained within a single load message, and every load message must contain the <CONFIGURATION_DATA> </CONFIGURATION_DATA> parameters.

Here is how the preceding example instruction would look, ready to be downloaded, formatted within the configuration file's Load Message record (spaces appear for readability only):

```
9 3 0 fs 000 fs 000 fs 11 fs 001 <CONFIGURATION_DATA
reboot="y"><DD_UPDATE><GROUP>CONSUMERFLOW</GROUP><NAME>ENV
ELOPEDEPOSIT</NAME><VALUE>YES,1,8</VALUE></DD_UPDATE></CONFIG
URATION_DATA>
```

Consult the ***NCR Aptra Edge Remote Configuration Guide, release 02.03.xx, January 11, 2006***, for more specific information on formatting instructional data.

Updating an XML file:

There are several XML files that may be updated as well. These include the following (default locations are indicated):

- ButtonToKEYMAP.xml – c:\Webatm\WebPages\scripts
- Coupon.xml – c:\Webatm
- Journal.xml – c:\Webatm
- Receipt.xml – c:\Webatm
- SoundEng.xml – c:\Webatm
- SoundSpa.xml – c:\Webatm

Just like Data Dictionary updates, XML file updates must be included within the <CONFIGURATION_DATA> </CONFIGURATION_DATA> block. Specific instructions are further nested within <XML_UPDATE> </XML_UPDATE> blocks. The initial <XML_UPDATE> command must include the complete path and filename of the XML file being modified. The filename clause is followed by another <XML_Update> </XML_Update> block, and then a nested <Update> </Update>, <Insert> </Insert>, or <Delete> </Delete> block, depending on the intent of the instruction.

The following is an example of an instructional update of the ButtonToKeyMap XML file as it might be constructed on the entry screen of the DCT.

```
<CONFIGURATION_DATA reboot="y">
  <XML_UPDATE
    FILENAME="C:\WebATM\WebPages\scripts\ButtonToKeyMap.xml">
    <XMLUpdate>
      <Update>
        <VALUE>YES,1,8</VALUE>
      </Update>
      <NodeKey>
        <Path>Document/FDKKey</Path>
      </NodeKey>
      <XML>
        <FDKKeyMap Button="1" FDKKey="1"/>
        <FDKKeyMap Button="2" FDKKey="2"/>
        <FDKKeyMap Button="3" FDKKey="3"/>
        <FDKKeyMap Button="4" FDKKey="4"/>
        <FDKKeyMap Button="5" FDKKey="5"/>
        <FDKKeyMap Button="6" FDKKey="6"/>
        <FDKKeyMap Button="7" FDKKey="7"/>
        <FDKKeyMap Button="8" FDKKey="8"/>
      </XML>
    </XMLUpdate>
  </XML_UPDATE>
</CONFIGURATION_DATA>
```

This example command affects the FDKKey Settings as defined within the specified XML file.

Note that a single Screen / Keyboard Data Load Message may contain multiple Data Dictionary update commands, multiple XML file update commands or a combination of both Data Dictionary and XML file update commands.

Consult the ***NCR Aptra Edge Remote Configuration Guide, release 02.03.xx, January 11, 2006***, for more specific information on formatting instructional data.

Creating XML Screen / Keyboard Data Load Message Records:

Selecting page 45 from the DCT main menu results in the Download Load Data / Access Configuration File submenu. Selecting Access Configuration File produces the individual config file record access screen. Selecting Screen will produce the Customer Screen palette. While this screen is normally used to construct and access customer screens, for Edge it will be used to construct and gain access to XML configuration load messages. The screen input is unrestricted, and free-format, meaning that it is the responsibility of the operator constructing the XML instructions to ensure the XML instructions are properly formatted.

Software Impacts

All BASE24-atm customers:

- Changes to AT60SRC.ATMFM, AT60SRC.ATMMM
- Changes to BA60MAKE.CUSTMACS, BA60MAKE.FLGSDOC

All BASE24-atm NCR 5XXX Device Handler customers:

- Changes to AT60N50.N50D, AT60N50.N50G, AT60N50.N50M, AT60N50.N50S

All BASE24-atm NCR 5XXX Device Handler Standard Aptra Edge customers:

- Restore the new AT60EN50.* subvolume (ATEN50FM, ATEN50MM, N50EDGE, N50EDGE, N50EDGES)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNEGE.

CUSTMACS Changes

There is a new compiler flag for the Edge module, atm_dh_n50_edge_on. Setting this flag to 'ON' will cause the Standard Aptra Edge device handler code to be compiled and bound into the main N50 device handler object.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

NCR Standard Aprta Edge Terminal devices come pre-configured with a standard set of XML language instructions. Customization of these XML instructions must be done if customers wish to modify that standard configuration. Customization can be accomplished either at the time of the terminal software build, and/or via terminal downloads of 'screen' records containing XML language instructions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- BASE24 NCR 5XXX Device Support Manual
- NCR Aprta Edge Remote Configuration Guide, rel 02.03.xx, January 11, 2006
- NCR Aprta Edge Standard NDC Message Format Guidelines, rev H, December 12, 2005
- NDC+ Programmer's Reference Manual, rel 7.0, August 2002

Preferred Transaction

This section describes the implementation requirements associated with the adding Preferred Transaction support to BASE24-atm.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Preferred Transaction enhancement.

The Preferred Transaction functionality is handled as 3 discreet messages. One message is used to set up the preferred transaction attributes. One message is to inquire on the existing attributes. The actual transaction is unchanged, except for an additional token that identifies it as a preferred transaction.

- A message to set up a Preferred Transaction profile is sent from the BASE24-eps IFX Device Handler as a Message to FI (Transaction Code 60). It has the transaction subtype token present with the subtype set to APS0.
- A message to inquire on the Preferred Transaction profile is sent from the BASE24-eps IFX Device Handler as a Card Review Transaction (Transaction Code 62). It has the transaction subtype token present with the subtype set to API0.
- A message to execute a Preferred Transaction profile is sent from the BASE24-eps IFX Device Handler with the standard transaction code (e.g., withdrawal, deposit, transfer, etc.) for the cardholder's preferred transaction. It has the transaction subtype token present with the subtype set to APT0.

This enhancement is not an add-on enhancement requiring a separate license. It is available to all BASE24-atm customers with the IFX Device Handler.

Prerequisites: The BASE24-eps IFX Device Handler is required.

Software Impacts

All BASE24 customers:

- Inline changes in BA60DDL. DDLATTKN, DDLBCNST, DDLFCAF, DDLFIDF, DDLGCAFX, and DDLGISO.
- Inline changes in the BA60AFT.RQCAFS, SCRNCASF, SVCAFS, RQIDFS, SCRNFIDF, SVIDFS, COBTKN, PITABLE and SECTBL.
- New requester and Make file, BA60AFT.RQIDFPTS and RQIDFPTM
- Changes to BA60AFT.BAAFTM and BAAFTMM for the new requester
- Changes to BA60SRC. ATTKNID, ATTKNCVS, and COBNAMES
- Changes in BA60EXT.OMFXG and OMFXS
- Changes in BA60REFR.REFRG and REFRS

All BASE24-atm customers:

- Inline changes in AT60AUTH.AUTHLIBS, AUTHS and AUTHG
- Inline changes in AT60HISO.ATHISOS
- Updated event (log) message manuals BA60LOGM.ATAUTH

All BASE24-fhm customers:

- Inline changes in FH60DDL.DDLFHSTM
- Inline changes in FH60FHM.FHMS
- Inline changes in FH60RPT.FHMRPTS
- Updated event (log) message manuals BA60LOGM.FHM

All BASE24-fhm CAF customers:

- Inline changes in FH60CAF.FHMCASF

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Cardholder Authorization File (CAF)

A new CAF screen 21 has been added for Preferred Transaction information. User security must be modified to allow operators to see this new screen.

This field indicates the account to be used in the preferred transaction.

Screen Name:	ACCT NUM
Screen:	CAF Screen 21
DDL Name:	CAF.PFRD-TXN.ACCT
Valid values:	Any account identifier, up to 19 alphanumeric characters. The default is spaces.

This field indicates the transaction code for the preferred transaction.

Screen Name: TRAN CODE
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.TRAN-CDE
Valid values: Valid transaction code identifiers. The default is spaces

This field indicates the 'from' account type for the preferred transaction.

Screen Name: FROM ACCT TYPE
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.FROM-ACCT-TYP
Valid Values: Valid account types. The default is spaces.

This field indicates the 'to' account type for the preferred transaction.

Screen Name: TO ACCT TYPE
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.TO-ACCT-TYP
Valid Values: Valid account types. The default is spaces.

This field indicates if a receipt is required for the preferred transaction.

Screen Name: RCPT OPTION
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.RCPT-OPT
Valid Values: N or Y. The default is N.

This field indicates the amount of the preferred transaction.

Screen Name: AMOUNT
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.AMT
Valid Values: Any numeric value up to 19 positions. The value is in whole units. No minor units are allowed. For example, for USD, the value of 10 indicates \$10.00 dollars. The default is zero.

This field indicates if the preferred transaction profile can be updated from the device.

Screen Name: PROFILE UPDATE IND
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.PRFL-UPDT-IND
Valid Values: N or Y. The default is Y.

This field indicates if American Disabilities Act (ADA) support at the terminal, if available.

Screen Name: ADA IND
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.PRFL-UPDT-IND
Valid Values: N or Y. The default is N.

This field indicates a market segment for the preferred transaction. It can be used by the terminal to display special messages.

Screen Name: MARKET SEG IND
Screen: CAF Screen 21
DDL Name: CAF.PFRD-TXN.MRKT-SEG-IND
Valid Values: Any 2 alphanumeric characters. The default is spaces.

A full description of these fields can be found in the **BASE24 Files Maintenance Manual**.

Institution Definition File (IDF)

Screens 5/6 of the IDF are used to indicate which segments should be added to Authorization files, such as the CAF. If it is required that the Preferred Transaction segment be added to CAF records for an FIID, the segment must be turned on by setting the segment to "Y" on this screen.

A new IDF screen 43 has been added for Preferred Transaction information. User security must be modified to allow operators to see this new screen.

This field indicates where to store data for the preferred transaction profile.

Screen Name: STORE DATA
Screen: IDF Screen 43
DDL Name: IDF.PFRD-TXN.ACCT
Valid values: 1 CAF only
2 Host only
3 CAF if Host is unavailable
4 Both CAF and Host
The default value is 1.

This field indicates where to retrieve data for the preferred transaction profile.

Screen Name: RETRIEVE DATA
Screen: IDF Screen 43
DDL Name: IDF.PFRD-TXN.TRAN-CDE
Valid values: 1 From the CAF
2 From the Host
3 From the CAF if Host is unavailable
The default value is 1.

A full description of these fields can be found in the **BASE24 Files Maintenance Manual**.

Transaction Code File (TCF)

The customer may want to set up unique descriptions for the transaction codes used in Preferred Transaction. These descriptions could be helpful when setting up TSRF records for each of the Transaction Subtypes implemented with this enhancement. The transaction codes used are:

- Message to FI (BASE24 Transaction Code 60) to ISO code of 91.
- Card Review Transaction (BASE24 Transaction Code 62) to ISO code of 98.

A full description of this file and fields can be found in the **BASE24 Files Maintenance Manual**.

Transaction Code / Subtype Relationship File (TSRF)

The customer must set up TSRF records for each of the Transaction Subtypes implemented with this enhancement. They are:

- API0 Preferred Transaction Inquiry
- APS0 Preferred Transaction Set-Up
- APT0 Preferred Transaction

A full description of this file and fields can be found in the **BASE24 Files Maintenance Manual**.

Token file (TKN)

The following BASE24-atm tokens have been added for this enhancement:

- The ATM Preferred Transaction Token (token ID “AM”) contains the information about the preferred transaction. The information matches the data in the CAF Preferred Transaction segment, except for an additional “on-us” field, set by the IFX device handler to indicate a particular transaction is “on-us”.

If the tokens for this enhancement are to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The BASE24-eps IFX Device Handler is required to perform Preferred Transactions. No BASE24-atm device handler supports this functionality at this time.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

There is a new segment on the CAF and IDF for Preferred Transaction data. In order for this segment to appear on the appropriate Pathway screens and in the data files, the Preferred Transaction segment must be set to "Y" in the SEG-TABLE section of BA60AFT.PITABLE.

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new CAF screen 21 and IDF screen 43.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new CAF and IDF screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

New segments were added to the CAF and IDF files. The conversion programs to convert these files from Release 5.3 to Release 6.0 version 8 were not modified to handle this segment, since no Release 5.3 system could have this segment present.

Cardholder Authorization File (CAF)

To add the new Preferred Transaction segment to a Release 6.0 version 8 CAF, the customer must perform a full file CAF refresh with this data present in the record.

Institution Definition File (IDF)

To add the new Preferred Transaction segment to a Release 6.0 version 8 IDF, each IDF record must be updated in Pathway after the new segment has been set to "Y" in the SEG-TABLE and the Pathway objects recreated with Make.

Host/Co-Network Impacts

Host impacts may be incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

The host or co-network may require modification to process the new ATM Preferred Transaction token. If the token for this enhancement is to be sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

It is possible to configure the processing of Preferred Transactions to store and receive preferred transaction profile data on the host rather than in the CAF. If this is the desired configuration, the host code must be modified to receive requests for this type of data, and return the ATM Preferred Transaction token, populated with the necessary information in response messages.

Super Extract Process

If the new token for this enhancement is to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to IDF and CAF records.

Refresh Process

Host code changes are required for all BASE24 users that utilize the Release 6.0 CAF Refresh file message format to provide the new Preferred Transaction segment in the CAF.

From Host Maintenance Process

Host code changes are required for all BASE24 users that utilize FHM to provide the new Preferred Transaction segment in the CAF.

User/Operational Impacts

Disk Impacts

The new CAF and IDF segments require additional disk space.

- The length of the CAF Preferred Transaction segment is 60 bytes. A new segment will be added for each CAF record in an FIID that has the segment set to Y on IDF screen 5/6.
- The length of the IDF Preferred Transaction segment is 10 bytes. A new segment will be added for each IDF record in network that has the segment set to Y in the SEG-TABLE of the PITABLE.

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Event Messages

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Shared NDC+ BNA Support

This section describes the implementation requirements associated with the Shared NDC+ BNA Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared NDC+ BNA Support enhancement.

Wincor Nixdorf has developed a new module for their ATMs to enhance their deposit capabilities. Wincor calls this new module the CashIn module. It is present on CRS and RM2 ATMs, and it is part of the Check Cash Deposit Module (CCDM) on the CCDM ATMs. This new module allows a bunch of notes (bills) and/or coins (if the CoinIn NOVOTEC Module is present) to be deposited into a secured part of the ATM. It can accept, count and verify a stack of bills and/or a number of coins, and can reject any that are invalid or unrecognizable (i.e., torn, dirty, etc.) before it sends the transaction request to the processing Host (BASE24). The bills and coins can be of different denominations as well as up to four different currencies that are supported by the ATM.

This enhancement is applicable to BASE24 systems running ATMs loaded with firmware that conforms to the Shared NDC+ CashIn specification created by Wincor Nixdorf. These ATMs will run off BASE24's NCR 5XXX Device Handler.

The ATM is configured with signature files. Each of these files provides the ATM with the information needed to recognize a particular denomination of note. The ATM is also configured with a Note Id List, which determines which notes of which denominations it can accept. Valid signature files must exist for particular note denominations before they are configured in the Note Id List, and the Note Id List must contain details of all note denominations for which signature files exist, although certain note denominations can be set as not being accepted. BASE24 also maintains a Note ID File (NIDF), detailing the denominations of the notes that can be accepted at the ATM, and this file is used by BASE24 to determine the transaction amount deposited at the ATM. The customer must ensure that these two files are kept in sync. Failure to do so will result in the cardholder being credited for an amount different from the amount of money that was deposited at the ATM.

At present, while the Wincor ATMs can be configured to support recycling of notes, BASE24 does not support recycling of notes at the ATM.

New commands available at the Device Control Terminal (DCT) will allow the customer to inquire as to the ATM's detailed configuration for accepting notes, change whether a particular note denomination is accepted or rejected, warmboot processes using the NIDFEMT and collect count information from the CashIn device.

This enhancement is provided in the form of two bindable extensions to the BASE24 NCR 5XXX Device Handler. The first extension, the Shared Bunch Note Acceptor (BNA) extension, is required to support the CashIn deposits. It will accept deposits made in any of the denominations of the terminal's currency, configured in the Note ID file as being accepted at the particular ATM. The second extension, the BNA Multiple Currency extension, is required if any ATM is to support the acceptance of more than just the terminal's currency. In this case the Note ID file will contain details of denominations in a maximum of four different currencies that will be accepted by the ATM.

Prerequisite: Wincor Nixdorf ATMs loaded with firmware conforming to Wincor Nixdorf's specification for CashIn are required to support the Shared NDC+ BNA enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLGSTM, DDLATTKN, DDLFOMF
- Minor inline changes to AFT files for the NIDF, BA60AFT.MEGATBL, SECTBL, CCAFMLM and SVMHLPS
- Changes to CUSTMACS and FLGSDOC
- Changes to Make files to include the new CUSTMACS flags, BA60SRC.BASEMM, BASEFM and BASEM
- Minor inline changes to BA60AFT.COBTKN, BA60SRC.ATTKNCVS and ATTKNID
- Minor inline changes to BA60DCT.RQDECTS and SVDPCTS
- Minor inline changes to BA60EMTB.BEMTBLM and EMTBRUN
- Minor inline changes to BA60RPT.OMFTBL
- Minor Inline changes to BA60MISC.BAMISCTD

All BASE24-atm customers:

- Inline changes to AT60DDL.DDLFTLF and DDLGTLFX
- Minor inline changes to AT60DDL.DDLFATD and DDLGATD
- New DDL for the NIDF, AT60DDL.DDLFNIDF
- Minor inline changes to AT60DDL.ATDDLMM and ATDDLMM
- Minor inline changes to AT60AFT.SCRNATD1, SVATDTBL and SVATDTS
- New Requester and Server code for the NIDF, AT60AFT.RQNIDFSS, RQNIDFXM, RQNIDFXS, SVNIDFTG, SVNIDFTS and SVNIDFTG
- Minor inline changes to Make files to include the NIDF, AT60AFT.ATAFTM and ATAFTMM
- Minor inline changes to Make files to include the new CUSTMACS flags, AT60SRC.ATMFM, ATMMM and ATMM
- Minor inline changes to AT60EXT.ATOMFXG, ATOMFXM and ATOMFXS

- New Extended Memory table build (EMTBUILD) code AT60EMTB.NIDEMTTG, NIDEMTTS and NIDEMTTM
- Minor inline changes to AT60EMTB.AEMTBM, AEMTBMM and AEMTDS
- Inline changes to AT60DCT.SCNDCOMM and RQDCOMMS
- Inline changes to AT60DCT.SVDCTG and SVDCTS
- Minor inline changes to AT60AFT.COBRCPT
- Minor Inline changes to BA60MISC.LCONFAT

All BASE24-atm MultiCurrency and Multiple Currency Dispense customers:

- Inline changes to AT60UMCD.ATMCDHLG and ATMCDHLS
- Updated event (log) message manual BA60LOGM.ATATMCDH

All BASE24-atm NCR 5XXX device handler customers:

- Changes to AT60N50.N50S and N50D
- Minor inline changes to AT60N50.N50G, N50NAMS and N50M
- Inline changes to AT60N50.N50DDL and N50DDL
- Changes to AT60N50.CNFG5070 and CNN50DOC
- Inline changes in AT60N50.RQ50STS, SCRNST
- Minor inline changes to AT60N50.SV50PRIS
- Minor inline changes to AT60N50.RQN50LS and SCRNN50L
- Inline changes to AT60N50.SVDN50S and SVDN50M
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX SSB device handler customers:

- Inline changes to AT60DN50.N50SSDS
- Minor inline changes to AT60DN50.N50SSDM.

All BASE24-atm NCR 5XXX device handler customers requiring the new Shared NDC+ BNA Support functionality:

- A new extension subvolume, AT60DWN5, with all of the files required to support the Shared NDC+ BNA Support functionality.

In addition to the above, any customer wishing to perform the Multiple Currency cash deposits must make sure they install the following, already existing, code:

- The ERF Requester/Server code(BA60AFT.RQERFS, RQERFM, SCRNERF, SVERFM and SVERFS)
- The ERFEMT Build code (BA60EMTB.ERFEMTTG, ERFEMTTS and ERFEMTTM)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNWNB.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the Shared NDC+ BNA module to the NCR 5XXX device handler.

- atm_dh_n50_shrd_bna_on

If Multiple Currency Cash Deposits are to be accepted, then set the following flag to true to bind the Shared NDC+ BNA MCU module to the NCR 5XXX device handler.

- atm_dh_n50_shrd_bna_mcu_on

Then set the following flag to true to bind the Multiple Currency Utilities into the NCR 5XXX device handler.

- atm_bna_mcu_on

LCONF Assign Changes

This section describes the assign records that are required to be added to the LCONF to support this enhancement. Details of the new assigns are provided in the LCONFAT file located on BA60MISC subvolume.

ATM-NIDF

The fully qualified file name of the Note ID file used by Shared NDC+ BNA. This assign is used and required by the NIDFEMT Build program.

ATM-NIDFEMT

The fully qualified file name of the Note ID EMT used by Shared NDC+ BNA. This assign is used and required by the NIDFEMT Build program and the NCR 5XXX device handler, if Shared NDC+ BNA is being supported.

In addition, if BNA Multiple Currency is being used, several params that already exist in a multiple currency environment will also be used. These following params therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

- ERFEMT
- ERF

LCONF Param Changes

This section describes the param records that are required to be added to the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFAT file located on BA60MISC subvolume.

ATM-NIDFEMT-MAX-TBL-SIZE

Identifies the maximum size in megabytes of the NIDFEMT extended memory table. This param is used to set the amount of memory allocated by the EMTBuild process.

The default value is 127 megabytes.

ATM-NIDFEMT-WARMIdentifier

Identifies the symbolic name of a process to receive the warmboot command when the NIDFEMT is warmbooted from the DCT.

The Device Control Terminal Server process retrieves all ATM-NIDFEMT-WARMIdentifier params when a network control operator issues a command to reinitialize the NIDF extended memory table. The process sends a warmboot command to each identified process.

When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another ATM-NIDFEMT-WARMIdentifier param. For example, ATM-NIDFEMT-WARM01 could contain the P1A^ N501 process and ATM-NIDFEMT-WARM02 could contain the P1A^ N502 process.

A maximum of 150 ATM-NIDFEMT-WARMIdentifier params can be used to identify the processes to be warmbooted from the DCT. However, each param can include any combination of wildcard characters (i.e., asterisks) and alphanumeric characters, enabling one ATM-NIDFEMT-WARMIdentifier param to identify several processes. For example, P1A^N* identifies all processes that start with P1A^N.

In addition, if BNA Multiple Currency is being used, several params that already exist in a multiple currency environment will also be used. These following params therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

- CURRENCY-CODE
- BASE-CURRENCY-CODE

Database Configuration

Token file (TKN)

The following BASE24-atm tokens have been added for this enhancement:

- The Shared BNA Counts Token (token ID “AS”) contains the note counts for the Bulk Note Acceptor (BNA) device. These counts can only be obtained from the ATM. The data in this token gets written to the BASE24-atm Transaction Log File (TLF) whenever the ATM is balanced. This token is created only for ATMs supporting the Shared NDC+ BNA enhancement.

- The BNA Multiple Currency Token (token ID “AT”) carries details of the separate deposit amounts for each of the currencies that were deposited at the ATM, if BNA Multiple Currency Cash deposits are supported. It is added by the Device Handler and used during the response processing to format the receipt.

If the tokens for this enhancement are to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

ATM Terminal Definition Files (ATD)

This section describes the new and enhanced fields in the ATD file that are required to support this enhancement.

The NOTE ID Profile is a new field on screen 2 of the ATD. It is only used by the Shared BNA enhancement.

Screen Field Name:	NOTE ID PROFILE
Screen Field Length:	16 alphanumeric characters
Range of Values:	Alphanumeric and special characters, no leading or embedded spaces allowed
Default Value:	Blank
DDL Name:	ATDS1.ATDS1-CORE.NIDF-PRFL

This field should be set up for each ATM that is to support the Shared BNA enhancement. It must contain the key value to be used in reading up the profile from the NIDF.

The existing Hardware field, on screen 1, is being amended with this enhancement to allow the customer to update the field. Previously the field was set by the device handler.

Screen Name: HARDWARE
Screen: ATD Screen 1
DDL Name: ATDD1-CORE.HDWR

Valid Values are: 0 – Unknown
 1 – Wincor
 2 – NCR
 3 – Diebold
 4 – Fujitsu
 5 – De La Rue
 6 – Triton
 7 – Olivetti

Default Value: 3 if the device type is 30 (Diebold)
 2 if the device type is 22 (NCR)
 0 for all other device types.

This needs to be set appropriately for the ATMs using the Shared BNA enhancement.

A full description of these fields can be found in the **BASE24 Files Maintenance Manual**.

NCR 5XXX State Screens (SCRNST)

Support was added for a new state type, the CashIn state (r). Customers wanting to take advantage of this state will need to make their own configuration changes. Example states and screens for the Shared BNA enhancement are available in the CNFG5070 file on the AT60N50 subvolume, in group 800.

Receipt file (RCPT)

New field type codes were added to the RCPT file for use when processing Multiple Currency Cash Deposits. The customer can configure these codes if wished, on the deposit receipt, to allow for exchange rate information to be produced.

The following new codes were defined:

- CA Converted amount – produces the amount for the particular currency deposited, once converted to the terminal's currency.
- OA Original amount – produces the amount in the currency that was deposited
- EA Exchange rate – produces the exchange rate used to convert the Original amount to the converted amount
- DC Produces the text "CONVERTED AMOUNT" on the receipt
- DO Produces the text "ORIGINAL AMOUNT" on the receipt
- DE Produces the text "EXCHANGE RATE" on the receipt

Each ATM can accept up to a maximum of four currencies, if Shared BNA Multiple currency support is implemented. Therefore, a suggested layout for the receipt is as follows:

```
@RECORD NO.#E^4 #SQ>>  
#DO#E^3#OACC$> #DE#EA> #DC#E^3#CACC$>  
#DO#E^3#OACC$> #DE#EA> #DC#E^3#CACC$>  
#DO#E^3#OACC$> #DE#EA> #DC#E^3#CACC$>  
#DO#E^3#OACC$> #DE#EA> #DC#E^3#CACC$>>  
#TT #A1$>  
@TO#E^5#TO>>  
#XL#E^4#LB$>  
#XA#E^4#AB$>>
```

Each of the 4 lines of new field codes above will print details for one of the currencies deposited. If less than four currencies are deposited, then the lines will only be printed for those currencies that were present.

See below for an example of a receipt for a BNA deposit of four different currencies, produced using the suggested template above.

1	
2	A C I D E M O
3	DATE TIME TERM
4	MM/DD/YY HH:MM S1AN5070
5	
6	123456700000003-003
7	
8	330 SOUTH 108TH AVE
9	
10	RECORD NO. 570
11	
12	ORIGINAL AMOUNT \$210.00
13	EXCHANGE RATE 1.000000
14	CONVERTED AMOUNT \$210.00
15	ORIGINAL AMOUNT CAD20.00
16	EXCHANGE RATE 1.063500
17	CONVERTED AMOUNT \$21.27
18	ORIGINAL AMOUNT MXN50.00
19	EXCHANGE RATE 4.596220
20	CONVERTED AMOUNT \$4.60
21	ORIGINAL AMOUNT GBP100.00
22	EXCHANGE RATE 2.091200
23	CONVERTED AMOUNT \$209.12
24	
25	DEPOSIT \$444.99
26	TO 013333
27	
28	LEDGER BAL \$3412.00

1	2	3
123456789012345678901234567890		

Exchange Rate File (ERF)

The ERF (and the corresponding ERF Extended Memory Table (ERFEMT)) are used by the Shared NDC+ BNA Multiple Currency extension if Multiple Currency cash deposits are to be accepted at any of the customer's ATMs. This file must contain the exchange rates to be used, and the ERFEMT must be successfully built from the ERF.

New Data Files

Note ID File (NIDF)

The BASE24-atm Note ID File (NIDF) contains in each record a set of note ids that will be accepted by the ATM. It is anticipated that several ATMs will share a similar setup of note ids, and so the records in the NIDF have a Note ID Profile as their key, allowing many ATD records to point to the same NIDF record.

Each NIDF record has details of the note id, currency code, and denomination of the note that is accepted. The NIDF File Maintenance sorts the note ids into ascending order in the NIDF. Note ids of 01 – 99 are valid, although BASE24 only supports a maximum of 45 note ids, so two screens are available to allow a maximum of 45 note ids to be configured. Note ids of 80 – 99 must be used to configure coins. Consecutive note ids do not have to be used. The denomination field is in whole units of the major currency unit if the note id is a note and not coin. For example, for USD, 10 would indicate \$10.00. It is whole units of the minor currency unit if the note id is for a coin. For example, for USD, 10 would indicate 10 cents.

The NIDF record also contains a modifier for each of the note ids. This indicates whether the note id should be accepted at the ATM or not. The LoadNoteIDs command from the DCT will load the appropriate NIDF record down to the ATM and allows the customer to remotely turn on or off acceptance of a particular note id that has a valid signature file at the ATM, and is in the ATM's Note ID List. Coin note ids (values of 80 – 99) will not be sent to the ATM as part of this command, as they cannot be updated remotely.

Note ID File Extended Memory Table (NIDFEMT)

The NIDFEMT is an Extended Memory Table built by the EMTBuild process from the NIDF. This allows the device handler to access the data in the NIDF in an efficient manner.

Template Changes

The customer must configure the AT60N50.N50NAMS file. This table contains the ncr_money_sym_tbl which provides a location to configure the currency symbol that should appear on the receipt, for any currency that the customer wishes to accept at an ATM supporting the Shared NDC+ BNA enhancement.

Device Configuration

NCR 5XXX Configuration File (CNFG5070)

The NCR 5XXX Configuration File (CNFGIX) requires new states and screens to support the new Shared NDC+ BNA enhancement. For more detailed information, see the CNN50DOC file on the AT60N50 subvolume. A default set of screens is provided, but these may require specific modifications depending upon the institution's requirements.

Load group 800 is the new SSB BNA load group, and is required to support the Shared NDC+ BNA Support enhancement. It must be loaded on top of base load group 0 and SSB DH load group 106.

ATM Registry settings

The ATM must have certain registry settings configured, to ensure that the Device Handler can correctly determine the status of the CashIn device.

These settings are as follows:

*[HKEY_LOCAL_MACHINE\SOFTWARE\Wincor
Nixdorf\ProTopas\CurrentVersion\CCOPEN\PROTOCOL\PARAMETER]*

"HW_CONF_22_SIG_DISPLAY"="00"
"HW_CONF_23_RESERVED"="00"
"HW_CONF_24_COIN_DISP"="00"
"HW_CONF_25_SYSTEM_DISPLAY"="02"
"HW_CONF_26_MED_ENT_IND"="00"
"HW_CONF_27_ENV_DISP"="00"
"HW_CONF_28_DOC_PROC_MOD"="00"
"HW_CONF_29_COIN_TAMP_IND"="00"
"HW_CONF_30_DOC_TAMP_IND"="00"
"HW_CONF_31_RESERVED"="00"
"HW_CONF_32_DIG_AUDIO_SERV"="00"
"HW_CONF_33_RESERVED"="00"
"HW_CONF_34_RESERVED"="20"
"HW_CONF_35_RESERVED"="00"
"HW_CONF_36_RESERVED"="00"
"HW_CONF_37_RESERVED"="00"

Please note, the values above need to be specifically set for the ATM Model that you have. Please contact Wincor Nixdorf for the specific values required for your ATM.

The following registry setting is required to ensure that the ATM can correctly respond to a COUNTS command within the correct timeframe:

*[HKEY_LOCAL_MACHINE\SOFTWARE\Wincor Nixdorf\ProTopas\
CurrentVersion\CCOPEN\PROTOCOL\NDC\H2TSendCinSupplyCounters]*

"VAL_FIELD_001"="VAL_WRONG_ATM_MODE,DC_MODE_TRAN_ACTIVE:20"

The 20 indicates the number of seconds that the ATM will wait to determine if the internal processing of the COUNTS command is complete. If the command has not finished being processed at that time, the command will be rejected. This time limit can be changed by the customer as required.

The following registry setting is required to ensure that if a coin module is used, the unsolicited or solicited messages regarding the coin module contain the correct device id that BASE24 expects:

*[HKEY_LOCAL_MACHINE\SOFTWARE\Wincor Nixdorf\ProTopas\
CurrentVersion\CCOPEN\PROTOCOL\NDC\T2HUnsolicitedCoinIn]*

"PREP_FIELD_006"="PREP_FIXED,r"

The following registry setting is required to ensure that a single cashin retract does not result in the ATM hitting the maximum allowed retracts:

*[HKEY_LOCAL_MACHINE\SOFTWARE\Wincor Nixdorf\ProTopas\
CurrentVersion\CCOPEN\CCCashInFW]*

"MaxRetract"="99"

This value can be set to the number of retracts that the customer determines is their maximum. If left to the default value of "-1" the first retract will be the maximum.

CashIn device

A CashIn device must be installed on every ATM implementing Shared NDC+ BNA.

Mega Table (MEGATBL) Changes

The NIDF has been added to the list of requesters that use the extended message format, in the MEGATBL file.

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new NIDF screens. This file will appear on the ATM Product Menu.

PATHCONF Changes

No PATHCONF file changes are required, as the new NIDF Server is bound into the Classic Control Server.

CRT Access Security Changes

Access to the new NIDF screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ATM Terminal Definition Files (ATD)

For users wishing to take the Shared NDC+ BNA Support enhancement, either the ATD conversion program must be run to convert the ATD, or the customer may modify the fields manually via PATHWAY for affected ATMs. If the conversion program is to be used, the customer must create the new ATD file using the ATDDLUP file.

The ATD conversion program must be run to convert the ATD. Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

The host or co-network may require modification to process the new BNA Multiple Currency token or Shared BNA Counts token. If the tokens for this enhancement are to be sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

Super Extract Process

If the new tokens for this enhancement are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24-atm users that want to use OMF Extract to accept and process information from the NIDF record.

User/Operational Impacts

BASE24 users must understand the concept of EMT Builds and the procedure to create an Extended Memory Table file (EMT). BASE24 users must understand the EMT Build Utility report that is generated by the EMT Build Utility. BASE24 users must also understand the concept of EMT warmboots and the procedure to warmboot an EMT. The NIDFEMT is used in this enhancement, along with the ERFEMT, if multiple currency cash deposits are being supported.

If changes are made to the NIDF, the NIDFEMT must be rebuilt, and any processes with the NIDFEMT open must be warmbooted in order to pick up these changes.

Disk Impacts

The TLF requires additional disk space as the new tokens for this enhancement can be included in the transaction records logged.

The new NIDF and NIDFEMT also require additional disk space. It is anticipated that most ATMs will be configured to share a particular profile in the NIDF; therefore the number of records to be stored in this file is expected to be small.

NCR 5XXX Device Handler – DCT Commands

The following command is available on the NCR 5XXX Commands Screen (screen 45) of the Device Control Terminal (DCT).

- **LOAD NOTEIDS**

When the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate a Load command to the ATM, to update its Note ID List to turn acceptance on or off for the note ids from the relevant record in the in the BASE24 NIDF.

The following command is available on the ATM Terminal Data Screen (screen 7) of the Device Control Terminal (DCT), as a SF3:

- **BNA STATUS**

When the operator performs a SF3, the DCT sends a command to the device handler to request configuration information from the ATM. Details of note ids configured at the ATM, and the status of the drums and cashin and coinin boxes is returned. The information is displayed via EMS, using many EMS events. Please note, as a large number of EMS events are generated by this command, it is suggested that it is used sparingly in a production environment.

The Balance to Current Cash Screen (screen 8) and the Balance to Standard Cash Screen (screen 9) now support the use of the field titled COUNTS, for the Shared NDC+ BNA Support enhancement.

- COUNTS

For an ATM equipped with a Shared BNA CashIn device, the BNA currency counts must be obtained from the device, before the ATM can be balanced. To obtain the BNA currency counts, the operator places the cursor in the COUNTS field and presses the F1 Function key. After obtaining the BNA currency counts, the operator can then balance the terminal as usual.

NCR 5XXX Device Handler – NCS/NCPCOM Commands

The following 9500 commands are supported in the NCR 5XXX device handler. They are entered at the BASE24 AFT NCS screen or through NCPCOM.

- LOADNOTEIDS
- LNOTEIDS

These 9500 commands are an alternative to the DCT's LOAD NOTEIDS command. They cause the device handler to initiate a Load command to the ATM, to update its Note ID List to turn acceptance on or off for the note ids from the relevant record in the in the BASE24 NIDF.

For more information refer to the LOAD NOTEIDS command in the preceding subsection (NCR 5XXX Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LOADNOTEIDS <terminal ID>" or
DELIVER PROCESS <process name>, "9500LNOTEIDS <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

- COUNTS

This 9500 command is an alternative to the DCT's COUNTS command. It causes the device handler to obtain the BNA currency counts before the ATM can be balanced. For more information refer to the COUNTS command in the preceding subsection (NCR 5XXX Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500COUNTS <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

The following command is also supported in the NCR 5XXX device handler. It is entered at the BASE24 AFT NCS screen or through NCPCOM. Please note that this is not a 9500 command.

- BNASTATUS

This command is an alternative to the DCT's BNA STATUS command. It causes the device handler to request configuration information from the ATM. For more information refer to the BNA STATUS command in the preceding subsection (NCR 5XXX Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "BNASTATUS <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Get Cnfg

A Get Cnfg command (previously supported by BASE24) must be performed from the DCT the first time that an ATM that will support the Shared NDC+ BNA Support enhancement is connected, to initialize the ATD additional offset for the CashIn device.

Load Procedures

Load procedures may have to be modified to load the new terminal groups described above under Database Configuration.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared NDC+BNA enhancement.

- ATMs running firmware compliant with Wincor Nixdorf's CashIn Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for CashIn release 1.41.

Wincor hardware is already compatible with the newly supported format. Since Wincor has made its standards available to other vendors, it is anticipated that other vendors will support it as well.

Reference Documents

- Wincor Nixdorf NDC/Diebold D91x Message Format Extension for CashIn (Release 1.41. 16th May 2007)

ACI Worldwide, Inc.

Shared NDC+ CPM Support

This section describes the implementation requirements associated with the Shared NDC+ CPM Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared NDC+ CPM Support enhancement.

Wincor Nixdorf has developed a new module for their ATMs to enhance their deposit capabilities. Wincor calls this new module the Check Cash Deposit Module (CCDM). This module allows a bundle of up to 50 checks to be deposited into a secured part of the ATM. It can scan each check's MICR data, attempt to read the amount (optional), store an image, or reject any checks that are invalid or unrecognizable (i.e., torn, dirty, etc.) or improperly inserted before it sends the transaction request to the processing Host (BASE24).

This enhancement is applicable to BASE24 systems running ATMs loaded with firmware that conforms to the Shared NDC+ CheckIn specification created by Wincor Nixdorf. These ATMs will run off BASE24's NCR 5XXX Device Handler.

This enhancement requires the N50 SSB Device Handler extension (AT60DN50 subvolume) to be present. Additionally, optional extensions that can be purchased to provide added check processing functionality are the SSB Enhanced ATM Application (AT60BN50 subvolume) and SSB Enhanced Check Processing Application (AT60CN50 subvolume).

At present, while the Wincor ATMs can be configured with an optional sidecar to dispense coins, this enhancement does not support the dispensing of coins from non-whole dollar amount checks.

Prerequisite: Wincor Nixdorf ATMs loaded with firmware conforming to Wincor Nixdorf's specification for CheckIn are required to support the Shared NDC+ CPM enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLATTKN
- Changes to CUSTMACS and FLGSDOC
- Minor inline changes to BA60AFT.COBTKN, BA60SRC.ATTKNCVS, BA60SRC.ATTKNID and BA60SRC.BACOUTLS

All BASE24-atm customers:

- Minor inline changes to Make files to include the new CUSTMACS flag, AT60SRC.ATMFM and AT60SRC.ATMMM
- Inline changes to AT60AFT.RQTLFS, SCRNTLF and SVTLFS
- Inline changes to AT60DCT.SCNDCOMM and RQDCOMMS
- Inline changes to AT60DCT.SCNDCURR and RQDCURRS
- Inline changes to AT60DCT.SCNDSTAN and RQDSTANS
- Inline changes to AT60SRC.ATCOUTLS
- Minor Inline changes to BA60MISC.LCONFAT

All BASE24-atm NCR 5XXX device handler customers:

- Changes to AT60N50.N50S and AT60N50.N50D
- Minor inline changes to AT60N50.N50G, N50SSBDS and N50M
- Inline changes to AT60N50.N50DDL and N50DDL
- Changes to AT60N50.CNFG5070 and CNN50DOC
- Inline changes in AT60N50.RQ50STS, SCRNST
- Minor inline changes to AT60N50.SV50PRIS
- Inline changes to AT60N50.SVDN50S
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX SSB device handler customers:

- Inline changes to AT60DN50.N50SSDS

All BASE24-atm NCR 5XXX SSB Enhanced ATM Application device handler customers:

- Inline changes to AT60BN50.N50SSBS
- Minor inline changes to AT60BN50.N50SSBM.

All BASE24-atm NCR 5XXX SSB Check Processing Application device handler customers:

- Inline changes to AT60CN50.N50SSCS
- Minor inline changes to AT60CN50.N50SSCM.

All BASE24-atm NCR 5XXX device handler customers requiring the new Shared NDC+ CPM Support functionality:

- A new extension subvolume, AT60CWN5, with all of the files required to support the Shared NDC+ CPM Support functionality.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNWNCI.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the Shared NDC+ CPM module to the NCR 5XXX device handler.

- atm_dh_n50_shrd_cpm_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that are required to be added to the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFAT file located on BA60MISC subvolume.

ATM-DH-MAX-TERMS-IN-CPLF-TBL

Identifies the maximum number of unique terminals that can exist at one time in the Check Proof List File (CPLF) extended memory table.

The default value is 1/5 of the value of the existing param, ATM-DH-MAX-TERMS.

ATM-DH-TERM-IN-CPLF-RETN-PRD

Identifies the length of time to retain a terminal in the Check Proof List File (CPLF) extended memory table before deleting. Unit of time is minutes.

The default value is 5 minutes.

Database Configuration

Token file (TKN)

The following BASE24-atm token has been added for this enhancement:

- The Check Bundle Token (token ID "AU") contains the total number of checks in the bundle that was deposited in the Check Processing Module (CPM) and the index of the check currently being processed. The data in this token gets written to the BASE24-atm Transaction Log File (TLF) by the Authorization process for each check processed in the bundle.

If the token for this enhancement is to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

NCR 5XXX State Screens (SCRNST)

Support was added for a new state type, the CheckIn state (q). Customers wanting to take advantage of this state will need to make their own configuration changes. Example states and screens for the Shared CPM enhancement are available in the CNFG5070 file on the AT60N50 subvolume, in groups 801, 802 and 803.

CPLF Extended Memory Table

This table is built by the device handler process. It provides the device handler with a storage location for CPLF records of checks that are marked to be returned to the cardholder after the last check is processed. If the cardholder does not take the checks, these records will be written to the CPLF file on disk.

Template Changes

Not applicable

Device Configuration

NCR 5XXX Configuration File (CNFG5070)

The NCR 5XXX Configuration File (CNFG5070) requires new states and screens to support the new Shared NDC+ CPM enhancement. For more detailed information, see the CNN50DOC file on the AT60N50 subvolume. A default set of states and screens is provided, but these may require specific modifications depending upon the institution's requirements.

Load group 801 is the new SSB DH CPM (Shared) – Check Handling load group, and is required to support the Shared NDC+ CPM Support enhancement. It must be loaded on top of base load group 0 and SSB DH load group 106.

Load group 802 is the new SSB DH CPM / BNA (Shared) – Check and Currency Handling load group, and is required to support both the Shared NDC+ CPM Support and Shared NDC+ BNA Support enhancements. It must be loaded on top of base load group 0, SSB DH load group 106, SSB DH BNA load group 800 and SSB DH CPM (Shared) load group 801.

Load group 803 is the new SSB DH CPM (Shared) – OCR Amount Recognition load group, and is required to support the Optical Character Recognition (OCR) of the courtesy amount area on the check where the amount is written in numeric form.

Note: This requires additional OCR software to be installed on the ATM. Wincor Nixdorf recommends using OCR scanning software packet A2IA CheckReader.

Load group 810 is the new SSB DH CPM (Shared) – Split Deposit load group, and is required to support split deposits in the Shared NDC+ CPM Support enhancement. It must be loaded on top of base load group 0, SSB DH load group 106, SSB Split Deposit load group 110 and SSB DH CPM (Shared) load group 801.

ATM Registry settings

The ATM must have certain registry settings configured, to ensure that the Device Handler will work as designed with the CheckIn device. Wincor Nixdorf made these registry changes and has kept a log of those that are required for the Shared NDC+ CPM Support enhancement to work properly. Contact Wincor Nixdorf for the ACI specific registry settings needed to support the CheckIn device.

The following registry settings were made by ACI and must be implemented if load group 801 is downloaded to the ATM:

Current check ID screen:

Parameter name: CHQ_CURRENT_SCRN
Section in registry: [..\LYNXCI\SCREEN]
Value: 855

Total number of checks screen:

Parameter name: CHQ_TOTAL_SCRN
Section in registry: [..\LYNXCI\SCREEN]
Value: 854

Check image size on screen:

Parameter name: CHQ_SIZE
Section in registry: [..\LYNXCI\SCREEN]
Value: 6,18

The following registry settings were made by ACI and must be implemented if load group 803 is downloaded to the ATM:

OCR amount screen:

Parameter name: CHQ_OCR_AMOUNT_SCRN
Section in registry: [..\LYNXCI\SCREEN]
Value: 858

OCR processing screen:
Parameter name: CHQ_OCR_PROCESSING_SCRN
Section in registry: [..\LYNXC\SCREEN]
Value: 857

CheckIn device

A CheckIn device must be installed on every ATM implementing Shared NDC+ CPM Support.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

The host or co-network may require modification to process the new Check Bundle token. If the token for this enhancement is to be sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

Super Extract Process

If the new token for this enhancement is to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF requires additional disk space as the new tokens for this enhancement can be included in the transaction records logged.

The CPLF requires additional disk space as CPLF records will be written to the CPLF file for all transactions involving checks retained by the ATM.

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on THE BA60LOGM SUBVOLUME.

Get Cnfg

A Get Cnfg (F2) command (already supported by BASE24) must be performed from the DCT screen 45 the first time that an ATM that will support the Shared NDC+ CPM Support enhancement is connected, to initialize the values in the ATD additional offset for the CheckIn device.

Load Procedures

Load procedures may have to be modified to load the new terminal groups described above under Database Configuration.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared NDC+ CPM enhancement.

- ATMs running firmware compliant with Wincor Nixdorf's CheckIn Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for CheckIn release 1.34.

Wincor hardware is already compatible with the newly supported format. Since Wincor has made its standards available to other vendors, it is anticipated that other vendors will support it as well.

Reference Documents

- Wincor Nixdorf NDC/Diebold D91x Message Format Extension for CheckIn (Release 1.34. 13th June 2007)

Tidel Dynamic Key Management

This section describes the implementation requirements associated with the adding Dynamic Key Management to the BASE24-atm Tidel Device Handler.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Dynamic Key Management enhancement.

The BASE24-atm Dynamic Key Management (DKM) add-on product enables the Tidel Device Handler to determine programmatically when the PIN and MAC keys need to be changed for a terminal. The BASE24-atm Dynamic Key Management (DKM) add-on product enhances transaction security by ensuring that keys are changed regularly without the need for manual intervention.

Prerequisite: This enhancement requires the installation of the Tidel MACing Support and Triple-DES Update enhancement that is also part of this version.

Software Impacts

All BASE24 customers:

- Changes to CUSTMACS and FLGSDOC

All BASE24-atm customers:

- Inline changes to AT60SRC.ATMFM and ATMMM to include the Tidel DKM flags

All BASE24-atm customers with a Tidel device handler:

- Inline changes in AT60TIDL.TIDELD, TIDELG, TIDELM, TIDELS ,TIDLDDLS and TIDLMAN
- Updated event (log) message manual BA60LOGM.ATTIDL

All BASE24-atm Tidel customers implementing the Dynamic Key Management Enhancement:

- New subvolume, AT60KTDL. The files on this subvolume include source files, global files and Make files required for the new functionality.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNTDKM.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

Set the following flag to true to bind the DKM module to the Tidel device handler.

- atm_dh_tidel_dkm_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Note that these modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

Different param records may be specified for each device type or device handler process, if desired.

ATM-DH-CONS-MAC-ERR-LMT

The number of consecutive message authentication code (MAC) errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of consecutive MAC errors.

ATM-DH-DKM-KEY-CHNG-INTRVL

This param specifies the maximum time allowed between key changes before the device handler downloads a new MAC and PIN key to the ATM. The value is in hours. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the length of time since the last key change.

ATM-DH-DKM-MAC-ERR-LMT

This param specifies the number of message authentication code (MAC) errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of MAC errors.

ATM-DH-DKM-MAC-TXN-LMT

This param specifies the number of message authentication code (MAC) generations or verifications that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of MAC generations or verifications.

ATM-DH-DKM-PIN-ERR-LMT

This param specifies the number of personal identification number (PIN) verification errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of PIN verification errors.

ATM-DH-DKM-PIN-TXN-LMT

This param specifies the number of personal identification number (PIN) verifications that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of PIN verifications performed.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

A new field was added to the Tidel device dependant data to save DKM counters.
No conversion program is required to initialize this data.

Other fields used for Dynamic Key Management were added to the TDF in Release 6.0 version 4. See the BASE24 Implementation Guide for version 4 for more information on these fields.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

This enhancement will automatically load new PIN and MAC keys to ATM terminals. Messages will be generated when the keys are loaded. Operators must be aware of this new functionality. Key load commands issued by operators will function as before. With this enhancement in place, the counters for transaction and error counts will be reset when an operator-initiated key load is performed.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on THE BA60LOGM SUBVOLUME.

Vendor Impacts

Not applicable

Reference Documents

ATM Communication Specification, NCR EasyPoint LLC, Revision: Q,
August 10, 2007

ACI Worldwide, Inc.

Tidel MACing Support and Triple-DES Update

This section describes the implementation requirements associated with the adding Message Authentication (MAC) Support and industry standard two key triple DES encryption to the BASE24-atm Tidel Device Handler.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Tidel MACing Support and Triple-DES Update.

The Tidel device handler has been enhanced to perform MACing. For both single and triple DES MACing, the MACing will be performed on the entire message.

Originally, triple-DES PIN encryption in the Tidel Device Handler used a proprietary three key encryption algorithm. With this enhancement, industry standard two key, EDE (encrypt-decrypt-encrypt) encryption is added.

Also, in this enhancement, the ability to download PIN and MAC keys in a financial response message was added. This was critical to the ability to automatically download keys in the Tidel Dynamic Key Management Enhancement.

More information on this functionality is provided in the Tidel Device Handler Manual located on AT60TIDL.TIDLMAN.

Software Impacts

All customers with a Tidel device handler:

- Inline changes in AT60TIDL.TIDLES and TIDLEG
- Inline changes in the AT60TIDL.RQTTIDLS, SCRNTIDL and SVTIDLTS
- Changes in the Tidel device dependent data (AT60TIDL.TIDLDDLS)
- Updated event (log) message manual BA60LOGM.ATTIDL

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Terminal Definition File (TDF)

The following new field has been added to the TDF Tidel Specific Screen 9, to configure MACing support. The field is used to determine whether Message Authentication will be performed on Financial Transaction Request/Response and Reversal messages to and from the ATM.

Screen Name: MAC SUPPORT TYPE
Screen: TDF Screen 9
DDL Name: DEV-INFO-TIDEL.MAC.MAC-TYP
Valid Values:
 00 - No MAC Support
 03 - Hardware MAC Support

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

BASE24 Triple-DES MACing and triple length MAC key support requires TSS. Additionally, Tidel requires device firmware compatibility (must be able to enter double length MAC keys at ATM).

Additional information on configuring the terminal to perform MACing is provided in the Tidel Device Handler Manual located on AT60TIDL.TIDLMAN.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Triple-DES MACing and double or triple length MAC key support requires TSS. All supported versions of TSS provide Triple-DES and double or triple length keys.

File Conversions

New fields were added to the Tidel device dependant data to support MACing.

Terminal Definition File (TDF)

The TDF conversion program must be run to convert the TDF. Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on THE BA60LOGM SUBVOLUME.

Vendor Impacts

Not applicable

Reference Documents

ATM Communication Specification, NCR EasyPoint LLC, Revision: Q,
August 10, 2007

ACI Worldwide, Inc.

Track 2 Context Data Removal

This section describes the implementation requirements associated with the Track 2 Context Data Removal enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Track 2 Context Data Removal enhancement.

When the track discretionary data is suppressed, the primary account number (PAN), expiration date and service code are retained. The rest of the track data is replaced with ASCII zeroes.

The BASE24-atm Diebold 10XX/478X device handler and the BASE24-atm NCR 5XXX device handler modules have been enhanced to offer the option to suppress discretionary data in the Track 2 data in the internal STM response message held in memory. The enhancement also ensures that track 1 and track 2 discretionary data does not reside in the temporary context areas of the ATD when the transaction has been completed or cancelled.

The BASE24-atm Authorization module has been enhanced to offer the option of removing the track 1 and track 2 discretionary data from the Track 2 data in the internal STM response if the transaction is authorized by either BASE24 or a Host.

In addition, response messages received from an interchange may have the discretionary data suppressed. When an interchange interface suppresses the discretionary data, the PAN, expiration date and service code are retained. The rest of the track data is replaced with ASCII zeroes. BASE24-atm Authorization must be able to process a response with a track 2 that may have the discretionary data suppressed.

Many interchanges will no longer send the Track 2 Data (P-35) data element on reversals, pre-auth completions, and advices. When the BASE24 interfaces and BIC ISO Interface receive these messages, the track2 field in BASE24's internal message must be manually constructed using the primary account number (PAN) (P-2) and the expiration date (P-14 or zeroes if P-14 is not present). When the track data is manually constructed, BASE24 will set the first character of the STM/PSTM Track2 field to "M".

The new LCONF parameter, ATM-DISCR-DATA-STM-SUPPRS, is used to indicate whether or not discretionary data suppression will be performed.

This enhancement will apply to those transactions that are authorized by BASE24, a host system, or an interchange.

Software Impacts

All BASE24 BIC ISO Host Interface customers:

- Inline changes to SW60BABI.BABICIG and BABICIS
- Updated event (log) message manual SW60BABI.BABILOGM

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG and AUTHLIBS
- Updated event (log) message manual BA60LOGM.ATAUTH

All BASE24-atm BIC ISO Host Interface customers:

- Inline changes to SW60ATBI.ATBICIS

All BASE24-atm ISO Host Interface customers:

- Inline changes to AT60HISO.ATHISOS

All BASE24-atm Diebold 10XX/478X device handler customers:

- Inline changes to AT601000.C1000G and C1000S
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm customers with the Diebold 10XX/478X Dial-Up device handler:

- Inline changes to AT60P100.C100DLUG and C100DLUS

All BASE24-atm customers with the Diebold 10XX/478X Mobile Top-up device handler:

- Inline changes to AT60T100.C100MTUS

All BASE24-atm NCR 5XXX device handler customers:

- Inline changes to AT60N50.N50G and N50S
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm customers with the NCR 5XXX Dial-Up device handler:

- Inline changes to AT60PN50.N50DLUS

All BASE24-atm customers with the NCR 5XXX Mobile Top-Up device handler:

- Inline changes to AT60TN50.N50MTUS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNTRK. Note that this SCN file also contains the SCNs needed to implement this enhancement in BASE24-pos.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added in the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DISCR-DATA-STM-SUPPRS

This param is used to indicate whether discretionary data, carried in the STM should be suppressed. Valid values are as follows:

Y = Yes, suppress discretionary data in the STM.

N = No, do not suppress discretionary data in the STM (default).

Database Configuration

External Message File (EMF)

The suppression of discretionary data can affect the content of messages exchanged between the host and the BASE24-atm or BASE24-pos ISO Host Interface process and between the co-networks and the BASE24-atm or BASE24-pos BIC ISO Interface process. This is true whether the data has already been suppressed in response messages that BASE24 receives from an interchange or host or the data is suppressed by BASE24.

As a result, users that prefer to send the card details to the back-end Host or co-network in the same format in which they were received should consider configuring EMF records that include the Primary Account Number (P-2) and Expiration Date (P-14) data elements and modify the default settings for the Track 2 (P-35) data element.

Member numbers are used by BASE24 to differentiate multiple cards issued with the same card number. Data Element 23, Card Sequence Number, is used to send and receive the member number if multiple cards are issued. Host systems and co-networks utilizing the member number other than zero should be configured to send and receive the Card Sequence Number in data element 23.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host systems may no longer receive full Track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternately, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35 (Track 2 Data). Hosts must be capable of processing response, reversal, and advice messages without the discretionary data or using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Wincor AKDS Support of NCR ATMs

This section describes the implementation requirements associated with the Wincor AKDS support of NCR ATMs enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Wincor AKDS support of NCR ATMs enhancement.

In Release 6.0 version 6, the Shared (Wincor) NDC+ AKDS Signatures and Shared (Wincor) 912 AKDS Signatures enhancements were introduced. These enhancements allowed customers to perform AKDS downloads from the NCR 5XXX NDC+ device handler or the Diebold 10XX/478X device handler to ATMs loaded with firmware that conformed to the Shared NDC+ Signature specification created by Wincor Nixdorf, and configured with Signature-based Encrypting PIN Pads (EPPs). These ATMs were typically Wincor or NCR ATMs.

It was discovered that a change was needed when the ATM being used was an NCR ATM. This is required because an NCR EPP uses only the modulus portion of the public key in transaction data and allows the exponent portion to be defaulted at the device, while the shared Wincor Nixdorf signature scheme uses an ASN.1 formatted public key that includes both the modulus and exponent.

BASE24 needs to be able to inform TSS (via the device type parameter passed in the TSS messages) that the ATM in question is an NCR, but that Wincor formatting needs to be done on the cryptographic data.

TSS, in release 07.4, has added a new device type ("03") to indicate the hybrid Wincor/NCR configuration. This Wincor AKDS support of NCR ATMs enhancement will use that new device type to allow a customer with NCR ATMs, running firmware that is compatible with Wincor Nixdorf's RKL Message Specification, to perform AKDS commands successfully on either the NCR 5XXX NDC+ or the Diebold 10XX/478X device handlers.

Prerequisites: The Transaction Security Services (TSS) version 07.4 must be installed prior to implementation of this enhancement.

The Shared (Wincor) 912 AKDS Signatures enhancement and/or the Shared (Wincor) NDC+ AKDS Signatures enhancement from Release 6.0 version 6 must be installed prior to installing this enhancement.

Software Impacts

All BASE24-atm customers:

- Minor inline changes to BA60DDL.DDLACNST

All BASE24-atm NCR 5XXX device handler Shared AKDS customers:

- Minor inline changes to AT60AWN5.N50WAKDS

All BASE24-atm Diebold 10XX/478X device handler Shared AKDS customers:

- Minor inline changes to AT60AW10.C10WAKDS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNWAKD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Setup for Transaction Security Services

Initial setup procedures must be carried out for TSS before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

In this case, you must use the NCR out-of-band procedures to generate and sign the root BASE24 system public key rather than the shared Wincor Nixdorf signature device out-of-band procedures.

The EPP must contain a public and private key pair, a signature of the EPP's public key, the Certificate Authority's public key, and a unique serial number. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Datasource (HSPKD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair. The TSS device type of WINCORNCR,WINNCR or "3" must be used.

The public key must be sent to the appropriate Certificate Authority to be verified and signed. In this case the Certificate Authority will be the NCR trusted center. The NCR trusted center must return BASE24's public key signature, as well as the NCR trusted center's own public key. These values must be entered into the Transaction Security Services database.

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

6: BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from Release 6.0 version 7.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to Release 6.0 version 8.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to Release 6.0 version 8.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Lesser Amount for Pre-Auth Purchase with Cash Back

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes for all BASE24 Release 6.0 version 8 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The POS Terminal Data File (PTD) has been modified in device dependent data. This data is not displayed but the PTD conversion program must be run to correctly initialize the new fields. For further information, see Track 2 Context Data Removal where the impacts of this enhancement are described later in Section 6.
- The ACI Standard Device Handler Configuration File (ACNF) has been modified to include additional values for fields on screens 2 through 9 for SPDH Message Data Element Encryption. The ACNF screen, requester and server have been modified to handle the new values. For further information, see the SPDH Message Data Element Encryption later in Section 6.

Software Impacts

Not applicable

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ACI Standard Device Configuration File (ACNF)

For customers converting from Release 5.3 to Release 6.0 version 8, the ACNF conversion program must be run to convert the ACNF. Running the conversion program is required for any BASE24-pos SPDH customer. Users must create the new ACNF file using the SPDHFUP file.

The ACNF conversion program on the PS60UC04 subvolume can be used to convert the ACNF records from the Release 5.3 format to the Release 6.0 version 8 format. The ACNF conversion program on the PS60UC04 subvolume can be used to convert the ACNF records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVACNFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/ out $s.#acncnv.t, name $acnm,nowait/ cnvacnfm  
-force -list -listreplace
```

Release 5.3 users should update the CNVACNFR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVACNFR obey file on the PS60UC0B subvolume with the appropriate information. CNVACNFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B file on the BA60UC0B subvolume for Release 6.0 version 7 customers.

POS Terminal Data File (PTD)

The PTD conversion program must be run to convert the PTD. Running the conversion program is required for any BASE24-pos customer utilizing the PTD in transaction processing. Users must create the new PTD file using the PSDDLUP file.

The PTD conversion program on the PS60UC04 subvolume can be used to convert the PTD records from the Release 5.3 format to the Release 6.0 version 8 format. The PTD conversion program on the PS60UC0B subvolume can be used to convert the PTD records from the Release 6.0 version 7 format to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPTDS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ptdcnv.t,name $ptdm,nowait/ cnvptdm –  
force –list –listreplace
```

Release 5.3 users should update the CNVPTDR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 version 7 users should update the CNVPTDR obey file on the PS60UC0B subvolume with the appropriate information. The CNVPTDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0B on the BA60UC0B subvolume for Release 6.0 version 7 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Lesser Amount for Pre-Auth Purchase with Cash Back

This section describes the implementation requirements associated with the Lesser Amount for Pre-Auth Purchase with Cash Back enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Lesser Amount for Pre-Auth Purchase with Cash Back enhancement.

BASE24-pos has been enhanced to extend lesser amount support to Pre-Auth Purchase with Cash Back transactions. When a Pre-Auth Purchase with Cash Back transaction is authorized for a lesser amount than originally requested, the cash back amount is reduced by the same quantity as the total transaction amount.

For example: A transaction is received for the purchase of \$100 in merchandise plus a cash back amount of \$50 (Amount1 = \$150).

#1 -- If the available balance on the account is \$120, the transaction will be authorized for \$120. However, in doing so, the cash back amount will be reduced to \$20. The merchandise purchase amount will remain \$100.

#2 – If the available balance on the account is \$80, the transaction will be authorized for \$80. The cash back amount will be reduced to \$0 while the resulting merchandise purchase amount will be \$80.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCN PCB.

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CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-pos Terminal Data file (PTD)

To utilize this enhancement, the AUTH LESSER AMT field on screen 6 of the PTD must be set to "Y" (Yes, allow transactions to be authorized for a lesser amount). Otherwise, BASE24 would not allow the transaction to be authorized unless the full amount in the request is available in the account.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with Release 6.0 version 8, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- APACS 70-2 Enhancements
- ATC Enhancements
- CAP Authentication
- Dynamic CAF
- Hypercom 3DES in Logon and Other Enhancements
- Immediate Availability PTD Static Data Changes
- POS Completion Message Configuration
- SPDH Message Data Element Encryption
- Stored Value BDM Support
- Stored Value Support in CRT Authorization
- Track 2 Context Data Removal
- Visa II Device Handler Enhancements

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements for future reference.

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APACS 70-2 Enhancements

This section describes the implementation requirements associated with the APACS 70-2 Enhancements for BASE24-pos.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the APACS 70-2 Enhancements, including the following:

This enhancement updates the BASE24-pos APACS device handler to APACS 70-2 standards. This includes a number of changes as outlined below.

- Support for Auxiliary Data records, including Electronic Commerce and Extended Terminal Capabilities and Attributes data records
- Support for Contactless chip and magnetic stripe transactions
- Support for Secure Internet Validation (SIV)
- Support for various items to bring the APACS device handler up to APACS 70 (October 2004) compliance.

Auxiliary Data Records:

There are several types of Auxiliary Data records. This enhancement will support two of these record types: Electronic Commerce (type 01), and Extended Terminal Capabilities and Attributes (type 06). Electronic Commerce Auxiliary Data records are needed for support of Secure Internet Validation (SIV). Extended Terminal Capabilities and Attributes Auxiliary Data records are required for Contactless chip and magnetic stripe transactions. The APACS device handler will map data passed into BASE24 via Auxiliary Data records to the appropriate BASE24 token. Auxiliary Data in response messages is not supported. For more information on Auxiliary Data Records, refer to the ***APACS Standard 70 Book 2 Card Acceptor to Acquirer Interface Standards, October 1, 2004.***

Contactless Chip and Magnetic Stripe Transactions:

Support has been added to the APACS device handler for Contactless transactions for both chip and magnetic stripe cards. Information regarding Contactless transactions is sent by the terminal in an Extended Terminal Capabilities and Attributes Auxiliary Data record. The BASE24-pos APACS device handler has been modified to process this Auxiliary Data, mapping the appropriate Point of Service Entry Mode value to the PSTM.

Visa and MasterCard have defined two new values for the Point of Service Entry Mode for Contactless Chip and Contactless Magnetic Stripe transactions. Contactless Chip transactions will be identified by a POS Entry Mode of '07'.

Contactless Magnetic Stripe transactions will be identified by a POS Entry Mode of '91'. The BASE24-pos APACS device handler has been modified to support these two new POS Entry Mode values. Note that BASE24-pos RTAU was enhanced to support Contactless Chip and Magnetic Stripe transactions in Release 6.0 version 7. BASE24 TSS is required for Contactless transactions, and was enhanced to support Contactless Chip and Magnetic Stripe transactions in version 06.4.

The APACS device handler determines a transaction is Contactless Chip or Contactless Magnetic Stripe by examining the data in the Extended Terminal Capabilities and Attributes Auxiliary Data record (if the record exists – if the record does not exist, then it is not a Contactless transaction). Two fields within this Auxiliary Data record, Terminal Attributes and Terminal Attributes Used, indicate a terminal's capabilities and capabilities used on this particular transaction, respectively. The APACS device handler sets the PSTM.PT-SRV-ENTRY-MDE to either "07" for Chip, or "91" for Magnetic Stripe, based on these values. For more information on Contactless transactions, refer to the ***APACS Standard 70 Book 2 Card Acceptor to Acquirer Interface Standards, October 1, 2004.***

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- **Secure Internet Validation:**

The APACS device handler has been modified to support Secure Internet Validation (SIV), or 3D Secure. 3D Secure is a method of authenticating the cardholder in an e-commerce transaction. Visa's method is known as Verified by Visa (VbV).

MasterCard's method is known as SecureCode. The APACS device handler maps Auxiliary Data record data (from an e-commerce record) to the appropriate BASE24 token. For Visa transactions, Auxiliary Data will be mapped to the TRANS-STAIN-XID-TKN token. For MasterCard transactions, Auxiliary Data will be mapped to the AUTHN-DATA-TKN. For more information on Secure Internet Validation, refer to the ***APACS Standard 70 Book 2 Card Acceptor to Acquirer Interface Standards, October 1, 2004.***

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- **APACS 70 (October 2004) Compliance Support:**

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- **Support additional values used to identify new types of e-commerce transactions:** Additional values have been added to the e^commerce^req^d define to support various e-commerce transactions based upon transaction type. This includes support for transactions such as Continuous Authority - Recurring Payment.
- **Support for POS-DATA1 token:** The APACS device handler will now set the RECUR-PMNT-IND field in the POS-DATA1 (CH) token, to 'R', for Recurring Payment - Continuous Authority transactions.
- **Support additional values to provide more accurate information to the card schemes:** The APACS device handler has been modified to map the cumulative values defined by the terminal type/capabilities fields

- **Support for field 15 in EFT request messages:** Field 15 is an optional field containing the terminal software level, and present in Financial Presentment messages. The APACS device handler has been modified to parse this field and, if sent, update the PTDF Software Release field (screen 2) with the value.
- **Support a Reason On-Line code '04' to identify an IC Fallback transaction:** The APACS device handler no longer identifies Fallback transactions by the capability of the terminal, and the method by which the card details were read/entered. The device handler now supports a more sure methodology, defined by APACS, using the Reason On-Line code
- **Support the correct value for Balance Enquiry transactions:** For the purposes of generating the Authorization Parameter, APACS has now defined balance inquiry transactions as type 'credit'. The APACS device handler's generation of the Authorization Parameter has been modified to account for this new definition for balance inquiry transactions.
- **Support for account funding transactions:** Support has been added to the APACS device handler for Account Funding transactions. Account Funding transactions can be processed as either Cash Advance or simple Purchase transactions. The APACS device handler will use the existing LCONF parameter FUND-TRAN-TO-CASH-ADV to determine how to process these transactions. Possible values are : "Y" = Process Funding Transactions as Cash Advance transactions; "N" = Process Funding Transactions as normal Purchase transactions. Processing as normal Purchase transactions is the default value
- **Treat the CID as a mandatory data element in EMV transactions:** The APACS device handler no longer treats the CID (EMV Tag '9F27') as an optional data element. It is now considered mandatory.
- **Support for two new subfields in the IC request data:** APACS has added two new fields to the IC Request Data: The Cryptogram Amount (EMV Tag '9F02'), and the Cryptogram Date and Time (EMV Tag '9A'). Since a subsequent Financial Request could contain different amount and date/time values from the original Authorization (pre-auth) message, these new fields – containing the original amount and date/time - provide the Financial Request with the correct amount and date/time values needed for issuer validation of the original ARQC. The APACS device handler now supports these two new IC Request Data fields.
- **Support additional validation on hexadecimal fields:** APACS specifications had defined a number of fields as Alpha when, in fact, they contain hexadecimal values. Where appropriate, the APACS device handler has been modified to validate these fields accordingly.
- **Support to ensure the CVM Results field is present in an EMV transaction:** The CVM Results field is optional for EMV transactions, but mandatory for Off-Line PIN. Therefore, the APACS device handler has been modified to treat this field as mandatory for EMV transactions.

- **Support for the CVM Results field in Off-Line Repeat Request messages:**
The APACS device handler has been modified to support the CVM Results field in Off-Line Repeat Request messages.
- **Support clarification of the Card Details:** In previous APACS specifications, the unit separator (US) prior to the Application PAN Sequence Number is sometimes defined as optional, and at other times as mandatory. Also, the Application PAN Sequence Number itself is sometimes defined as variable length, and at other times as fixed length. This has now been clarified as follows. The US is mandatory if any subsequent sub data elements are present. The Application PAN Sequence Number is optional, but if present is of a fixed length. The APACS device handler has been modified to support this clarification.

Software Impacts

All BASE24-pos APACS customers:

- Changes to PS60HPDH.APACSDHG
- Changes to PS60HPDH.APACSDHS

All BASE24-pos APACS EMV customers:

- Changes to PS60IAPD.APDHEMVG
- Changes to PS60IAPD.APDHEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNA702.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that is required to be added to the LCONF to support this enhancement. Details of the param are provided in the BASE24 Logical Network Configuration File manual. A sample of the param is included in the LCONFPS file located on BA60MISC subvolume.

There are no new LCONF parameters for this enhancement. However, an existing parameter, **FUND-TRAN-TO-CASH-ADV**, is being used in conjunction with device handler processing of Account Funding transactions.

Valid values are as follows:

“Y” = Process Funding Transactions as Cash Advance transactions

“N” = Process Funding Transactions as normal Purchase transactions.

Default = Processing as normal Purchase transactions.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- APACS Common Attachment to Standards 29, 30, 40, 50, Version 15, January 1, 2000.
- APACS Standard 70 Book 2 Card Acceptor to Acquirer Interface Standards, October 1, 2004.
- APACS Standard 70 Book 2 Card Acceptor to Acquirer Interface Standards, October 1, 2006

ATC Enhancements

This section describes the implementation requirements associated with the ATC Enhancements for BASE24-pos.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ATC Enhancements.

This enhancement modifies the ATC logic provided with the Contactless Chip and Magnetic Stripe enhancement delivered in Release 6.0 version 7.

The three ATC enhancements provided are:

- Add ATC limit checking capabilities to the Contactless Chip/Magnetic Stripe enhancement. This check will allow customers to set a range of allowed ATC values when comparing the incoming ATC with the ATC kept on the CAF Base segment.
- Add a token to allow the ATC data to be managed on the host.
- Enhance ATC Second Card logic to be consistent the ATC Second Card logic for EMV.

Prerequisite: This enhancement requires the installation of the POS Router/Auth DCV extension module. The minimum version of TSS required for the implementation of this enhancement is version 06.4.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCPF and DDLPSTKN.
- Inline changes to BA60AFT.RQCPFS, SCRNCPPF and SVCPPFS.
- Inline changes to BA60SRC.PSTKNCVS and PSTKNID

All BASE24-pos customers:

- Inline changes to PS60RTAU.RTAUG and ROUTERS
- Updated event (log) message manual BA60LOGM.PSAUTH

All Base24-pos customers with Dynamic Card Verification

- Inline changes to PS60QRTA.RTAUDCVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNATC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that is required to be added to the LCONF to support this enhancement. Details of the param are provided in the BASE24 Logical Network Configuration File manual. A sample of the param is included in the LCONFPS file located on BA60MISC subvolume.

CAF-SCND-CRD-PROMOTE — A flag indicating whether the second card is automatically promoted to the primary card. If this value is set to Y, BASE24 automatically promotes the second card to primary when the first transaction is received from the second card. The current primary card is removed from the system. If this value is set to N, BASE24 does not automatically promote the second card to primary, requiring the institution to do so manually. This is not a new param and is already in use for EMV processing.

Valid values are as follows:

Y = Yes, automatically promote the second card to primary.

N = No, do not automatically promote the second card to primary.

Param Default: N

Database Configuration

Card Prefix File (CPF)

CPF screen 11 has been modified to include the following new field:

This field identifies the range of allowed ATC values when comparing to the value on the BASE24 database. Valid values are 0 to 9999. The default value is 0. If set to 0, no limit checking is performed.

Field Name: ATC LIMIT

Screen: CPF Screen 3

DDL Name: CPF.BASE.ATC-LMT

Note: If a customer expects to have an interchange stand-in to approve Dynamic Card Verification Digits (DCVD), this field should be set to a value large enough to cover the number of transactions that might be approved by the interchange. This will allow the ATC in an incoming transaction to be accepted as valid, even though BASE24 has not seen several transactions that were approved by the interchange.

Token file (TKN)

The following BASE24-pos token has been added for this enhancement:

- The POS Data 2 Token (token ID “A5”) includes the updated ATC value if the DCVD was verified. If DCVD verification fails, no ATC value is included in the token if it is present. If the token is not already present, it will not be added. No further processing is done in BASE24 on the ATC value included in this token. It can be sent to the host for further processing if desired.

If the tokens for this enhancement are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

ATC data sent in the new POS-DATA2-TKN token can be used by the host to manage the ATC value. No further processing of the data in this token will happen on BASE24.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to CPF records.

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

CAP Authentication

This section describes the implementation requirements associated with the CAP Authentication enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the CAP Authentication enhancement.

MasterCard launched the Chip Authentication Program (CAP) as part of its OneSMART suite of EMV-based applications to provide a common authentication mechanism for its Member banks. CAP aims to address the requirements of authentication in the e-Banking and Card Not Present (CNP) channels, e.g. e-Commerce and MO/TO. Visa has taken the decision to adopt the same specification and has sub-licensed the CAP solution from MasterCard. Their version is called Visa Dynamic Passcode Authentication (DPA). BASE24 has been enhanced to support both the MasterCard and Visa implementations of CAP.

The BASE24 solution provides maximum flexibility in implementing a CAP Token Validation Service (CTVS). The following options are available:

- A stand-alone solution based on CAP-only cards
- A solution involving the existing EMV card application to be used for CAP Token authentication
- Multi-application payment/CAP cards for maximum configurability, while maintaining a single card for the customer

The BASE24-pos CTVS is supported via the existing (non-financial) Card Verification transaction type, identified by a BASE24-pos transaction code of '16'. The CAP Token (and associated data) is included in the existing Authentication Data token, identified by a BASE24 Token Id of 'CE'. BASE24-pos invokes the Transaction Security Services (TSS) to validate the CAP Token, using data included in the Authentication Data token, and other data configured at the Card and Card Prefix level on BASE24. BASE24-pos also performs other authorization functions on the card verification transaction, both existing, e.g. card status checking, and new, e.g. limit checking on CAP Token validation failures.

The BASE24 enhancements needed to fulfill the requirements are as follows:

- Support additional data at the Card and Card Prefix level.
- Support enhanced authorization processing, specifically CAP Token validation via TSS and limit checking on CAP Token validation failures.

The POS Router/Authorization module controls the required routing/authorization and cryptographic processing, invoking a new POS Router/Auth extension module (RTAUCAPS), new CAP procedures within the EMV security utilities module (CAMUTILS), and TSS.

These enhancements are described in more detail below.

Support Additional Data at the Card and Card Prefix Level

The following new data elements have been added to the EMV segment in the Card Prefix File (CPF) record layout:

- CAP Profile. This value is used by TSS to retrieve the data required to validate the CAP Token in an authorization message.
- CAP ATC Update. This value indicates which ATC field on the CAF should be updated on a CAP Token validation request. If the CAP application is separate from the EMV application on the card, then the CAP ATC field should be updated. If the CAP application is part of the EMV application on the card, then the EMV ATC field or the Base ATC field should be updated (depending on which ATC field is used in EMV processing).
- CAP Token Validation Option. This value allows BASE24-pos to authorize a CAP Token validation request, even if Level 3 (online/offline) authorization is configured for the card. If BASE24-pos validates the CAP Token, then there would be no requirement to forward the request to an issuer host system, because there is no further authorization processing to be performed on the transaction.
- Bad CAP Token Validation Limit. The maximum number of consecutive failed CAP Token validation attempts permitted. When this limit is reached for a card, any subsequent requests for CAP Token validation are declined without the validation being performed. A value of zero disables the limit check.

These new fields are maintained via the CPF AFT screen.

The following new data elements have been added to the EMV segment in the Cardholder Authorization File (CAF) record layout:

- CAP ATC for the “primary” card. The ATC in the last approved CAP Token validation transaction performed on the “primary” card.
- CAP ATC for the “secondary” card. The ATC in the last approved CAP Token validation transaction performed on the “secondary” card.
- Bad CAP Token Validation Accumulator for the “primary” card. The number of consecutive failed CAP Token validation attempts performed on the “primary” card. This value is reset to zero when a CAP Token is validated successfully for the card.
- Bad CAP Token Validation Accumulator for the “secondary” card. The number of consecutive failed CAP Token validation attempts performed on the “secondary” card. This value is reset to zero when a CAP Token is validated successfully for the card.
- CAP Application PAN Sequence Number for the “primary” card. The APSN for the “primary” card identified by the PAN/Member Number in the CAF record.
- CAP Application PAN Sequence Number for the “secondary” card. The APSN for the “secondary” card identified by the PAN/Member Number in the CAF record.
- CAP Derivation Key Index for the “primary” card. The DKI for the “primary” card identified by the PAN/Member Number in the CAF record.
- CAP Derivation Key Index for the “secondary” card. The DKI for the “secondary” card identified by the PAN/Member Number in the CAF record.
- Bad CAP Token Override Flag. If the Bad CAP Token Validation Limit (on the CPF) has been reached, then this flag allows the CAP Token validation to be performed. If the validation is successful, then the Bad CAP Token Validation Accumulator will be reset to zero. This mechanism is the only way that the Bad CAP Token Validation Accumulator can be reset once the limit has been reached, without deleting/re-adding or replacing the CAF record.

The CAP ATC fields and the Bad CAP Token Validation Accumulator fields are display-only, but the other new fields are maintained via the CAF AFT screen, Refresh, and From Host Maintenance (FHM).

Support Enhanced Authorization Processing

CAP Authentication support within BASE24-pos is provided by a separately licensable module, bound into POS Router Authorization. The core module has calls to dummy procedures to allow the CAP module to be bound-in, if it is purchased. There are also new security utilities in BA60IEMV.CAMUTILS that are required to support CAP Authentication.

The new CAP procedures are in a new subvolume: PS60JRТА. Make files to support the extension module are located on this subvolume. The Make file on PS60RTAU was modified to bind in the new files. New CUSTMACS flags were defined to add the CAP functionality into POS Router Authorization.

A CAP Token validation request (a Card Verification request including CAP Token data in the Authentication Data token) may be received by the BASE24-pos Router/Authorization module. Standard authorization processing (e.g. card status checking, expiration date checking) is performed if configured for the card, and the required transaction data is available.

For Level 2 (offline) and Level 3 (online/offline) authorization where a CAF is used in authorization processing, the new POS Router/Authorization CAP extension module is invoked. Otherwise, no CAP processing is performed.

The CAP extension module performs the following processing:

- If the Authentication Data token is not present or the Authentication Data token does not contain CAP Token data, then perform no further CAP processing.
- If the CAP Token Validation Option field in the CPF indicates that BASE24-pos should not validate the CAP Token, or the CAP Profile field in the CPF indicates that CAP Token validation is not supported for the prefix, then perform no further CAP processing.
- If the value of the Bad CAP Token Validation Accumulator on the CAF has reached the value of the Bad CAP Token Validation Limit on the CPF, then check the Bad CAP Token Override Flag on the CAF. If the flag indicates that the limit should not be overridden, then decline the transaction by setting an appropriate response code, and perform no further CAP processing.

- Invoke the new CAP security procedures, which perform the following processing:
 - Format and send TSS request message '60' using transaction data from the Authentication Data token and configuration data from the CPF (e.g. the CAP Profile), and the CAF (e.g. the last ATC seen from the card).
 - Receive and process TSS response message '60', returning the result code and the derived ATC to the calling application (the POS Router/Authorization CAP extension module).
- If the CAP Token could not be validated, e.g. due to an HSM error, then decline the transaction by setting an appropriate response code, and perform no further CAP processing.
- Determine whether the CAP Token validation was successful or unsuccessful.
 - If the CAP token failed validation, then increment the Bad CAP Token Validation Accumulator on the CAF record.
 - If the CAP token validation was successful, then update the appropriate ATC and reset the Bad CAP Token Validation Accumulator on the CAF record.
- Set an appropriate response code in the PSTM and return control to the POS Router/Authorization module.

It is unlikely that BASE24-pos would have to forward a CAP Token validation request to an issuer Host for authorization, because there would be no point in including the BASE24 system in the transaction path. However, this configuration is possible, and will be supported. If an issuer Host system performs the CAP processing, then the POS Router/Authorization CAP extension module will be invoked to update the CAP fields on the CAF when processing the response message:

- If the response message indicates that the CAP Token validation failed, then update the appropriate ATC and increment the Bad CAP Token Validation Accumulator on the CAF record.
- If the response message indicates that the CAP Token validation succeeded, then update the appropriate ATC and reset the Bad CAP Token Validation Accumulator on the CAF record.

Standard processing ensures that the transaction is written to the POS Transaction Log File (PTLF), and that a response message is returned to the acquiring interface process or device handler module.

Items to consider for CAP Authentication:

- Modifications are required to the Transaction Security Service (TSS) for this enhancement. A new TSS command message for CAP Token validation ('60') is used by BASE24. Another new TSS command message for extracting the APSN from the CAP Token ('61') has also been implemented, but is not used by BASE24.
- Although CAP functionality is based on the EMV infrastructure, there is little similarity in the authorization processing performed for an EMV payment transaction and a CTVS transaction. Because of this, a CAP Token validation transaction does not use the EMV Tokens, and does not invoke logic in EMV extension modules. However, a BASE24 customer must implement the BASE24 EMV extension modules in order to retrieve the CAP data added to the EMV segments in the CPF and CAF. It is assumed that BASE24 customers who implement CAP Token validation also support EMV payment processing (because the CAP solution "leverages" the EMV technology).
- It is assumed that no more than two physical cards identified by the same PAN/Member Number combination will be in circulation at any one time. This assumption allows the APSN for up to two physical cards to be configured on a single CAF record.
- The maximum length of a CAP Token supported by BASE24 is 18 bytes, which is consistent with the BASE24-eps solution.
- The BASE24 solution will provide support for CAP processing on CCD/CPA EMV cards, assuming that there is no impact on CAP that is specific to the CCD/CPA EMV implementation.
- If the CAP functionality provided by a card is part of the EMV application, then it is assumed that the same BASE24 system is used both to validate CAP Tokens and to authorize financial transactions. The reason for this limitation is that the same CAF record must be used to maintain the ATC for the application; otherwise, TSS will not be able to derive the ATC required in CAP Token validation.

- Because a CAF record is required, CAP Token validation is performed by BASE24-pos only if Positive Authorization (Type 2) or Positive Authorization with Balance (Type 3) is configured for the card. Otherwise, the CTVS request is processed as a “standard” card verification transaction.
- No new BASE24-pos response codes have been implemented to identify error conditions specific to CAP processing, e.g. CAP Token validation failure.
- The BASE24 solution does not support ATC checking. There is no point in checking that the ATC derived by TSS is greater than the last ATC value stored on the CAF, because the logic used by TSS to derive the ATC assumes that the value is greater anyway.
- The BASE24 solution does not check that the ATC derived by TSS is not greater than the sum of the last ATC value stored on the CAF and a “tolerance limit” configured on BASE24. There seems little point in providing this functionality, because any number of CAP Tokens could be generated and discarded by the cardholder before a validation request is sent to BASE24.
- BASE24-pos already utilizes card verification transactions in its Secure Internet Validation (SIV) extension product. Therefore, the CAP solution cannot be implemented in a BASE24 system which already includes support for SIV.
- The BASE24 solution does not provide support for offline (batch) CAP transactions.
- The BASE24 solution does not provide support for the validation of the CAP Token as a “pre-screen” function. There is no point in supporting this, because the Host system would have no further authorization processing to perform if the CAP Token were to be validated successfully.

NOTE: If the customer does not perform regular full file refreshes of the CAF, the BASE and POS changes in the BASE24-pos Dynamic CAF optional enhancement must be installed prior to, or in conjunction with, the implementation of the CAP Authentication enhancement. For more information refer to the BASE24-pos Dynamic CAF optional enhancement later in this document.

Prerequisite: This enhancement requires the installation of the POS Router/Auth EMV extension module. The minimum version of TSS required for the implementation of this enhancement is version 07.4.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCAF, DDLFCPF, DDLGCAFX, DDLGISO, DDLGTSS, and DDLPSTKN
- Inline changes to BA60AFT.SCRNCAF, RQCAFS, SVCAFS, SCRNCPPF, RQCPFS, SVCPPFS and SECTBL

All BASE24-pos customers:

- Inline changes to PS60SRC.POSFM and POSMM
- Inline changes to PS60RTAU.AUTHLIBS, ROUTERS, RTAUD, RTAUG and RTAUM.
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos EMV customers:

- Inline changes to BA60IEMV.CAMUTILG and CAMUTILS
- Inline changes to PS60IRTA.RTAUEMVS.

All BASE24-from host maintenance customers:

- Inline changes to FH60RPT.FHMRPTS

All BASE24-from host maintenance CAF customers:

- Inline changes to FH60CAF.FHMCASF

All BASE24-pos customers implementing the CAP Authentication Enhancement:

- New subvolume, PS60JRTA, with files for the CAP Authentication extension.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNCAP.

CUSTOMACS Changes

Set the following flag to true to bind the CAP Authentication extension module to the POS Router/Authorization process:

- pos_cap_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields relevant to CAP Authentication support.

Card Prefix File (CPF)

A new screen (#13) has been added to the CPF in support of the CAP Authentication enhancement. User security must be modified to allow operators to see this new screen. The new fields listed in this section have been added this screen.

This field identifies the CAP group to which this record belongs. If this field is not blank, the Device Handler/Router/Authorization module uses the value of this field to read the record in TSS. If this field contains blanks, then CAP Token verification is not performed. The default value is spaces.

Screen Name: CAP GROUP

Screen: CPF Screen 13

DDL Name: CPF.EMV.CAP-GRP

Valid Values are: Any combination of alphanumeric characters and leading and trailing spaces.

Default: Spaces

This field identifies the CAF ATC field to update in CAP Token verification transactions.

Screen Name: CAP ATC UPDATE

Screen: CPF Screen 13

DDL Name: CPF.EMV.CAP-ATC-UPDT

Valid Values are: 0 (CAP) and 1 (EMV).

Default: 0 (CAP)

This field is used to specify the maximum number of consecutive failed CAP Token validation attempts permitted. A value of zero disables the limit check.

Screen Name: BAD CAP TOKEN VALIDATION LIMIT

Screen: CPF Screen 13

DDL Name: CPF.EMV.BAD-CAP-TKN-CHK-LMT

Valid Values are: Any numeric value in the range 0000 – 9999.

Default: 0000

This field identifies whether or not CAP Token verification transactions are sent to the host.

Screen Name: HOST CAP TOKEN VALIDATION OPTION
Screen: CPF Screen 13
DDL Name: CPF.EMV.HOST-CAP-TKN-OPT
Valid Values are: 0 (Approve, don't send to Host), 1 (Send, approve if Host is down), and 2 (Send, deny if Host is down).
Default: 0 (Approve, don't send to Host)

A full description of these fields can be found in the ***BASE24 Files Maintenance Manual***.

Cardholder Authorization File (CAF)

The following additional new fields have been added to the CAF Screen 13 to configure CAP Authentication support.

This field will allow the CAP Token validation to be performed even if the Bad CAP Token Validation Limit (on the CPF) has been reached. If the validation is successful, then the Bad CAP Token Accumulator will be reset to zero. This mechanism is the only way that the Bad CAP Token Validation Accumulator can be reset once the limit has been reached, without deleting/re-adding or replacing the CAF record.

Screen Name: BAD CAP TOKEN OVERRIDE FLAG
Screen: CAF Screen 13
DDL Name: CAF.EMV.BAD-CAP-TKN-OVRRD-FLG
Valid Values are: N (no) and Y (yes).
Default: N (no)

The CAP Application PAN Sequence Number of the primary card.

Screen Name: CAP APSN (primary card data)
Screen: CAF Screen 13
DDL Name: CAF.EMV.CAP-DATA.APSN
Valid Values are: spaces or any numeric value in the range 00 – 99.
Default: spaces

The CAP Application PAN Sequence Number of the secondary card.

Screen Name: CAP APSN (secondary card data)
Screen: CAF Screen 13
DDL Name: CAF.EMV.SCND-CRD-DATA.CAP-DATA-2.APSN
Valid Values are: spaces or any numeric value in the range 00 – 99.
Default: spaces

The CAP Derivation Key Index of the primary card.

Screen Name: CAP DKI (primary card data)
Screen: CAF Screen 13
DDL Name: CAF.EMV.CAP-DATA.DKI
Valid Values are: Any numeric value in the range 00 – 99.
Default: 00

The CAP Derivation Key Index of the secondary card.

Screen Name: CAP DKI (secondary card data)
Screen: CAF Screen 13
DDL Name: CAF.EMV.SCND-CRD-DATA.CAP-DATA-2.DKI
Valid Values are: Any numeric value in the range 00 – 99.
Default: 00

In addition, FORCE ONLINE FALLBACK CHECK and EMV ISSUER have been moved from Screen 11 to Screen 13.

A full description of these fields can be found in the ***BASE24 Files Maintenance Manual***.

Token File (TKN)

The Authentication Data Token (token ID “CE”) is utilized for this enhancement. A value of “02” in the Authentication Indicator Flag within the Authentication Data token, indicates the token is in the CAP token format.

Template Changes

Not applicable

Device Configuration

The initial requirement is to support CAP Token validation request messages received from an acquiring Host system via the HISO interface. However, because the BASE24 solution relies on an existing transaction type (card verification) and an existing token (Authentication Data Token), it is possible for customers to utilize any BASE24-pos acquiring interface or device handler that supports the card verification transaction and the Authentication Data token, e.g. SPDH.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry has been added for the new CPF Screen 13.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new CAF screen must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) version 07.4 or greater must be installed in order to use this enhancement.

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Cardholder Authorization File (CAF)

For customers converting from Release 5.3 to Release 6.0 version 8, the CAF conversion program must be run to convert the CAF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

No separate conversion program was created to move the CAF from version 7 to version 8. Customers must do a full file refresh using the new CAF layout. This will initialize the new CAF fields to appropriate values.

Host/Co-Network Impacts

Super Extract Process

If the token (i.e., Authentication Data Token) used for this enhancement is to be included in the PTLF Extracts, BASE24 users must consider the format of the CAP token data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to CPF and CAF records.

Refresh Process

Host code changes are required for all BASE24 users that utilize the Release 6.0 CAF Refresh file message format to provide the new CAP fields in the CAF EMV segment.

From Host Maintenance Process

Host code changes are required for all BASE24 users that utilize FHM to provide the new CAP fields in the CAF EMV segment.

Acquirer Host Systems

The initial requirement is to support CAP Token validation request messages received from an acquiring Host system via the HISO interface. To allow BASE24-pos to perform CAP Authentication, the acquirer Host must send a request message containing a card verification processing code and the Authentication Data Token.

Card Management Systems

Card management systems must be capable of providing the additional card data required by BASE24.

User/Operational Impacts

The full ATC does not have to be included in a CAP authentication request message. However, when fewer ATC digits are sent in a transaction, it is more likely that the full ATC value in the database will get out of sync with the full ATC kept on the card. Once they are out of sync, it is unlikely they will ever match up and the CAP Token sent will never be verified. ACI recommends sending the maximum number of ATC digits possible in the CAP token data. The new Dynamic Cardholder Authorization File (CAFD) will attempt to ensure that the correct ATC is maintained by BASE24. The issuer must recognize there is greater chance that the value will be wrong when fewer digits are included in the transaction data.

Performance Impacts

CAP authentication takes the place of ARQC verification in the (EMV) transaction path. The performance impact of doing CAP authentication should be the same as doing standard card authentication in an EMV transaction.

Vendor Impacts

Not applicable

Reference Documents

- Visa International - Dynamic Passcode Authentication – Card & Reader Implementation Guide, January 2005
- MasterCard International - Chip Authentication Program Functional Architecture, Final Draft, Version 3.0.2, December 2006
- APACS - CAP User Interface Specification – UK Implementation, version 1.01, 27th June 2006
- APACS - Specification for the UK Minimum Implementation of the MasterCard Chip Authentication Program Functional Architecture, version 1.11, 27th June 2006

Dynamic CAF

This section describes the implementation requirements associated with the Dynamic CAF enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Dynamic CAF enhancement.

In contactless magnetic stripe and Chip Authentication Program (CAP) transactions, an Application Transaction Counter (ATC) is used in the card authentication processing. Typically, however, only part of the ATC is included in the transaction message received by BASE24-pos. The idea is that the authorization system, e.g. BASE24, maintains the ATC on its cardholder database, which allows the full ATC to be derived during transaction processing.

BASE24 maintains cardholder data on the Cardholder Authorization File (CAF), and is the obvious file to hold these ATCs. However, a full file Refresh of the CAF will wipe out these values, meaning that it will no longer be possible to derive the complete ATC for a contactless magnetic stripe or CAP transaction. Therefore, a mechanism is required to ensure that the ATCs are retained by BASE24, even when a full CAF Refresh is performed.

A new Dynamic Cardholder Authorization File (CAFD), keyed by PAN and member number, has been implemented in BASE24 to store dynamic card data that must be retained following a full CAF Refresh. The CAFD is read and updated during BASE24-pos transaction processing, but is not affected by Refresh or FHM.

Software Impacts

All BASE24 customers:

- Changes to BA60SRC. BASEFM, BASEM, and BASEMM
- A new subvolume, BA60CAFD, for the Dynamic Cardholder Authorization File (CAFD) Maintenance program
- Changes to BA60SRC.BACOUTLS
- Changes to BA60DDL.BADDLM and BADDLMM
- A new file BA60DDL.DDLFCAFD
- Inline changes to BA60DDL.DDLFIDF
- Inline changes to BA60AFT.SCRNIDF, RQIDFS, SVIDFS, SVCAFM, and SVCAFS

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS, AUTHS, ROUTERS, RTAUEMVD, RTAUG and RTAUM.
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVS, RTAUEMVM

A list of the Base files affected by this enhancement can be found in Section 4.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNCADF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields relevant to Dynamic CAF support.

Institution Definition File (IDF)

The following additional new field has been added to the IDF Screen 1 to configure Dynamic CAF support.

The name of the Dynamic Cardholder Authorization File (CAFD) for this institution.

Screen Name: CAFD NAME

Screen: IDF Screen 1

DDL Name: IDF.BASE.CAFD-NAME

A full description of this field can be found in the ***BASE24 Files Maintenance Manual***.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The IDF conversion program must be run to convert the Institution Definition File (IDF). Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to IDF records.

User/Operational Impacts

Disk Impacts

Customers deciding to implement support for the CAFD will require additional disk space. The CAFD record length means that 32 records will be held in each 4K disk block.

Performance Impacts

When the CAFD is accessed during transaction processing, an additional disk read and a corresponding update is performed. Note, however, that the CAFD is accessed only when processing the following types of transaction:

- CAP Token validation requests;
- Contactless magnetic stripe requests;
- EMV authorization requests and advices, where the ATC CHECK TYPE on Screen 11 of the CPF is set to '9' (use ATC located in the Base segment).

The CAFD is accessed by BASE24-pos only if Positive Authorization (Type 2) or Positive Authorization with Balance (Type 3) is configured for the card and when a CAFD file name has been configured in the IDF.

Operational Impacts

Data from the CAFD is displayable on the CAF screens, but the file from which the data is obtained is transparent to the user. A small batch program, the CAFD Maintenance program (which may be run at any time), has been written to "clean up" old records in the CAFD, as there is no other mechanism for deleting records. Note that the CAF Server reads CAFD records only, i.e. the CAF Server cannot add, delete or update CAFD records. Records are only added via online transaction processing.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Hypercom 3DES in Logon and Other Enhancements

This section describes the implementation requirements associated with the Hypercom 3DES in Logon and Other Enhancements.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Hypercom 3DES in Logon and Other Enhancements.

BASE24-pos has been enhanced to support the following enhancements to the Hypercom device handler (HPDH):

- Support for double length PIN keys in logon response messages.
- Support for an offline Transaction Certificate (TC) in 0320 Batch Upload messages.
- Support for additional fields in the Terminal Configuration Table.
- Support for Contactless EMV and Magnetic Stripe transactions.

Support for Double Length Keys in a Logon Response Message:

The HPDH has been enhanced to support double length PIN keys in “0810” Logon Response messages. Single length MAC keys in “0810” Logon Response messages will still be supported.

Current HPDH processing supports single length PIN and MAC keys in “0810” Logon Response messages. These single length keys are carried in Private Use Field 62. This enhancement is to support double length PIN keys in “0810” Logon Response messages. Double length keys will be carried in Private use Field 63. If Field 63 is not present, the single length keys in Field 62 will be used by the terminal. If Field 63 is present, the key fields of Field 62 will be ignored.

Double length keys require TSS support, which requires use of an HSM. However, Hypercom MACing is proprietary and currently supported by BASE24 via software only. Therefore, double length MAC keys are not supported at this time.

Support for Offline Transaction Certificates in 0320 Batch Upload Messages:

The HPDH has been enhanced to support EMV transactions which were approved at the terminal in offline mode and forwarded as part of a batch upload.

Currently, if the terminal supports sending the Transaction Certificate (TC), this field is not logged to the PTLF. This enhancement will enable the HPDH to map the Transaction Certificate (from field DE55, Tag 9F27 – CID, Tag 9F26 – Application Cryptogram) to the EMV Request token where it will be logged to the PTLF.

Support for Terminal Configuration Table Changes:

The HPDH was enhanced to support two new fields in the Terminal Configuration Table which is downloaded to the terminal during the Hypercom terminal download sequence. This table resides on the Hypercom T7 Configuration File (HPTC). The two new fields added to this table were a 4 byte Check Service Flags field and a 1 byte Options 8 field, for a total of 5 bytes.

Support for Contactless EMV and Magnetic Stripe transactions:

The HPDH was enhanced to support Contactless EMV and Magnetic Stripe transactions from a Hypercom terminal. According to Hypercom's v3.44 message specification, the Point of Service (POS) Entry Mode is not differentiated for Contactless EMV and Magnetic Stripe transactions. For both cases, a value of 91 is defined for the POS Entry Mode. The differentiation is made by sending field 55 for Contactless EMV transactions and not sending field 55 for Contactless Magnetic Stripe transactions. For Contactless EMV transactions where field 55 is sent, the POS Entry Mode will be mapped to a value of 07 in the PSTM.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60HPDH.HPDHDDL, HPDHEMVD, HPDHG, HPDHS, HPDHLD, SCRNHPT, SVHPTCS
- Inline changes to PS60IHPD.HPDHEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNH3D.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Hypercom T7 Configuration File (HPTC) screen 4:

Field Name: EARLY AMT (EMV)

DDL Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ERLY-AMT-EMV

A code indicating whether there is an early amount entry for EMV transactions.
Valid values are:

0 = No early amount entry for EMV transactions.

1 = Yes, there is an early amount entry for EMV transactions.

Field Name: ALLOW PIN BYPASS

DDL Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ALW-PIN-BYP

A code indicating whether PIN bypasses are allowed from the terminal. Valid values are:

0 = Do not allow PIN bypass.

1 = Allow PIN bypass.

Field Name: ICC EMV ENABLED

Data Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.ICC-EMV-ENBL

A code indicating whether the terminal is ICC EMV enabled. Valid values are:

0 = ICC EMV is disabled.

1 = ICC EMV is enabled

Field Name: TERM WITH SCR
DDL Name: HPTC.CONFIGURATION-PARAMS.OPTIONS-8.TRM-SCR
A code indicating whether the terminal is SCR enabled. Valid values are:

0 = Terminal is not SCR enabled.
1 = Terminal is SCR enabled.

For more information on setting these fields, refer to the following vendor manual:
Hypercom T7 & ICE System Point of Sale Software Initialization Parameters, v1.24a,
July 31, 2006.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

If the BASE24 user would like to download the Terminal Configuration Table to the Hypercom terminal, the HPTC screen 4 must be updated with the appropriate values prior to the download.

Vendor Impacts

Not applicable

Reference Documents

- Hypercom T7 & ICE System Point of Sale Software Initialization Parameters, v1.24a, July 31, 2006.
- Hypercom The System Logon Transaction Support of Double Length Key, v2006-01-31, September 12, 2005.
- Hypercom Message Specification "The System", v3.44, May 2, 2005.

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Immediate Availability PTD Static Data Changes

This section describes the implementation requirements associated with the Immediate Availability of PTD Static Data Changes enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Immediate Availability of PTD Static Data Changes enhancement.

BASE24-pos has been enhanced to address changes to the static data in the PTD (PTDS1) file. When static data is updated via AFT, a device handler will not retrieve this data from disk if it already has the record loaded in memory until a warmboot occurs. This enhancement is optional and enabled by an LCONF parameter to improve performance. When the LCONF parameter is enabled, SPDH has been enhanced to check for changes to static data by comparing the timestamp of the PTDD1 and the PTDS1 in memory. If a change in timestamps is detected, the PTDS1 record is removed from memory and re-read from disk. This will allow changes to terminal records to be made available immediately for online transaction processing without needing a warmboot.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60SPDH.SPDHG, SPDHM, SPDHS
- Inline changes to PS60SRC.PUTLM, PUTLS
- New event (log) message manual BA60LOGM.PSUTLS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNCPTDS.

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CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that is required to be added to the LCONF to support this enhancement. Details of the param are provided in the BASE24 Logical Network Configuration File manual. A sample of the param is included in the LCONFPS file located on BA60MISC subvolume.

POS-DH-PTD-STATIC-REPL

This param specifies whether the PTD static record should be replaced in extended memory if it is found that the static record has been updated via AFT. This is an optional param. Valid values are “Y” and “N”. If the param is not found or is invalid, the default of “N” is used.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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POS Completion Message Configuration

This section describes the implementation requirements associated with the POS Completion Message Configuration enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the POS Completion Message Configuration enhancement.

The COMPLETION REQUIRED TO HOST flag on IPCF screen #2 is defined as “a flag indicating whether a completion is required to be sent to the host for this transaction.”

BASE24-pos has been enhanced to provide new values for the COMPLETION REQUIRED TO HOST flag. These new values will be used to indicate that completions are to be sent to the host only if BASE24 performed the authorization. Customers will now be able to configure the system to either send all BASE24 authorized transactions, including declines or to only send approved BASE24 authorized transactions.

As part of the BASE24-pos Completion Messages Configuration enhancement, implemented in Release 6.0 Version 7, the IPCF has been modified to provide for a default value of “0” (Do Not Send 0220 Messages) for the Completion Required field. The IPCF requester and server have been modified to process this change. Please refer to the “BASE24 Release 6.0 Version 7 Implementation Guide” for further details of this enhancement.

Software Impacts

All BASE24 customers:

- Changes to comments in BA60DDL.DDLGPSTM

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS, AUTHS, and RTAUG

All BASE24-pos ISO Host Interface customers:

- Inline changes to PS60HISO.PSHISOG and PSHISOS

A complete list of the files affected by the POS Completion Message Configuration enhancement can be found in the SCN file BA60UD0B.SCNCOMP.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Issuer Processing Code File (IPCF)

Field Name: COMPLETION REQUIRED TO HOST
Screen: IPCF Screen 2
DDL Name: IPCF.COMPL-REQ

A code used to control the production of 0220 financial transaction advice messages. Valid values are:

- 0 = Do not send 0220 messages
- 1 = Send 0220 messages for approved transactions
- 4 = Send 0220 messages for all transactions
- 5 = Send 0220 message for approved BASE24 authorized transactions only
- 6 = Send 0220 message for all BASE24 authorized transactions
(approved/denied)

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

If the BASE24 user wishes to enable the new value of “5” or “6”, then the host system must be capable of accepting and processing the appropriate approved or non-approved advices. The host system must also be prepared to handle the increase in message traffic if applicable.

User/Operational Impacts

If the BASE24 user wishes to enable the new value of “5” or “6”, then the COMPLETION REQUIRED TO HOST field on IPCF screen 2 must be set appropriately.

Performance Impacts

If the new value of “5” or “6” is implemented, there will be an increase or decrease in the number of messages sent to the Host depending upon the previous value. The Host must be prepared to handle the increase in message traffic if applicable.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

SPDH Message Data Element Encryption

This section describes the implementation requirements associated with the SPDH Message Data Element Encryption enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH Message Data Element Encryption enhancement.

Numerous BASE24-pos customers have requested that the SPDH protect not only PIN data, but other sensitive data fields as well, in messages exchanged between POS devices and the SPDH. The SPDH Message Data Element Encryption enhancement enables customers to protect these data fields by employing either *full message encryption* or *configurable message encryption*. Message data is single-DES or triple-DES encrypted under the Data Encryption Key (FID I, DID j), which supports 16-byte (single-length), 32-byte (double-length) and 48-byte (triple-length) keys.

Full message encryption allows the customer to encrypt all optional data fields in an ACI Standard POS Message, with the exception of those listed in Table 1. Full message encryption is enabled by setting the Data Encryption Type field in the POS Terminal Data File (PTD) to “03” (full message encryption).

FID	Description	Reason
G	Authentication Code	Created using FID H (Authentication Key)
H	Authentication Key	Already encrypted under the MAC terminal master key
I	Data Encryption Key	Cannot be used to encrypt itself.
M	PIN Communications Key	Already encrypted under the PIN terminal master key
b	PIN/Customer	Already encrypted under FID M (PIN Communications Key)
c	PIN/Supervisor	Already encrypted under FID M (PIN Communications Key)

Table 1 – Optional data fields that are never encrypted under the Data Encryption Key

Configurable message encryption allows the customer to encrypt specific optional data fields in an ACI Standard POS Message, with the exception of those listed in Table 1. Configurable message encryption is enabled by setting the Data Encryption Type field in the PTD to “04” (configurable message encryption). The optional data fields to be encrypted are specified in the ACNF request and response field map records. If an optional data field is to be encrypted, an “E” is placed in the one-byte MSG field corresponding to the optional data field’s FID for the specific transaction type. Although specific optional data fields can be encrypted, specific optional data subfields cannot. All subfields under FIDs 6, 7, 8 or 9 are encrypted, or none.

Whether using full message encryption or configurable message encryption, the format of the ACI Standard POS Message is the same. The message begins with the standard message header, which is never encrypted. Following the standard message header is the unencrypted data area, which contains all the optional data fields and subfields that are not encrypted under the data encryption key. Next comes the *encryption separator* (hexadecimal 1F), a new separator character that is used to mark the beginning of the encrypted data area. The encrypted data area extends from the encryption separator to the end of the message, and contains all the optional data fields and subfields that are encrypted under the data encryption key. Field separator characters (hexadecimal 1C) and field identifiers (FIDs), which preface optional data fields, are included in the encrypted data. Likewise, record separator characters (hexadecimal 1E) and subfield identifiers (subFIDs), which preface optional data subfields, are included. The encryption separator itself is not encrypted, nor is the end-of-text character at the end of the message, if present. Figure 1 illustrates the format of an encrypted ACI Standard POS Message.

Standard Message Header	Unencrypted Data Area	E S	Encrypted Data Area	E T X
-------------------------	-----------------------	--------	---------------------	-------------

Figure 1 – Format of an encrypted ACI Standard POS Message

By grouping optional data fields and subfields into the encrypted data area, the SPDH can encrypt or decrypt the data with a single call to the host security module (HSM).

A long-standing requirement of the ACI Standard POS Message is that all data must be in ASCII format, not binary. Furthermore, some HSMs cannot process binary data reliably if it contains certain special characters. For instance, the Atalla NSP cannot encrypt/decrypt binary data containing the special characters "<", "#", "^", ">", or CR. For these reasons, encrypted data in the ACI Standard POS Message is required to be in ASCII hexadecimal format when full message encryption or configurable message encryption is performed. The Data Type field in the device's Channel Security Configuration (CSEC) record must be set to "Unpacked ASCII hexadecimal".

In Release 6.0 Version 3, available balance encryption was added to the SPDH as part of the POS Enhancements. The SPDH Message Data Element Encryption enhancement is integrated in a way that retains the original functionality of available balance encryption. If available balance encryption is enabled, the Available Balance field (FID J) is encrypted individually, as it was before. If full or configurable message data encryption is enabled, the Available Balance field is grouped with other optional data fields in the encrypted data area of the ACI Standard POS Message, rather than encrypted individually.

The Data Encryption Type field in the PTD takes on the new and redefined values shown in Table 2.

Value	New Definition	Previous Definition
00	No encryption	No encryption
01	Available balance encryption – ASCII	ASCII encryption
02	Available balance encryption – binary	Binary encryption
03	Full message encryption	
04	Configurable message encryption	

Table 2 – PTD Data Encryption Type

Message encryption is allowed for SPDH transactions of any message type, such as financial transactions (message type = F) and administrative transactions (message type = A). Message encryption is also allowed for transactions of any message subtype, such as online transactions (message subtype = O) and store-and-forward transactions (message subtype = S).

Message encryption is allowed for all SPDH transaction types except one: The download transaction message (transaction code 90) cannot be encrypted.

The Transaction Security Services (TSS) version 07.4 or greater must be installed in order to use this enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLPCNST

All BASE24-pos customers:

- Inline changes to PS60AFT.SVPTDTBL and SVPTDTS
- Inline changes to PS60DDL.DDLFPTD
- Inline changes to PS60SETL.SETLS

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHS, RQACNFS, SPDHD, SPDHDDLs and SVACNFS
- Updated event (log) message manual BA60LOGM.PSSPDH

All BASE24-pos ACI Standard POS Device Handler (SPDH) Data Encryption customers

- Inline changes to PS60DSPD.SPDHDEG and SPDHDES

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNMDEE.

CUSTMACS Changes

Set the following Make flag to true to bind the Data Encryption module to the SPDH.

- pos_dh_spdh_de_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ACI Standard Device Configuration File (ACNF)

The following field has been added to the end of the ACNF's Field Data record definition (FLD-DATA). The addition of this field enables the user to specify which optional data fields (FIDs) are encrypted in device request and response messages, when configurable message encryption is used. See Device Configuration for more information.

Screen Name:	<Transaction Type> MSG
Screen:	ACNF Request Field Map Screens 02 through 09
	ACNF Response Field Map Screens 02 through 09
DDL Name:	ACNF.FLD-DATA.ENCRYPT-FLDS

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

BASE24-pos Terminal Data file (PTD)

Depending on the type of data encryption supported by the device, the Data Encryption Type field in the PTD must be set to one of the following values.

- 00 = No encryption
- 01 = Available balance encryption – ASCII
- 02 = Available balance encryption – binary
- 03 = Full message encryption
- 04 = Configurable message encryption

ACI Standard Device Configuration File (ACNF)

No data modifications are needed in the ACNF if full message encryption is used. If configurable message encryption is used, data in the ACNF should be modified as follows.

For each optional data field that is to be encrypted in a device request or response message, place an “E” in the one-byte MSG field corresponding to the optional data field’s ID (FID) for the specific transaction type. The value “E” means the optional data field will be included in a message and it will be encrypted under the data encryption key. Following is an example.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	BASE24-POS		SPDH ACNF MAINT		lgnt	fiid	yy/mm/dd	hh:mm 02 OF 03
2	TERMINAL GROUP: _____			RECORD TYPE: _		RECORD ID: __		
3								
4	TRANSACTION		FIELD MAP (REQUEST)					
5								
6	ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789							
7	NORM PURCH	MSG	N	Y	N	N	N	N
8		MAC	N	N	N	N	N	N
9	PRE AUTH	MSG	N	Y	N	N	N	N
10		MAC	N	N	N	N	N	N
11	PRE AUTH C	MSG	N	Y	N	N	N	N
12		MAC	N	N	N	N	N	N
13	PH/MAIL	MSG	N	Y	N	N	N	N
14		MAC	N	N	N	N	N	N
15	MERCH RET	MSG	N	Y	N	N	N	N
16		MAC	N	N	N	N	N	N
17								
18								
19								
20								
21	***** BASE24 *****							
22	NEW PAGE: __	FILE DESTINATION: _____		NEW LOGICAL NETWORK ID: _____				
23	F12-HELP							
24	<message text -----> <No>							

In this example, FID B (Amount 1) is configured to be included, but not encrypted, in device request messages for normal purchase, pre-authorization, pre-authorization completion, mail/phone order, and merchandise return transactions (MSG field = “Y”). FID q (Track 2/Customer) is configured to be included and will also be encrypted under the data encryption key (MSG field = “E”). The remaining FIDs are not included in the request messages.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) version 07.4 or greater must be installed in order to use this enhancement.

File Conversions

The ACNF conversion program must be run to convert the ACI Standard Device Configuration File (ACNF). Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

Message data encryption may impact device performance due to the extra processing required to encrypt data in outgoing messages and to decrypt data in incoming messages.

Message data encryption may impact BASE24 system performance due to extra I/O with the HSM. Each message from the POS device that contains encrypted data will require one extra I/O with the HSM to decrypt the data. Each message to the POS device that contains encrypted data will require one extra I/O with the HSM to encrypt the data.

Vendor Impacts.

POS device software must be modified to encrypt and decrypt data in the ACI Standard POS Message.

A new separator character has been defined to delimit the encrypted portion of an ACI Standard POS Message:

- Encryption Separator – Hexadecimal 1F (Octal 37)

The encryption separator marks the beginning of encrypted data in the message. All data between the encryption separator and the end of the message (not including the end-of-text character) is considered to be encrypted under the data encryption key. The following rules must be observed when encrypting data in the ACI Standard POS Message:

- The standard message header must not be encrypted. Only optional data fields and subfields may be encrypted.
- All unencrypted optional data fields and subfields must immediately follow the standard message header.
- The encryption separator must immediately follow the last unencrypted optional data field or subfield.
- All encrypted optional data fields and subfields must immediately follow the encryption separator.
- Field separator characters (hexadecimal 1C) and field identifiers (FIDs), which preface optional data fields, are included in the encrypted data.
- Record separator characters (hexadecimal 1E) and subfield identifiers (subFIDs), which preface optional data subfields, are included in the encrypted data.
- If an end-of-text (ETX) character is appended to the message, it must not be encrypted.

The following examples illustrate the placement of the encryption separator and encrypted data. **Bold text** represents data encrypted under the data encryption key. Plain text represents unencrypted data. The symbols used in the examples are defined below.

[std msg hdr] = standard message header
[es] = encryption separator character (hexadecimal 1F)
[fs] = field separator character (hexadecimal 1C)
[fid] = field identifier (A-Z, a-z, 0-9)
[field] = optional data field associated with the FID
[rs] = record separator character (hexadecimal 1E)
[sfid] = subfield identifier (A-Z, a-z, 0-9)
[subfield] = optional data subfield associated with the SFID
[etx] = End-of-text character (hexadecimal 03)

The following message is unencrypted.

[std msg hdr][fs][fid][field][fs][fid][field][etx]

All optional data fields are encrypted in the following message.

[std msg hdr][es]**[fs][fid][field][fs][fid][field]**[etx]

All optional data fields are encrypted in the following message. There is no ETX.

[std msg hdr][es]**[fs][fid][field][fs][fid][field]**

All optional data fields and subfields are encrypted in the following message.

[std msg hdr][es]**[fs][fid][field][fs][fid 6][rs][sfid][subfield]**[etx]

One optional data field is unencrypted in the following message. The rest are encrypted.

[std msg hdr][fs][fid][field][es]**[fs][fid][field][fs][fid][field]**[etx]

Reference Documents

Not applicable

Stored Value BDM Support

This section describes the implementation requirements associated with the Stored Value Bulk Data Maintenance (BDM) Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Optional Enhancement Stored Value BDM Support enhancement.

The Bulk Database Maintenance (BDM) process in BASE24-card Release 6.0 Version 8 allows BASE24-card customers the ability to add and delete mass quantities of card records. The BASE24-stored value extension to BDM provides the ability to add, delete, and update large numbers of Stored Value (SV) account records, as well as activating and deactivating SV cards and accounts.

Software Impacts

All BASE24 customers

- Inline changes to BA60AFT.SECTBL, SVMHLPS, RQCAFS, SVCAFS

All BASE24-card customers:

- Inline changes to CT60AFT.CTAFTM, CTAFTMM, CTALGOS, RQCOFS, SCRNCOF, SVAGENM, SVAGENS, SVCMFM and SVCMFS
- Inline changes to CT60DDL.CTDDLMM and CTDDLMM

All BASE24-card Stored Value customers:

- New files (CT60AFT.RQBDMM, RQBDMS, SCRNBDM, SCRNSBDM, BDMG, BDMM, BDMS and BDMTOGM)
- New file (CT60DDL.DDLFBMIF)

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNSTVB.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

This section describes the param record that is required to be added to the LCONF to support this enhancement. Details of the param are provided in the BASE24 Logical Network Configuration File manual. A sample of the param is included in the LCONFCT file located on BA60MISC subvolume.

CMS-BDM-OMF

The fully qualified OMF filename. This assign is required.

CMS-BDM-RPT

BDM output report location. This assign is required.

LCONF Param Changes

CMS-BDM-OMF-AUDIT

Indicates the extent of BDM OMF auditing. Values: 0 - no audit, 1 - audit record and header information, 2 - header information only, no record. (default = 1 if invalid or not present). This param is optional.

CMS-BDM-OMF-RETENTION-PERIOD

Indicates the number of days to retain BDM OMF files before purging them. Minimum value is 3, maximum value is 99 (default=7 if invalid or not present). This param is optional.

Database Configuration

This section describes fields, within BASE24-card files, relevant to Stored Value BDM support. Field settings depend on desired configuration.

BASE24-card Card Operator Function File (COF)

Screen Name: OUTPUT TYPE
Screen: COF Screen 1
DDL Name: CMS-PARAMS.OUTPUT-TYPE
Valid Values: R = Report
 L = Label
 D = Disk

Screen Name: OUTPUT DISK FILE
Screen: COF Screen 1
DDL Name: CMS-PARAMS.RECOVERY-FILE
Valid Values: Valid disk filename format; required if Output Type = "D"

Screen Name: FILE PRI EXT
Screen: COF Screen 1
DDL Name: CMS-PARAMS.FILE-PRI-EXT
Valid Values: 5-digit numeric
Default Value 4; required if Output Type = "D"

Screen Name: FILE SEC EXT
Screen: COF Screen 1
DDL Name: CMS-PARAMS.FILE-SEC-EXT
Valid Values: 5-digit numeric
Default Value 4; required if Output Type = "D"

Screen Name: NUMBER OF ACCOUNTS
Screen: COF Screen 1
DDL Name: CMS-PARAMS.NUM-ACCTS
Valid Values: 7-digit numeric
Default Value 0

BASE24-base Server Menu Help (SVMHLP)

The Menu Help Server has been modified to allow help access for the BDM screen. The BDM screen has been incorporated into the CMS-TABLE resulting in a mandatory increase in the number of entries for that table (i.e., CMS-ENTRY, CMS-ENTRIES-L). Please refer to section 4 for more details.

New Data Files

BASE24-card Bulk Database Management File (BDM)

This new file allows BASE24-card customers the ability to add and delete mass quantities of card records. The BASE24-stored value extension to BDM also provides the ability to add, delete, and update large numbers of SV account records, as well as activating and deactivating SV cards and accounts.

Field Name: SYMBOLIC NAME
Screen: BDM Screen 1
DDL Name: WS-SYMBOLIC-NAME
Valid Values: 16-byte alphanumeric
Default Value: Blanks

Field Name: USER
Screen: BDM Screen 1
DDL Name: WS-NCP-USER-ID
Valid Values: 16-byte alphanumeric
Default Value: Blanks

Field Name: PASSWORD
Screen: BDM Screen 1
DDL Name: WS-NCP-PASSWORD
Valid Values: 16-byte alphanumeric
Default Value: Blanks

Field Name: FIID
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.FIID
Valid Values: 4-byte alphanumeric
Default Value: Blanks

Field Name: CARD SEED
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.SEED-PAN
Valid Values: 19-byte numeric
Default Value: Blanks

Field Name: CARD TYPE
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.CARD-TYP
Valid Values: 2-byte alphanumeric;
Card Type "SD" (BDM Card Seed) not allowed.
Default Value: Blanks

Field Name: CARD TYPE DESCRIPTION
Screen: BDM Screen 1
DDL Name: None
Valid Values: Card type description from COBNAMES CARD-TYPE-TABLE,
corresponding to Card Type value entered.
Default Value: "()"

Field Name: BULK MAINT INPUT FILE
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.INPUT-FNAME
Valid Values: Valid disk filename
Default Value: Blanks

Field Name: RANGE BEGIN
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.RANGE-BEG
Valid Values: 19-digit numeric
Default Value: Blanks

Field Name: END
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.RANGE-END
Valid Values: 19-digit numeric
Default Value: Blanks

Field Name: NUM RECS
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.NUM-RECS
Valid Values: 8-byte numeric
Default Value: Blanks

Field Name: AMT TO DEBIT/CREDIT
Screen: BDM Screen 1
DDL Name: BDM-STARTUP-CMD.AMT
Valid Values: 17-bytes numeric
Default Value: Blanks

Field Name: D/C
Screen: BDM Screen 1
Additional Notes: Field title and value are only displayed if SV is supported.
Required on Alter Balance functions, not valid for others.
DDL Name: BDM-STARTUP-CMD.AMT (sign)
Valid Values: D = Debit; C = Credit
Default Value: Blank

Field Name: ACTIVATION STATE
Screen: BDM Screen 1
Additional Notes: Field title and value are only displayed if SV is supported. Not valid for functions other than Add SV Card and Activate, but only required on Activate functions.
DDL Name: BDM-STARTUP-CMD.CRD-ACTVT-ST
Valid Values: 3-byte alphanumeric
Default Value: Blanks

Bulk Maintenance Input File (BMIF)

The Bulk Maintenance Input File, optionally generated from the BASE24-card Card Operator Functions (COF) screen as the Output Disk File, is used as input to the Bulk Database Maintenance (BDM) module. It contains one record for each PAN the user wishes BDM to generate a non-Stored Value CAF record for. In a BASE24-stored value environment, BDM will use the BMIF to add both CAF records and corresponding PBF records, using the BMIF PAN as the PBF account number.

RECORD BMIF. FILE IS "BMIF" ENTRY-SEQUENCED.

* Primary Account Number used by the Bulk Database Maintenance (BDM)
* process when creating CAF records (as well as associated PBF records
* in a Stored Value environment). This number will be used in
* combination with a seed record in the appropriate file(s) to populate
* an environment where BASE24 is the database of record.

03 PAN TYPE *.

* Reserved for future use.

03 USER-FLD1 PIC X(1).

* Sign to indicate whether the value in the AMT field indicates a debit
* or credit. This field is only valid in Alter Balance functions.

* Valid values are:

*

* "-" - Debit amount

* "+" or " " - Credit amount

03 DB-CR-IND PIC X(1).

* Amount used by BASE24-stored value on Add SV Card and Alter Balance
* functions. On Add SV Card functions, this field contains a balance
* that, for the PBF record being added, supercedes the balance
* specified in the PBF seed record. In an Alter Balance function, this
* field represents the amount to increment or decrement the PBF balance
* by for an individual record, superseding the amount specified in the
* BDM startup message.
*
* This field should be blank-filled if BDM is to default to the balance
* in the PBF seed record (for Add SV Card functions) or the increment/
* decrement amount in the BDM startup message (for Alter Balance
* functions). It is not used by the BASE24-card Card Operator Function
* (COF) user interface.
03 AMT PIC X(17).

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entry added to the SECTBL to support BDM.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Stored Value Support in CRT Authorization

This section describes the implementation requirements associated with the Stored Value Support in CRT Authorization enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Optional Enhancement Stored Value Support in CRT Authorization.

This enhancement added Stored Value transaction support to CRT Authorization. Additional transactions will be made available to CRT operators to support stored value/prepaid card implementations. New transactions will provide the capability to perform card activations, replenish funds to a card/account, and also allow the redemption of funds from a card/account.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS, RTAUG
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos CRT Authorization customers:

- Inline changes to BP60CRTA.DDLFCSEC, RQCSECS, RQOLTS, RQPRFS, SCRNCSEC, SVCSECS, CRTAUDHS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0B.SCNSTVC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields, within BASE24-pos, relevant to Stored Value Support in CRT Authorization. Field settings depend on desired configuration.

BASE24-pos CRT Auth Security File (CSEC)

Screen Name: CARD ACTIVATION
Screen: CSEC Screen 1
DDL Name: CSEC.TRAN-SEC-ACCESS.CARD-ACTV
Valid Values: Y = Yes, this CRT authorization operator can send this type of transaction from the Online Transaction screen.
N = No, this CRT authorization operator is not allowed to perform this type of transaction from the Online Transaction screen.

Screen Name: REPLENISHMENT
Screen: CSEC Screen 1
DDL Name: CSEC.TRAN-SEC-ACCESS. REPLENISH
Valid Values: Y = Yes, this CRT authorization operator can send this type of transaction from the Online Transaction screen.
N = No, this CRT authorization operator is not allowed to perform this type of transaction from the Online Transaction screen.

Screen Name: FULL REDEMPTION
Screen: COF Screen 1
DDL Name: CSEC.TRAN-SEC-ACCESS.FULL-REDEMP
Valid Values: Y = Yes, this CRT authorization operator can send this type of transaction from the Online Transaction screen.
N = No, this CRT authorization operator is not allowed to perform this type of transaction from the Online Transaction screen.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Added three new transactions for Stored Value Support. "CARD ACTIVATION", "REPLENISHMENT" and "FULL REDEMPTION".

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide, Inc.

Track 2 Context Data Removal

This section describes the implementation requirements associated with the Track 2 Context Data Removal enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Track 2 Context Data Removal enhancement.

When the track discretionary data is suppressed, the primary account number (PAN) and expiration date are retained. The rest of the track data is replaced with ASCII zeroes.

The BASE24-pos ACI Standard POS device handler (SPDH) module has been enhanced to offer the option to suppress discretionary data in the Track 2 data in the internal PSTM response message held in memory. The enhancement also ensures that the member number is retained in the device dependent temporary context area of the PTD and is available for reversal and advice processing.

The BASE24-pos Authorization module has been enhanced to offer the option of removing the track 1 and track 2 discretionary data from the Track 2 data in the internal PSTM response if the transaction is authorized by either BASE24 or a Host. In addition, response messages received from an interchange may have the discretionary data suppressed. When an interchange interface suppresses the discretionary data, the PAN and expiration date are retained. The rest of the track data is replaced with ASCII zeroes. BASE24-pos Authorization must be able to process a response with a track 2 that may have the discretionary data suppressed.

Many interchanges will no longer send the Track 2 Data (P-35) data element on reversals, pre-auth completions, and advices. When the BASE24 interfaces and BIC ISO Interface receives these messages, the track2 field in BASE24's internal message must be manually constructed using the primary account number (PAN) (P-2) and the expiration date (P-14 or zeroes if P-14 is not present). When the track data is manually constructed, BASE24 will set the first character of the PSTM Track2 field to "M".

The new LCONF parameter, POS-DISCR-DATA-PSTM-SUPPRS, is used to indicate whether or not discretionary data suppression will be performed.

This enhancement will apply to those transactions that are authorized by BASE24, a host system, or an interchange.

Software Impacts

All BASE24 BIC ISO Host Interface customers:

- Inline changes to SW60BABI.BABICIG and BABICIS
- Updated event (log) message manual SW60BABI.BABILOGM

All BASE24-pos customers:

- Inline changes to PS60RTAU.RTAUG, ROUTERS, and AUTHLIBS
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos BIC ISO Host Interface customers:

- Inline changes to SW60PSBI.PSBICIS

All BASE24-pos ISO Host Interface customers:

- Inline changes to PS60HISO.PSHISOS

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHS and SPDHDDL
- Updated event (log) message manual BA60LOGM.PSSPDH

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNTRK. Note that this SCN file also contains the SCNs needed to implement this enhancement in BASE24-atm.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that is required to be added to the LCONF to support this enhancement. Details of the param are provided in the BASE24 Logical Network Configuration File manual. A sample of the param is included in the LCONFPS file located on BA60MISC subvolume.

POS-DISCR-DATA-PSTM-SUPPRS

This param is used to indicate whether discretionary data, carried in the PSTM should be suppressed. Valid values are as follows:

Y = Yes, suppress discretionary data in the PSTM.

N = No, do not suppress discretionary data in the PSTM (default).

Database Configuration

External Message File (EMF)

The suppression of discretionary data can affect the content of messages exchanged between the host and the BASE24-pos ISO Host Interface process and between the co-networks and the BASE24-pos BIC ISO Interface process. This is true whether the data has already been suppressed in response messages that BASE24 receives from an interchange or host or the data is suppressed by BASE24.

As a result, users that prefer to send the card details to the back-end Host or co-network in the same format in which they were received should consider configuring EMF records that include the Primary Account Number (P-2) and Expiration Date (P-14) data elements and modify the default settings to “M” (mandatory) for the Track 2 (P-35) data element.

Member numbers are used by BASE24 to differentiate multiple cards issued with the same card number. Data Element 23, Card Sequence Number, is used to send and receive the member number if multiple cards are issued. Host systems and co-networks utilizing the member number other than zero should be configured to send and receive the Card Sequence Number in data element 23.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The PTDF conversion program must be run to convert the POS Terminal Data File (PTDF). Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

Host systems may no longer receive full Track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternately, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35 (Track 2 Data). Hosts must be capable of processing response, reversal, and advice messages without the discretionary data or using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Visa II Device Handler Enhancements

This section describes the implementation requirements associated with the Visa II Device Handler Enhancements.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Visa II Device Handler Enhancements.

Gift Card and Prepaid Card Transactions

The Visa II Device Handler has been enhanced to support Gift Card and Prepaid Card Transactions. This enhancement integrates with the Stored Value functionality that is present in BASE24-pos. Table 1 shows the new Visa II gift card and prepaid card transaction codes, their required record formats (message formats), and their corresponding BASE24-pos transaction codes.

V2 Tran Code	Description	Record Format	B24 Tran Code	Description
5L	Credit/stored value balance inquiry	D, E	17	Balance inquiry
G3	Gift card balance inquiry	T, U	17	Balance inquiry
G4	Gift card purchase/redemption	T, U	10	Normal purchase
G5	Gift card return/refund	T, U	14	Merchandise return
G6	Gift card add value/load card	T, U	27	Replenishment
G7	Gift card decrease value/unload card	T, U	28	Full redemption
P1	Prepaid card activation	D, E	25	Card activation
P2	Prepaid card activation reversal	D, E	25	Card activation
P3	Prepaid card load	D, E	27	Replenishment
P4	Prepaid card load reversal	D, E	27	Replenishment
R1	Prepaid card activation	T, U	25	Card activation
R2	Prepaid card activation reversal	T, U	25	Card activation
R3	Prepaid card load	T, U	27	Replenishment
R4	Prepaid card load reversal	T, U	27	Replenishment

Table 1 – Visa II / BASE24-pos Transaction Codes

Support for record formats T (request), U (response) and V (confirmation) has been added to the Visa II Device Handler.

The Visa II Device Handler appends the Stored Value Token (U2) to the request PSTM of all gift card and prepaid card transactions. Router/Auth requires this token to be present for some stored value transaction types.

The Visa II Device Handler returns Group III Version 200 (Gift Card Information) to the device for approved gift card transactions. Included in Group III Version 200 is the Available Balance, which is obtained from the card balance field of the Stored Value Token. The Available Balance is the amount remaining on the card following the transaction. If the authorizer did not provide a card balance, the Available Balance is not included in Group III Version 200.

Gift card transactions can be authorized for a lesser amount when the requested authorization amount exceeds the available gift card balance. Also included in Group III Version 200 is the Amount Actually Used from Card, which is the portion of the total transaction amount that is actually covered by the remaining balance of the gift card, when the total transaction amount exceeds the remaining balance. If the transaction is fully covered by the card's remaining balance, this field's value is zero.

Extended Address Verification Data

The Visa II Device Handler has been enhanced to support Extended Address Verification Data as required by American Express. Extended Address Verification Data contains detailed information about the cardholder, including billing and shipping names, addresses and phone numbers. By including Extended Address Verification Data in request messages, transaction acquirers can receive better interchange discount rates.

The Visa II Device Handler obtains Extended Address Verification Data from Group III Version 029 of the transaction request message. A portion of the data (Cardholder Billing Street Address and Cardholder Billing Postal Code) is placed in the Address and Zip Code fields of the request PSTM. The remainder of the data is placed in the AMEX Private Use Data Token (CX).

If the device sent Extended Address Verification Data in Group III Version 029 and also sent AVS Data in the Cardholder Identification Data field, the Visa II Device Handler overwrites the AVS Data (Cardholder Billing Street Address and Cardholder Billing Postal Code) received in the Cardholder Identification Data field with the Extended Address Verification Data received in Group III Version 029.

Card-Level Results

The Visa II Device Handler has been modified to return Group III Version 034 (Card-Level Results) to the device when the 0210 response PSTM contains the Switch Common Data Token (BY) and when the card level product ID is present in the token. Table 2 shows the Card-Level Result values.

Note: ^ = blank character

Value	Description	D^	Visa Infinite
A^	Visa Traditional	E^ - F^	Reserved
B^	Visa Traditional Rewards	G^	Visa Business Card
C^	Visa Signature	H^	Visa Check Card

I^	Visa Commerce	G1	Visa Signature Business
J^	Reserved	G2	Visa Business Check Card
K^	Visa Corporate Card	J1	Visa General Prepaid
L^	Reserved	J2	Visa Prepaid Gift Card
M^	MasterCard/Eurocard and Diners	J3	Visa Prepaid Healthcare
N^ - P^	Reserved	J4	Visa Prepaid Commercial
Q^	Private Label	K1	Visa GSA Corporate T&E
R^	Proprietary Card	Q1	Private Label Prepaid
S^	Visa Purchasing Card	S1	Visa Purchasing with Fleet
T^	Reserved/Interlink	S2	Visa GSA Purchasing
Value	Description	S3	Visa GSA Purchasing with Fleet
U^	Visa TravelMoney	V1	Reserved
W^ - Z^	Reserved	AX	American Express
0^ - 9^	Reserved	DI	Discover

Table 2 – Card-Level Results

Software Impacts

All BASE24-pos Visa II Device Handler customers:

- Inline changes to PS60VVDH.VVDHDDL, VVDHG and VVDHS
- Updated event (log) message manual BA60LOGM.PSV2DH

A complete list of the files affected by these enhancements can be found in the SCN file BA60UD0B.SCNV2DH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

The Visa II Device Handler adds the following BASE24 tokens to the PSTM:

- AMEX Private Use Data Token (token ID CX)
- Stored Value Token (token ID U2)

If these tokens should not be logged to the PTLF or the ILF, the appropriate token records must be configured in the Token file.

If these tokens are to be extracted with the PTLF or ILF, sent to the host through HISO, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

New event (log) messages were created as a result of these enhancements. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

To utilize these enhancements, POS devices must support Gift Card and Prepaid Card Transactions, Extended Address Verification Data and Card-Level Results as defined in the TSYS External Interface Specifications, Authorization Record Formats (EIS 1080).

Reference Documents

TSYS External Interface Specifications, Authorization Record Formats (EIS 1080), January 2007

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7: BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from Release 6.0 version 7.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to Release 6.0 version 8.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to Release 6.0 version 8.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with Release 6.0 version 8, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-teller.

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8: BASE24-remote banking

Section 8 discusses the changes and enhancements made to BASE24-remote banking in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing BASE24-remote banking users must consider when upgrading from Release 6.0 version 7. This section covers changes made to the following products within the BASE24 family:

- BASE24-telebanking
- BASE24-billpay
- BASE24 Kernel/IEP

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-remote banking users, along with the associated conversion and implementation requirements that all BASE24-remote banking users must address.
- Optional enhancements describe the changes that affect only those BASE24-remote banking users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-remote banking users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24-remote banking users need to consider when upgrading to Release 6.0 version 8.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-remote banking users must address in order to upgrade to Release 6.0 version 8.

There are no general enhancements inherent to BASE24-remote banking.

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Optional Enhancements

This section identifies the BASE24-remote banking enhancements that can be enabled or disabled with Release 6.0 version 8, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-remote banking.

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9: BASE24 Interchange Interfaces

Section 9 discusses the changes and enhancements made to BASE24 Interchange Interfaces in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from Release 6.0 version 7.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 9 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to Release 6.0 version 8.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to Release 6.0 version 8.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- PCI - Audit Enhancement

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

The following switch interface related conversion programs will be needed:

- American Express Global Network (AEGN) Interface ANEG
- EPS-Net (EURO) Interface ENRLF
- MasterCard Debit Switch (MDS) Interface MLF

Software Impacts

For all American Express AEGN Interface customers:

- New modules MLA0608M, MLA0608R and MLA0608S

For all EPS-Net (EURO) Interface customers:

- New modules MLE0608M, MLE0608R and MLE0608S

For all MasterCard MDS Interface customers:

- New modules MLM0608M, MLM0608R and MLM0608S

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

American Express Global Network (AEGN) Interface ANEG

The ANEG conversion program must be run to convert the ANEG. Running the conversion program is required for any BASE24 AEGN Interface customer that sends hot card update messages to AMEX GNS.

The ANEG conversion program on the SW60AEGN subvolume can be used to convert the ANEG records to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the MLA0608S file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#mlacnv.t,name $mlam,nowait/ mla0608m –  
force –list –listreplace
```

Users should update the MLA0608R obey file on the SW60AEGN subvolume with the appropriate information. The MLA0608R should then be obeyed to execute the conversion program.

EPS-Net (EURO) Interface ENRLF

The ENRLF conversion program must be run to convert the ENRLF. Running the conversion program is required for any BASE24 EURO Interface customer that sends hot card update messages to EPS-Net.

The ENRLF conversion program on the SW60EURO subvolume can be used to convert the ENRLF records to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the MLE0608S file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#mlecnv.t,name $mlam,nowait/ mle0608m –  
force –list –listreplace
```

Users should update the MLE0608R obey file on the SW60EURO subvolume with the appropriate information. The MLE0608R should then be obeyed to execute the conversion program.

MasterCard Debit Switch (MDS) Interface MLF

The MLF conversion program must be run to convert the MLF. Running the conversion program is required for any BASE24 MDS Interface customer that sends hot card update messages to MDS.

The MLF conversion program on the SW60MDS subvolume can be used to convert the MLF records to the Release 6.0 version 8 format. If modifications to the conversion program are necessary, the source code may be modified by editing the MLM0608S file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#mlmcnv.t,name $mlmm,nowait/ mlm0608m  
-force -list -listreplace
```

Users should update the MLM0608R obey file on the SW60MDS subvolume with the appropriate information. The MLM0608R should then be obeyed to execute the conversion program.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

PCI – Audit Enhancement

This section describes the implementation requirements associated with the PCI Audit enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI Audit enhancement.

BASE24 must support the auditing of user access to sensitive information, in accordance with the PCI Data Security Standards.

The PCI Data Security Standards state that that the system must implement automated audit trails for all system components to reconstruct events, including: “10.2.1 All individual user access to cardholder data.”

Requirement 10.2.1 applies to any access - read, update, create, or delete. BASE24 auditing needs to be enhanced to address user reads of cardholder data. This is logging of activity by human users – there is no requirement to log accesses that occur as part of transaction processing.

This enhancement impacts the following interfaces:

- American Express Global Network
- Banknet
- Deluxe ISO Generic Gateway
- EPS-Net
- FDR ISO
- JCB ISO
- MAC MASM
- MasterCard Debit Switch
- Publix
- Shazam ISO
- SPAN2
- VisaNet ISO

Refer to the interface’s respective manual and updated guide for further details on the implementation of this enhancement.

For more complete details, refer to the PCI – Audit enhancement within Section 4 of this document.

Software Impacts

All BASE24 AEGN Interface customers:

- Inline changes to SW60AEGN.RQANEGS, SVANEGS

All BASE24 BNET Interface customers:

- Inline changes to SW60BNET.SVBNRLFS

All BASE24 DDSI Interface customers:

- Inline changes to SW60DDSI.RQDNEGS, SVDMLFS, RQDNMLFS, SVDNMLFS, RQDPMLFS, SVDPMLFS

All BASE24 EURO Interface customers:

- Inline changes to SW60EURO.RQENRLFS, SVENRLFS

All BASE24 FDRI Interface customers:

- Inline changes to SW60FDRI.RQFNEGS, SVFNEGS

All BASE24 JCBI Interface customers:

- Inline changes to SW60JCBI.SVJBRLFS

All BASE24 MACI Interface customers:

- Inline changes to SW60MACI.RQMMLFS, SVMMLFS

All BASE24 MDS Interface customers:

- Inline changes to SW60MDS.RQMNEGS, SVMNEGS

All BASE24 PBLX Interface customers:

- Inline changes to SW60PBLX.RQNLFS, SVNLF

All BASE24 SHZM Interface customers:

- Inline changes to SW60SHZM.RQSNEGS, SVSMLFS

All BASE24 SPN2 Interface customers:

- Inline changes to SW60SPN2.RQSSDFS, SVSSDFS

All BASE24 VISA Interface customers:

- Inline changes to SW60VISA.SVRLFS, RQVSDFS, SVVSDFS

A complete list of the files affected by the PCI – Audit enhancement can be found in the SCN file BA60UD0B.SCNAUDT for the PCI – Audit enhancement.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with Release 6.0 version 8, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- PCI - Masking on Pathway
- PCI - Masking on Reports
- Track 2 Context Data Removal

BASE24 Interchange Interface users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding Release 6.0 version 8 changes, conversion requirements, and implementation requirements for future reference.

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PCI – Masking on Pathway

This section describes the implementation requirements associated with the PCI Masking on Pathway enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI Masking on Pathway enhancement.

All customers who wish to “mask” sensitive information that is currently displayed on the BASE24 screens (in accordance with the PCI Data Security Standards) will need to implement this enhancement.

This enhancement impacts the following interfaces:

- American Express Global Network
- Banknet
- Deluxe ISO Generic Gateway
- EPS-Net
- FDR ISO
- JCB ISO
- MAC MASM
- MasterCard Debit Switch
- Publix
- Shazam ISO
- SPAN2
- VisaNet ISO

Refer to the interface’s respective manual and updated guide for further details on the implementation of this enhancement.

Software Impacts

All BASE24 AEGN Interface customers:

- Inline changes to SW60AEGN.AEGNS, AMEXDDL, RQANEGS, SVANEGS

All BASE24 BNET Interface customers:

- Inline changes to SW60BNET.SCRNBRLF, RQBNRLFS, SVBNRLFS

All BASE24 DDSI Interface customers:

- Inline changes to SW60DDSI.DDSIS, RQDNEGS, SVDMLFS, RQDNMLFS, SVDNMLFS, RQDPMLFS, SVDPMFLFS

All BASE24 EURO Interface customers:

- Inline changes to SW60EURO.EUROS, EURODDL, RQENRLFS, SVENRLFS

All BASE24 FDRI Interface customers:

- Inline changes to SW60FDRI.FDRIS, RQFNEGS, SVFNEGS

All BASE24 JCBI Interface customers:

- Inline changes to SW60JCBI.RQJBRLFS, SVJBRLFS

All BASE24 MACI Interface customers:

- Inline changes to SW60MACI.RQMMLFS, SVMMLFS

All BASE24 MDS Interface customers:

- Inline changes to SW60MDS.MDSS, MDSDDL, RQMNEGS, SVMNEGS

All BASE24 PBLX Interface customers:

- Inline changes to SW60PBLX.RQNLFS, SVNLF

All BASE24 SHZM Interface customers:

- Inline changes to SW60SHZM.RQSNEGS, SVSMLFS

All BASE24 SPN2 Interface customers:

- Inline changes to SW60SPN2.SCRNSSDF, RQSSDFS, SVSSDFS

All BASE24 VISA Interface customers:

- Inline changes to SW60VISA.RQRLFS, SVRLFS, SCRNVSDF, RQVSDFS, SVVSDFS

For complete details, refer to the PCI – Masking on Pathway enhancement within Section 4 of this document.

A complete list of the files affected by the PCI – Masking on Pathway enhancement can be found in the SCN file BA60UD0B.SCNMASK for the PCI – Masking on Pathway enhancement.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

American Express Global Network (AEGN) Interface ANEG

The ANEG conversion program must be run to convert the AMEX GNS ANEG file. Details of this conversion program can be found in the **General Changes** earlier in Section 9.

EPS-Net (EURO) Interface ENRLF

The ENRLF conversion program must be run to convert the EPS-Net ENRLF file. Details of this conversion program can be found in the **General Changes** earlier in Section 9.

MasterCard Debit Switch (MDS) Interface MLF

The MLF conversion program must be run to convert the MasterCard MLF file. Details of this conversion program can be found in the **General Changes** earlier in Section 9.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

PCI – Masking on Reports

This section describes the implementation requirements associated with the PCI Masking on Reports enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI Masking on Reports enhancement.

All customers who wish to “mask” sensitive information that is currently displayed on the switch reports (in accordance with the PCI Data Security Standards) will need to implement this enhancement.

This enhancement impacts the following interfaces:

- American Express International
- BIC ISO
- BIC ANSI
- Banknet
- DIAS
- LINK LIS5
- MAC MASM
- MasterCard Debit Switch
- NEN/Excel ISO
- SPAN2
- VisaNet ISO

Refer to the interface’s respective manual and updated guide for further details on the implementation of this enhancement.

Software Impacts

All BASE24 AMXI Interface customers:

- Inline changes to SW60AMXI.RP09AMIS

All BASE24 BIC ISO Interface customers:

- Inline changes to SW60BABI.RP09ISOS

All BASE24 BIC ANSI Interface customers:

- Inline changes to SW60BIC.RP09BICS

All BASE24 BNET Interface customers:

- Inline changes to SW60BNET.RP09BNTS

All BASE24 DIAS Interface customers:

- Inline changes to SW60DIAS.RP09DIAS

All BASE24 LINK Interface customers:

- Inline changes to SW60LINK.RP09LNKS

All BASE24 MACI Interface customers:

- Inline changes to SW60MACI.RP09MACS

All BASE24 MDS Interface customers:

- Inline changes to SW60MDS.RP09MDSS

All BASE24 NEN Interface customers:

- Inline changes to SW60NEN.RP09NENS

All BASE24 SPN2 Interface customers:

- Inline changes to SW60SPN2.RP09SPNS

All BASE24 VISA Interface customers:

- Inline changes to SW60VISA.RP09VISS

For complete details, refer to the PCI – Masking on Reports enhancement within Section 4 of this document.

A complete list of the files affected by the PCI – Masking on Reports enhancement can be found in the SCN file BA60UD0B.SCNRMASK for the PCI – Masking on Reports enhancement.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

Track 2 Context Data Removal

This section describes the implementation requirements associated with the Track 2 Context Data Removal enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Track 2 Context Data Removal enhancement.

When the track discretionary data is suppressed, the primary account number (PAN) and expiration date are retained. The rest of the track data is replaced with ASCII zeroes.

Two new LCONF parameters, ATM-DISCR-DATA-STM-SUPPRS and POS-DISCR-DATA-PSTM-SUPPRS, are used to indicate whether or not discretionary data suppression will be performed on the STM/PSTM before it is sent into BASE24 from the BIC Interface process.

The suppression of discretionary data can affect the content of messages exchanged between the co-networks and the BIC ISO Interface process. This is true whether the data has already been suppressed in response messages that BASE24 receives from an interchange or host or the data is suppressed by BASE24.

Software Impacts

All BASE24 BIC ISO Interface customers:

- Inline changes to SW60BABI.BABICIG and BABICIS
- Updated event (log) message manual SW60BABI.BABILOGM

All BASE24-atm BIC ISO Interface customers:

- Inline changes to SW60ATBI.ATBICIS

All BASE24-pos BIC ISO Interface customers:

- Inline changes to SW60PSBI.PSBICIS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0B.SCNTRK.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added in the LCONF to support this enhancement. These modifications are provided in the LCONFPS and LCONFAT files located on the BA60MISC subvolume.

ATM-DISCR-DATA-STM-SUPPRS

This param is used to indicate whether discretionary data, carried in the STM should be suppressed. Valid values are as follows:

Y = Yes, suppress discretionary data in the STM.

N = No, do not suppress discretionary data in the STM (default).

POS-DISCR-DATA-PSTM-SUPPRS

This param is used to indicate whether discretionary data, carried in the PSTM should be suppressed. Valid values are as follows:

Y = Yes, suppress discretionary data in the PSTM.

N = No, do not suppress discretionary data in the PSTM (default).

Database Configuration

External Message File (EMF)

The suppression of discretionary data can affect the content of messages exchanged between the co-networks and the BASE24-pos BIC ISO Interface process. This is true whether the data has already been suppressed in response messages that BASE24 receives from an interchange or the data is suppressed by BASE24.

As a result, users that prefer to send the card details to the co-network in the same format in which they were received should consider configuring EMF records that include the Primary Account Number (P-2) and Expiration Date (P-14) data elements and modify the default settings to "M" (mandatory) for the Track 2 (P-35) data element.

Member numbers are used by BASE24 to differentiate multiple cards issued with the same card number. Data Element 23, Card Sequence Number, is used to send and receive the member number if multiple cards are issued. Co-networks utilizing the member number other than zero should be configured to send and receive the Card Sequence Number in data element 23.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Co-networks may no longer receive full Track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternately, co-networks may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35 (Track 2 Data). Co-networks must be capable of processing response, reversal, and advice messages without the discretionary data or using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

10: BASE24 Value-Added Products

Section 10 discusses the changes and enhancements made to the BASE24 value-added products in Release 6.0 version 8, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from Release 6.0 version 7.

The value-added products discussed in this section include the following:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory Voucher Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 10 must be used in conjunction with Section 4 to identify the changes that BASE24 value-added product users need to consider when upgrading to Release 6.0 version 8.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all BASE24 Value-added product users must address in order to upgrade to Release 6.0 version 8.

There are no general enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory Voucher Manager

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with Release 6.0 version 8 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory Voucher Manager

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A: Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with Release 6.0 version 8 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for Release 6.0 version 8. Refer to the specified page numbers for explanations of how these assigns can or must be set.

ATM Dynamic Currency Conversion

ERF	5-35
ERFEMT	5-35

ATM Dynamic Currency Conversion

DCCD	5-35
DCCDEMT	5-35

Shared NDC+ BNA Support

ATM-NIDF	5-94
ATM-NIDFEMT	5-94
ERF	5-94
ERFEMT	5-94

Stored Value BDM Support

CMS-BDM-OMF	6-66
CMS-BDM-RPT	6-66

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PARAMS

The following params were added or modified for Release 6.0 version 8. Refer to the specified page numbers for explanations of how these params can or must be set.

APACS 70-2 Enhancements

FUND-TRAN-TO-CASH-ADV6-17, 6-18

ATC Enhancements

CAF-SCND-CRD-PROMOTE6-22

ATM Dynamic Currency Conversion

DCC-CRNCY-RATE-SOURCE5-35

DCCDEMT-MAX-TBL-SIZE5-36

DCCDEMT-WARMIdentifier5-36

ERFEMT-MAX-TBL-SIZE..... 5-36

ERFEMT-WARMIdentifier 5-36

Configurable AUTH for Rear Balancing Transactions

ATM-REAR-ADMN-TRAN-PIN-ALWD5-48

Diebold 912 AKDS Signature Support

ATM-DH-1000-AKDS-SERIAL-NUM-VRFY5-52

ATM-DH-AKDS-TSS-TIMER.....5-52

Enhanced Mobile Top-Up Routing Support

SPLIT-TXN-RTG-DEST5-59, 5-60, 5-61, 5-63

Immediate Availability PTD Static Data Changes

POS-DH-PTD-STATIC-REPL6-50

PCI - Masking on Pathway

AFT-PAN-DIGITS.....4-38, 4-41

Shared NDC+ BNA Support

ATM-NIDFEMT-MAX-TBL-SIZE.....5-95

ATM-NIDFEMT-WARMIdentifier5-95

BASE-CURRENCY-CODE.....5-95

CURRENCY-CODE5-95

Shared NDC+ CPM Support

ATM-DH-MAX-TERMS-IN-CPLF-TBL.....5-111

ATM-DH-TERM-IN-CPLF-RETN-PRD5-111

Stored Value BDM Support

CMS-BDM-OMF-AUDIT6-66

CMS-BDM-OMF-RETENTION-PERIOD6-66

Tidel Dynamic Key Management

ATM-DH-CONS-MAC-ERR-LMT	5-118
ATM-DH-DKM-KEY-CHNG-INTRVL	5-118
ATM-DH-DKM-MAC-ERR-LMT	5-118
ATM-DH-DKM-MAC-TXN-LMT	5-119
ATM-DH-DKM-PIN-ERR-LMT	5-119
ATM-DH-DKM-PIN-TXN-LMT	5-119

Track 2 Context Data Removal

ATM-DISCR-DATA-STM-SUPPRS	5-127, 5-129, 9-23, 9-24
POS-DISCR-DATA-PSTM-SUPPRS.....	6-77, 6-79, 9-23, 9-24

B: Host/CO-Network Impacts

Appendix B contains a list of Release 6.0 version 8 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

ATC Enhancements.....	6-24
CAP Authentication.....	6-37
Dynamic CAF.....	6-41
Multi-language Indicator for Statement Print	5-19
PCI – Audit Enhancement.....	4-26
PCI – Masking on Pathway.....	4-43
PCI – Masking on Reports.....	4-53
POS Completion Message Configuration	6-55
Preferred Transaction	5-89
Shared NDC+ BNA Support	5-103
Shared NDC+ CPM Support.....	5-115
Track 2 Context Data Removal (atm)	5-130
Track 2 Context Data Removal (pos).....	6-80

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C: CUSTMACS File Changes

Appendix C contains a list of Release 6.0 version 8 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 8 code.

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Enhancements

The following enhancements have CUSTMACS impacts:

ATM Dynamic Currency Conversion

atm_dcc_on.....	5-34
atm_dh_c1000_dcc_on.....	5-34
atm_dh_n50_dcc_on.....	5-34

CAP Authentication

pos_cap_on.....	6-32
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NCR Standard APTRA Edge NDC+ Support

atm_dh_n50_edge_on	5-75, 5-80
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Shared NDC+ BNA Support

atm_bna_mcu_on	5-94
atm_dh_n50_shrd_bna_mcu_on.....	5-94
atm_dh_n50_shrd_bna_on	5-94

Shared NDC+ CPM Support

atm_dh_n50_shrd_cpm_on.....	5-111
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SPDH Message Data Element Encryption

pos_dh_spdh_de_on.....	6-60
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Tidel Dynamic Key Management

atm_dh_tidel_dkm_on.....	5-118
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D: Re-Sequenced Files and New Files

Appendix D consists of the following information:

- No files were resequenced as part of Release 6.0 version 8.
- The tables later in this section illustrating the new files added to BASE24 for the Release 6.0 version 8 enhancements.

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New Files Added to BASE24 with Version 8

The following table lists the files added by the Release 6.0 version 8 enhancements:

Subvol	Filename	Language	Enhancement Name
AT60AFT	RQNIDFSS	N/A	Shared NDC+ BNA Support
	RQNIDFXM	N/A	Shared NDC+ BNA Support
	RQNIDFXS	SCOBOL	Shared NDC+ BNA Support
	SVNIDFTG	N/A	Shared NDC+ BNA Support
	SVNIDFTM	N/A	Shared NDC+ BNA Support
	SVNIDFTS	TAL	Shared NDC+ BNA Support
AT60CWN5	ATCWN5FM	N/A	Shared NDC+ CPM Support
	ATCWN5M	N/A	Shared NDC+ CPM Support
	ATCWN5MM	N/A	Shared NDC+ CPM Support
	N50WCPMG	TAL	Shared NDC+ CPM Support
	N50WCPMM	N/A	Shared NDC+ CPM Support
	N50WCPMS	TAL	Shared NDC+ CPM Support
AT60DDL	DDLFNIDF	DDL	Shared NDC+ BNA Support
AT60DDLD	ATD0B	DDL	Shared NDC+ BNA Support
AT60DWN5	ATDWN5FM	N/A	Shared NDC+ BNA Support
	ATDWN5M	N/A	Shared NDC+ BNA Support
	ATDWN5MM	N/A	Shared NDC+ BNA Support
	N50WBNAG	N/A	Shared NDC+ BNA Support
	N50WBNAM	N/A	Shared NDC+ BNA Support
	N50WBNAS	N/A	Shared NDC+ BNA Support
	N50WBNMG	TAL	Shared NDC+ BNA Support
	N50WBNMM	N/A	Shared NDC+ BNA Support
	N50WBNMS	TAL	Shared NDC+ BNA Support
AT60EMTB	NIDEMTTG	N/A	Shared NDC+ BNA Support
	NIDEMTTM	N/A	Shared NDC+ BNA Support
	NIDEMTTS	TAL	Shared NDC+ BNA Support
AT60EN50	ATEN50FM	N/A	NCR Standard APTRA Edge NDC+ Support
	ATEN50MM	N/A	NCR Standard APTRA Edge NDC+ Support
	N50EDGE G	TAL	NCR Standard APTRA Edge NDC+ Support
	N50EDGE M	N/A	NCR Standard APTRA Edge NDC+ Support
	N50EDGE S	TAL	NCR Standard APTRA Edge NDC+ Support
AT60KTDL	ATTDLFM	N/A	Tidel Dynamic Key Management
	ATTDLMM	N/A	Tidel Dynamic Key Management
	TIDLDKMG	TAL	Tidel Dynamic Key Management
	TIDLDKMM	N/A	Tidel Dynamic Key Management
	TIDLDKMS	TAL	Tidel Dynamic Key Management
AT60R100	ATR100FM	N/A	ATM Dynamic Currency Conversion
	ATR100MM	N/A	ATM Dynamic Currency Conversion
	C100DCCG	TAL	ATM Dynamic Currency Conversion
	C100DCCM	N/A	ATM Dynamic Currency Conversion
	C100DCCS	TAL	ATM Dynamic Currency Conversion
	C1DCNAMS	TAL	ATM Dynamic Currency Conversion

Re-Sequenced Files and New Files

Subvol	Filename	Language	Enhancement Name
AT60RATH	ATRATHFM	N/A	ATM Dynamic Currency Conversion
	ATRATHMM	N/A	ATM Dynamic Currency Conversion
	AUTHDCCG	TAL	ATM Dynamic Currency Conversion
	AUTHDCCM	N/A	ATM Dynamic Currency Conversion
	AUTHDCCS	TAL	ATM Dynamic Currency Conversion
AT60RN50	ATRN50FM	N/A	ATM Dynamic Currency Conversion
	ATRN50MM	N/A	ATM Dynamic Currency Conversion
	N50DCCG	TAL	ATM Dynamic Currency Conversion
	N50DCCM	N/A	ATM Dynamic Currency Conversion
	N50DCCS	TAL	ATM Dynamic Currency Conversion
	N5DCNAMS	TAL	ATM Dynamic Currency Conversion
AT60UC0B	AAREAD0B	N/A	ATM Dynamic Currency Conversion Shared NDC+ BNA Support Tidel MACing Support And Triple-DES Update
	ATUC0BFM	N/A	ATM Dynamic Currency Conversion Shared NDC+ BNA Support Tidel MACing Support And Triple-DES Update
	ATUC0BM	N/A	ATM Dynamic Currency Conversion Shared NDC+ BNA Support Tidel MACing Support And Triple-DES Update
	ATUC0BMM	N/A	ATM Dynamic Currency Conversion Shared NDC+ BNA Support Tidel MACing Support And Triple-DES Update
	CNVATDM	N/A	ATM Dynamic Currency Conversion Shared NDC+ BNA Support
	CNVATDR	N/A	ATM Dynamic Currency Conversion Shared NDC+ BNA Support
	CNVATDS	TAL	ATM Dynamic Currency Conversion Shared NDC+ BNA Support
	CNVTD FM	N/A	Tidel MACing Support And Triple-DES Update
	CNVTD FR	N/A	Tidel MACing Support And Triple-DES Update
	CNVTD FS	TAL	Tidel MACing Support And Triple-DES Update
BA60AFT	RQDCCDM	N/A	ATM Dynamic Currency Conversion
	RQDCCDS	COBOL	ATM Dynamic Currency Conversion
	RQIDFPTM	N/A	Preferred Transaction
	RQIDFPTS	COBOL	Preferred Transaction
	SCRNDCCD	N/A	ATM Dynamic Currency Conversion
	SVDCCDM	N/A	ATM Dynamic Currency Conversion
	SVDCCDS	COBOL	ATM Dynamic Currency Conversion
BA60CAFD	BACAFDFM	N/A	Dynamic CAF
	BACAFDM	N/A	Dynamic CAF
	BACAFDMM	N/A	Dynamic CAF
	CAFDM	N/A	Dynamic CAF
	CAFDMR	TACL	Dynamic CAF
	CAFDM S	TACL	Dynamic CAF PCI - Masking on Reports

Subvol	Filename	Language	Enhancement Name
BA60DDL	DDLFCAFD	DDL	Dynamic CAF
	DDLFDCCD	DDL	ATM Dynamic Currency Conversion
BA60EMTB	DCDEMTTG	TAL	ATM Dynamic Currency Conversion
	DCDEMTTM	N/A	ATM Dynamic Currency Conversion
	DCDEMTTS	TAL	ATM Dynamic Currency Conversion
BA60LOGM	PSUTLS	N/A	Immediate Availability PTD Static Data Changes
BA60UC0B	AAREAD0B	N/A	ATC Enhancements ATM Dynamic Currency Conversion CAP Authentication Dynamic CAF Enhanced Mobile Top-Up Routing Support PCI - Masking on Pathway PCI - Masking on Reports Preferred Transaction Shared NDC+ BNA Support Skipping HISO SAF SPDH Message Data Element Encryption Tidel MACing Support And Triple-DES Update Track 2 Context Data Removal
	BAUC0BFM	N/A	ATC Enhancements CAP Authentication Dynamic CAF Enhanced Mobile Top-Up Routing Support PCI - Masking on Pathway PCI - Masking on Reports Preferred Transaction
	BAUC0BM	N/A	ATC Enhancements CAP Authentication Dynamic CAF Enhanced Mobile Top-Up Routing Support PCI - Masking on Pathway PCI - Masking on Reports Preferred Transaction Skipping HISO SAF
	BAUC0BMM	N/A	ATC Enhancements CAP Authentication Dynamic CAF Enhanced Mobile Top-Up Routing Support PCI - Masking on Pathway PCI - Masking on Reports Preferred Transaction Skipping HISO SAF
	CNVCPFM	N/A	ATC Enhancements CAP Authentication Enhanced Mobile Top-Up Routing Support
	CNVCPFR	TACL	ATC Enhancements CAP Authentication Enhanced Mobile Top-Up Routing Support

Re-Sequenced Files and New Files

Subvol	Filename	Language	Enhancement Name
BA60UC0B	CNVCPFS	TAL	ATC Enhancements CAP Authentication Enhanced Mobile Top-Up Routing Support
	CNVICFEM	N/A	PCI - Masking on Reports
	CNVICFER	TACL	PCI - Masking on Reports
	CNVICFES	TAL	PCI - Masking on Reports
	CNVICFM	N/A	PCI - Masking on Reports
	CNVICFR	TACL	PCI - Masking on Reports
	CNVICFS	TAL	PCI - Masking on Reports
	CNVIDFM	N/A	Dynamic CAF PCI - Masking on Reports Preferred Transaction
	CNVIDFR	TACL	Dynamic CAF PCI - Masking on Reports Preferred Transaction
	CNVIDFS	TAL	Dynamic CAF PCI - Masking on Reports Preferred Transaction
	CNVSAFM	N/A	Skipping HISO SAF
	CNVSAFR	TACL	Skipping HISO SAF
	CNVSAFS	TAL	Skipping HISO SAF
	CNVSECM	N/A	PCI - Masking on Pathway
	CNVSECR	TACL	PCI - Masking on Pathway
	CNVSECS	TAL	PCI - Masking on Pathway
BA60UD0B	SCNA702	N/A	APACS 70-2 Enhancements
	SCNAGIL	N/A	Agilis 2.0 Enhancements
	SCNATC	N/A	ATC Enhancements
	SCNAUDT	N/A	PCI - Audit Enhancement
	SCNCAFD	N/A	Dynamic CAF
	SCNCAP	N/A	CAP Authentication
	SCNCARB	N/A	Configurable AUTH for Rear Balancing Transactions
	SCNCBC	N/A	Card before Cash on Fast Cash
	SCNCOMP	N/A	POS Completion Message Configuration
	SCNDAKD	N/A	Diebold 912 AKDS Signature Support
	SCNDCC	N/A	ATM Dynamic Currency Conversion
	SCNEAID	N/A	NCR NDC+ Increase of Secondary EMV AIDs
	SCNEDGE	N/A	NCR Standard APTRA Edge NDC+ Support
	SCNEMTP	N/A	Enhanced Mobile Top-Up Routing Support
	SCNGINQ	N/A	Enhanced Generic Inquiry
	SCNH3D	N/A	Hypercom 3DES in Logon and Other Enhancements
	SCNHSAF	N/A	Skipping HISO SAF
	SCNMASK	N/A	PCI - Masking on Pathway
	SCNMDEE	N/A	SPDH Message Data Element Encryption
	SCNMSTAT	N/A	NCR NDC+ Additional M-Status Codes
	SCNNGBLS	N/A	NCR NDC+ Device Handler Global Space Issue
	SCNPCB	N/A	Lesser Amount for Pre-Auth Purchase with Cash Back
	SCNPSWD	N/A	PCI - Initial User Password

Subvol	Filename	Language	Enhancement Name
BA60UD0B	SCNPTDS1	N/A	Immediate Availability PTD Static Data Changes
	SCNPTXN	N/A	Preferred Transaction
	SCNRMSK	N/A	PCI - Masking on Reports
	SCNRPAK	N/A	Refresh Support of PauseACK Improvements
	SCNSTMT	N/A	Multi-language Indicator for Statement Print
	SCNSTVB	N/A	Stored Value BDM Support
	SCNSTVC	N/A	Stored Value Support in CRT Authorization
	SCNTDKM	N/A	Tidel Dynamic Key Management
	SCNTMAC	N/A	Tidel MACing Support And Triple-DES Update
	SCNTRK	N/A	Track 2 Context Data Removal
	SCNV2DH	N/A	Visa II Device Handler Enhancements
	SCNWAKD	N/A	Wincor AKDS support of NCR ATMs
	SCNWNBN	N/A	Shared NDC+ BNA Support
	SCNWNCI	N/A	Shared NDC+ CPM Support
CT60AFT	BDMG	TAL	Stored Value BDM Support
	BDMM	N/A	Stored Value BDM Support
	BDMS	TAL	Stored Value BDM Support
	BDMTOGM	N/A	Stored Value BDM Support
	RQBDM	N/A	Stored Value BDM Support
	RQBDM	TAL	Stored Value BDM Support
	SCRNBDM	Screen File	Stored Value BDM Support
	SCRNSBDM	Screen File	Stored Value BDM Support
CT60DDL	DDLFBMIF	DDL	Stored Value BDM Support
PS60JRТА	PSJRTAFM	N/A	CAP Authentication
	PSJRTAMM	N/A	CAP Authentication
	RTAUCAPG	TAL	CAP Authentication
	RTAUCAPM	N/A	CAP Authentication
	RTAUCAPS	TAL	CAP Authentication
PS60UC04	CNVACNFS		SPDH Message Data Element Encryption
PS60UC0B	AAREAD0B	N/A	SPDH Message Data Element Encryption Track 2 Context Data Removal
	CNVACNFM	N/A	SPDH Message Data Element Encryption
	CNVACNFR	N/A	SPDH Message Data Element Encryption
	CNVACNFS	TAL	SPDH Message Data Element Encryption
	CNVPTDM	N/A	Track 2 Context Data Removal
	CNVPTDR	TACL	Track 2 Context Data Removal
	CNVPTDS	TAL	Track 2 Context Data Removal
	PSUC0BFM	N/A	SPDH Message Data Element Encryption Track 2 Context Data Removal
	PSUC0BM	N/A	SPDH Message Data Element Encryption Track 2 Context Data Removal
	PSUC0BMM	N/A	SPDH Message Data Element Encryption Track 2 Context Data Removal
SW60AEGN	MLA0608M	TAL	PCI - Masking on Pathway
	MLA0608R	TAL	PCI - Masking on Pathway
	MLA0608S	TAL	PCI - Masking on Pathway

Re-Sequenced Files and New Files

Subvol	Filename	Language	Enhancement Name
SW60EURO	MLE0608M	TAL	PCI - Masking on Pathway
	MLE0608R	TAL	PCI - Masking on Pathway
	MLE0608S	TAL	PCI - Masking on Pathway
SW60MDS	MLM0608M	TAL	PCI - Masking on Pathway
	MLM0608R	TAL	PCI - Masking on Pathway
	MLM0608S	TAL	PCI - Masking on Pathway

The following table lists the same files but by version 8 enhancement name:

Enhancement Name	Subvol	Filename	Language
Agilis 2.0 Enhancements	BA60UD0B	SCNAGIL	N/A
APACS 70-2 Enhancements	BA60UD0B	SCNA702	N/A
ATC Enhancements	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVCPFM	N/A
		CNVCPFR	TACL
		CNVCPFS	TAL
	BA60UD0B	SCNATC	N/A
ATM Dynamic Currency Conversion	AT60R100	ATR100FM	N/A
		ATR100MM	N/A
		C100DCCG	TAL
		C100DCCM	N/A
		C100DCCS	TAL
		C1DCNAMS	TAL
	AT60RATH	ATRATHFM	N/A
		ATRATHMM	N/A
		AUTHDCCG	TAL
		AUTHDCCM	N/A
		AUTHDCCS	TAL
	AT60RN50	ATRN50FM	N/A
		ATRN50MM	N/A
		N50DCCG	TAL
		N50DCCM	N/A
		N50DCCS	TAL
		N5DCNAMS	TAL
	AT60UC0B	AAREAD0B	N/A
		ATUC0BFM	N/A
		ATUC0BM	N/A
		ATUC0BMM	N/A
		CNVATDM	N/A
		CNVATDR	N/A
		CNVATDS	TAL
	BA60AFT	RQDCCDM	N/A
		RQDCCDS	COBOL
		SCRNDCCD	N/A
		SVDCCDM	N/A
		SVDCCDS	COBOL
	BA60DDL	DDLFDCCD	DDL
	BA60EMTB	DCDEMTTG	TAL
		DCDEMTTM	N/A
		DCDEMTTS	TAL
	BA60UC0B	AAREAD0B	N/A
	BA60UD0B	SCNDCC	N/A

Re-Sequenced Files and New Files

Enhancement Name	Subvol	Filename	Language
CAP Authentication	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVCPFM	N/A
		CNVCPFR	TACL
		CNVCPFS	TAL
	BA60UD0B	SCNCAP	N/A
	PS60JRTA	PSJRTAFM	N/A
		PSJRTAMM	N/A
		RTAUCAPG	TAL
		RTAUCAPM	N/A
		RTAUCAPS	TAL
Card before Cash on Fast Cash	BA60UD0B	SCNCBC	N/A
Configurable AUTH for Rear Balancing Transactions	BA60UD0B	SCNCARB	N/A
Diebold 912 AKDS Signature Support	BA60UD0B	SCNDAKD	N/A
Dynamic CAF	BA60CAFD	BACAFDFM	N/A
		BACAFDM	N/A
		BACAFDMM	N/A
		CAFDM	N/A
		CAFDMR	TACL
		CAFDFS	TACL
	BA60DDL	DDLFCADF	DDL
	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVIDFM	N/A
		CNVIDFR	TACL
		CNVIDFS	TAL
	BA60UD0B	SCNCAFD	N/A
Enhanced Generic Inquiry	BA60UD0B	SCNGINQ	N/A
Enhanced Mobile Top-Up Routing Support	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVCPFM	N/A
		CNVCPFR	TACL
	CNVCPFS	TAL	
	BA60UD0B	SCNEMTP	N/A
Hypercom 3DES in Logon and Other Enhancements	BA60UD0B	SCNH3D	N/A
Immediate Availability PTD Static Data Changes	BA60LOGM	PSUTLS	N/A
	BA60UD0B	SCNPTDS1	N/A
Lesser Amount for Pre-Auth Purchase with Cash Back	BA60UD0B	SCNPCB	N/A
Multi-language Indicator for Statement Print	BA60UD0B	SCNSTMT	N/A
NCR NDC+ Additional M-Status Codes	BA60UD0B	SCNMSTAT	N/A

Enhancement Name	Subvol	Filename	Language
NCR NDC+ Device Handler Global Space Issue	BA60UD0B	SCNNGBLS	N/A
NCR NDC+ Increase of Secondary EMV AIDs	BA60UD0B	SCNEAID	N/A
NCR Standard APTRA Edge NDC+ Support	AT60EN50	ATEN50FM	N/A
		ATEN50MM	N/A
		N50EDGE GEG	TAL
		N50EDGEM	N/A
		N50EDGES	TAL
	BA60UD0B	SCNEDGE	N/A
PCI - Audit Enhancement	BA60UD0B	SCNAUDT	N/A
PCI - Initial User Password	BA60UD0B	SCNPSWD	N/A
PCI - Masking on Pathway	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVSECM	N/A
		CNVSECR	TACL
		CNVSECS	TAL
	BA60UD0B	SCNMASK	N/A
	SW60AEGN	MLA0608M	TAL
		MLA0608R	TAL
		MLA0608S	TAL
	SW60EURO	MLE0608M	TAL
		MLE0608R	TAL
		MLE0608S	TAL
	SW60MDS	MLM0608M	TAL
		MLM0608R	TAL
		MLM0608S	TAL
PCI - Masking on Reports	BA60CAFD	CAFDMS	TACL
	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVICFEM	N/A
		CNVICFER	TACL
		CNVICFES	TAL
		CNVICFM	N/A
		CNVICFR	TACL
		CNVICFS	TAL
		CNVIDFM	N/A
		CNVIDFR	TACL
		CNVIDFS	TAL
	BA60UD0B	SCNRMSK	N/A
POS Completion Message Configuration	BA60UD0B	SCNCOMP	N/A

Re-Sequenced Files and New Files

Enhancement Name	Subvol	Filename	Language
Preferred Transaction	BA60AFT	RQIDFPTM	N/A
		RQIDFPTS	COBOL
	BA60UC0B	AAREAD0B	N/A
		BAUC0BFM	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVIDFM	N/A
		CNVIDFR	TACL
		CNVIDFS	TAL
	BA60UD0B	SCNPTXN	N/A
Refresh Support of PauseACK Improvements	BA60UD0B	SCNRPAK	N/A
Shared NDC+ BNA Support	AT60AFT	RQNIDFSS	N/A
		RQNIDFXM	N/A
		RQNIDFXS	SCOBOL
		SVNIDFTG	N/A
		SVNIDFTM	N/A
		SVNIDFTS	TAL
	AT60DDL	DDLFNIDF	DDL
	AT60DDLD	ATD0B	DDL
	AT60DWN5	ATDWN5FM	N/A
		ATDWN5M	N/A
		ATDWN5MM	N/A
		N50WBNAG	N/A
		N50WBNAM	N/A
		N50WBNAS	N/A
		N50WBNMG	TAL
		N50WBNMM	N/A
		N50WBNMS	TAL
	AT60EMTB	NIDEMTTG	N/A
		NIDEMTTM	N/A
		NIDEMTTS	TAL
	AT60UC0B	AAREAD0B	N/A
		ATUC0BFM	N/A
		ATUC0BM	N/A
		ATUC0BMM	N/A
		CNVATDM	N/A
		CNVATDR	N/A
		CNVATDS	TAL
	BA60UC0B	AAREAD0B	N/A
	BA60UD0B	SCNWNBN	N/A
Shared NDC+ CPM Support	AT60CWN5	ATCWN5FM	N/A
		ATCWN5M	N/A
		ATCWN5MM	N/A
		N50WCPMG	TAL
		N50WCPMM	N/A
		N50WCPMS	TAL
	BA60UD0B	SCNWNCI	N/A

Enhancement Name	Subvol	Filename	Language
Skipping HISO SAF	BA60UC0B	AAREAD0B	N/A
		BAUC0BM	N/A
		BAUC0BMM	N/A
		CNVSAFM	N/A
		CNVSAFR	TACL
		CNVSAFS	TAL
	BA60UD0B	SCNHSAF	N/A
SPDH Message Data Element Encryption	BA60UC0B	AAREAD0B	N/A
	BA60UD0B	SCNMDEE	N/A
	PS60UC04	CNVACNFS	
	PS60UC0B	AAREAD0B	N/A
		CNVACNFM	N/A
		CNVACNFR	N/A
		CNVACNFS	TAL
		PSUC0BFM	N/A
		PSUC0BM	N/A
		PSUC0BMM	N/A
Stored Value BDM Support	BA60UD0B	SCNSTVB	N/A
	CT60AFT	BDMG	TAL
		BDMM	N/A
		BDMS	TAL
		BDMTOGM	N/A
		RQBDMM	N/A
		RQBDMS	TAL
		SCRNBDM	Screen File
		SCRNSBDM	Screen File
	CT60DDL	DDLFBMIF	DDL
Stored Value Support in CRT Authorization	BA60UD0B	SCNSTVC	N/A
Tidel Dynamic Key Management	AT60KTDL	ATTDLFM	N/A
		ATTDLMM	N/A
		TIDLDKMG	TAL
		TIDLDKMM	N/A
		TIDLDKMS	TAL
	BA60UD0B	SCNTDKM	N/A
Tidel MACing Support And Triple-DES Update	AT60UC0B	AAREAD0B	N/A
		ATUC0BFM	N/A
		ATUC0BM	N/A
		ATUC0BMM	N/A
		CNVTD FM	N/A
		CNVTD FR	N/A
		CNVTD FS	TAL
	BA60UC0B	AAREAD0B	N/A
	BA60UD0B	SCNTMAC	N/A

Re-Sequenced Files and New Files

Enhancement Name	Subvol	Filename	Language
Track 2 Context Data Removal	BA60UC0B	AAREAD0B	N/A
	BA60UD0B	SCNTRK	N/A
	PS60UC0B	AAREAD0B	N/A
		CNVPTDM	N/A
		CNVPTDR	TACL
		CNVPTDS	TAL
		PSUC0BFM	N/A
		PSUC0BM	N/A
		PSUC0BMM	N/A
Visa II Device Handler Enhancements	BA60UD0B	SCNV2DH	N/A
Wincor AKDS support of NCR ATMs	BA60UD0B	SCNWAKD	N/A

E: Moved, Renamed, Deleted Files

Appendix E contains a list of files that were Moved, Renamed, or Deleted in BASE24 Release 6.0 Version 8.

No files were moved, renamed or deleted in Release 6.0 version 8.

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F: Regional Specific Information

Appendix F contains the information that is required to implement the changes for Release 6.0 version 8 that are specific to each region.

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